Improved Statistical Inference for Expectile-Based Risk Measures

An Optimal Pooling Approach in the Heavy-Tailed Regime

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May 30, 2026

Abstract

[**TODO:** Write a ∼200-word abstract once Chapter 3 is in shape. Outline of the abstract: (i) expectiles as the only coherent and elicitable law-invariant risk measure; (ii) the problem of combining tail-index and extreme-quantile information from several samples (distributed inference, multi-study evidence synthesis); (iii) our pooled extreme-expectile estimator, its joint CLT via plug-in into the expectile–quantile asymptotic equivalence, and the variance-/AMSE-optimal weights tailored to the expectile target; (iv) Weissman-style extrapolation to very-extreme expectile levels; (v) outlook on relaxing the iid assumption.]

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Chapter 1

Introduction

1.1 Motivation

Quantitative risk management routinely needs to summarise the tail of a loss distribution by a single number. The industry-standard summary has long been the Value-at-Risk (VaR), defined as a high quantile of the loss distribution. VaR is intuitive and easy to compute, but it suffers from two well-known shortcomings that have shaped the search for alternatives. First, VaR is not subadditive: it can deem a diversified portfolio more risky than its components, contradicting the diversification principle and disqualifying VaR from the class of *coherent* risk measures in the sense of Artzner et al. (1999). Expected Shortfall (ES) corrects this defect and has progressively replaced VaR in regulatory practice. However, ES is not *elicitable* (Gneiting 2011): no scoring function has its expectation uniquely minimised at the true ES, so ES forecasts cannot be ranked or backtested directly.

Expectiles are the only law-invariant risk measure that is simultaneously coherent (for confidence levels $\tau \geq 1/2$) and elicitable (Bellini et al. 2014). Originally introduced by Newey and Powell (1987), the τ-expectile of a random loss X is defined as the minimiser of an asymmetric squared loss:

$$\xi_\tau(X) \;=\; \operatorname*{arg\,min}_{\theta \in \mathbb{R}} \; \mathbb{E}\Big[\,|\tau - \mathbb{1}\{X \leq \theta\}|\;(X-\theta)^2\,\Big]\,.$$

Setting $\tau = 1/2$ recovers the mean; setting $\tau \to 1$ moves the expectile into the right tail. Because the asymmetric loss is squared rather than absolute, expectiles — unlike quantiles — are sensitive to the *magnitude* of tail losses, not just to their frequency. Ehm et al. (2016) place expectiles within a broader theory of consistent scoring functions and Choquet representations.

This thesis addresses inference for expectiles at levels τ close to one, in two situations that are increasingly relevant in practice:

(i) the loss distribution is heavy-tailed, in the sense of having an extreme-value index $\gamma > 0$, so that the empirical asymmetric-least-squares estimator becomes unstable as τ approaches one and extreme-value theory becomes the natural framework;

(ii) the data are spread across several independent samples — whether because of distributed computation, regulatory data-sharing restrictions, or natural heterogeneity between studies — and inference must be conducted by communicating only summary statistics between sources.

1.2 Extreme expectile inference

The central bridge between expectile estimation and classical extreme-value theory is the *expectile–quantile asymptotic equivalence*. Under heavy-tailed assumptions with extreme-value index $\gamma \in (0, 1/2)$,

$$\frac{\xi_\tau(X)}{Q_X(\tau)} \xrightarrow{\tau\uparrow 1} \psi(\gamma) \;:=\; (\gamma^{-1} - 1)^{-\gamma}, \tag{1.1}$$

where $Q_X(\tau)$ is the τ-quantile of X. Two families of estimators arise naturally from (1.1). The *asymmetric-least-squares* (LAWS) estimator minimises the empirical asymmetric criterion over the data; the *quantile-based* (QB) estimator substitutes consistent estimators of γ and of an extreme quantile of X into the asymptotic-equivalence formula. Their asymptotic theory at intermediate levels and via Weissman-type extrapolation to very-extreme levels was developed for iid heavy-tailed data by Daouia, Girard, et al. (2018) and subsequent work, and extended to the dependent setting by Davison et al. (2023); to the multivariate setting by Padoan and Stupfler (2022). The relevant machinery is reviewed in Chapter 2.

1.3 Pooled inference across samples

When m samples are available, with heterogeneous sizes and possibly heterogeneous tails, the natural question is how best to combine them. Daouia, Padoan, et al. (2024) develop an optimal pooling framework for the tail index γ and for extreme quantiles. Their estimators take the form of weighted arithmetic combinations (for the tail index) and weighted *geometric* combinations (for extreme quantiles), the latter specifically adapted to the multiplicative-power structure of the Weissman extrapolation. They derive the joint asymptotic normality of the per-sample estimators across m samples, allowing for asymptotic dependence between samples driven by a tail-copula, and identify the variance-optimal and AMSE-optimal pooling weights in closed form. In the important special case of *distributed inference* — where data are iid both within and across machines — the cross-sample tail-copula vanishes, off-diagonal entries of the joint asymptotic covariance collapse to zero, and the pooled estimators recover the asymptotic efficiency of the unfeasible Hill / Weissman estimator built from the combined sample.

The framework of Daouia, Padoan, et al. (2024) is the methodological foundation of this thesis. The framework does not address expectiles; the question of how to combine m pairs $(\hat\gamma_j, \hat Q_j(p_j))$ into a coherent inference procedure for an extreme expectile ξ_{τ_n} has not been studied.

1.4 Research question and scope

The thesis answers the following question:

> *Given m independent samples of iid heavy-tailed observations, with each sample j contributing point estimates $(\hat{\gamma}_j, \hat{Q}_j(p_j))$ together with their joint asymptotic covariance, how should one combine these summary statistics to perform inference and Weissman-style extrapolation for an extreme expectile of the common underlying distribution?*

The scope is restricted as follows:

- the data within each sample are univariate iid (no time-series dependence), matching the distributed-inference setting of Daouia, Padoan, et al. (2024);
- the loss distribution is heavy-tailed with extreme-value index $\gamma \in (0, 1/2)$ — the standard range that guarantees the existence of the expectile and the validity of the asymptotic equivalence (1.1);
- the input to the pooling procedure is summary statistics, not raw data, in line with practical distributed-inference constraints.

1.5 Contributions

1. We define a *pooled extreme-expectile estimator* at intermediate levels by feeding the pooled tail-index estimator $\hat{\gamma}_n(\boldsymbol{\omega}_\gamma)$ and the pooled weighted-geometric Weissman extreme-quantile estimator $\widehat{q}_n^{\star}(\,\cdot\,; \boldsymbol{\omega}_q)$ of Daouia, Padoan, et al. (2024) into the asymptotic-equivalence formula (1.1).
2. We derive the joint asymptotic distribution of the pooled inputs, accounting for the cross-source structure (which is diagonal because the samples are independent) and for the within-source dependence between $\hat{\gamma}_j$ and $\hat{Q}_j$ induced by the use of overlapping top order statistics.
3. Applying the delta method through ψ to this joint pooled CLT, we obtain the asymptotic normality of the pooled intermediate-level expectile estimator together with explicit forms of its asymptotic bias and variance.
4. We characterise the *variance-optimal* and *AMSE-optimal* pooling weights tailored to the pooled expectile estimator. These need not coincide with the weights optimal for the tail index or for the extreme quantile alone, owing to the non-trivial dependence between the two pieces of information.
5. We extend the construction to very-extreme levels $\tau_n' \uparrow 1$ with $n(1-\tau_n') \to c < \infty$ via a Weissman-style extrapolation of the pooled intermediate-level estimator, and derive the corresponding central limit theorem by feeding the joint intermediate-level limit of the previous contribution into the plug-in CLT of Davison et al. (2023, Proposition C.6). As in the single-sample case, the extrapolated estimator inherits the asymptotic variance of the pooled tail-index estimator.

6. We construct asymptotic confidence intervals for extreme expectiles and a test for *expectile-tail-homogeneity* across the m samples, modelled on the tail-quantile-homogeneity test of Daouia, Padoan, et al. (2024) and on the GLS-weighted equality-of-means test of Padoan and Stupfler (2022, Section 3.3).

1.6 Outline of the thesis

Chapter 2 collects the technical background: regular variation and the second-order condition, the Hill and Weissman estimators, expectiles and their characterisation, single-sample iid extreme-expectile inference, and the optimal pooling framework of Daouia, Padoan, et al. (2024). Chapter 3 develops the pooled estimator at intermediate expectile levels and the associated asymptotic theory. Chapter 4 treats the Weissman-style extrapolation to very-extreme expectile levels. Chapter 5 discusses limitations, the link to Expected Shortfall, and avenues for future work, including extensions to time-series dependence and multivariate observations. Detailed proofs are gathered in Appendix A.

Chapter 2

Background

Outline of the chapter

This chapter collects the technical ingredients on which the thesis builds: regular variation and the second-order condition; the Hill and Weissman estimators for the tail index and extreme quantiles; expectiles and their characterisation as a coherent and elicitable risk measure; single-sample iid extreme-expectile inference, with the plug-in central limit theorem that the contribution chapters take as their workhorse; and the optimal pooling framework of Daouia, Padoan, et al. (2024). Nothing in this chapter is original; the contributions begin in Chapter 3.

2.1 Regular variation and the second-order condition

Throughout the thesis we work with a real-valued random loss X that has continuous distribution function F and survival function $\overline{F} = 1 - F$. Given an iid sample $X_1, \ldots, X_n$ from F, the corresponding order statistics are denoted by $X_{1:n} \leq X_{2:n} \leq \cdots \leq X_{n:n}$.

Heavy-tailed asymptotics are governed by the language of regular variation; the reference is Haan and Ferreira (2006, Appendix B.1).

Definition 2.1 (Regular variation)**.** A Lebesgue measurable function $f : (0, \infty) \to \mathbb{R}$ that is eventually positive is *regularly varying at infinity with index* $\alpha \in \mathbb{R}$, written $f \in \mathrm{RV}_\alpha$, if for every $y > 0$,

$$\lim_{t\to\infty} \frac{f(ty)}{f(t)} \;=\; y^{\alpha}.$$

The functions in RV_0 are called *slowly varying*.

The first-order condition under which all of our asymptotics are taken asserts that F has a Pareto-type tail.

Condition 2.2 (First-order condition)**.** There exists $\gamma > 0$ such that $\overline{F} \in \mathrm{RV}_{-1/\gamma}$, or equivalently the *tail quantile function*

$$U(t) \;:=\; \inf\Big\{x \in \mathbb{R} \,:\, \tfrac{1}{\overline{F}(x)} \geq t\Big\}, \qquad t > 1,$$

satisfies $U \in \mathrm{RV}_\gamma$.

Throughout, for a continuous distribution function F, we write

$$Q_X(\tau) \;:=\; F^{\leftarrow}(\tau) = \inf\{x \in \mathbb{R} : F(x) \geq \tau\}, \qquad 0 < \tau < 1.$$

With the tail quantile $U(t) = F^{\leftarrow}(1 - 1/t)$, $t > 1$, this gives

$$Q_X(\tau) = U((1-\tau)^{-1}), \qquad Q_X(1-p) = U(1/p).$$

Condition 2.2 is equivalent to saying that X lies in the Fréchet max-domain of attraction with extreme-value index γ (Haan and Ferreira 2006, Theorem 1.2.1). The parameter $\gamma > 0$ is the *extreme-value index* of F; it controls how slowly $\overline{F}$ decays at infinity. Standard heavy-tailed distributions encountered in risk management satisfy Condition 2.2, including the Pareto, Burr, Student-t, Fréchet, and F-distribution families (Haan and Ferreira 2006, §2.6).

For the asymptotic-normality statements that drive the inferential theory we need a quantitative refinement of Condition 2.2 controlling the rate of convergence in Definition 2.1. The standard refinement is the second-order condition.

Condition 2.3 (Second-order condition $\mathcal{C}_2(\gamma, \rho, A)$)**.** There exist $\gamma > 0$, a second-order parameter $\rho \leq 0$, and a measurable auxiliary function A of constant sign with $A(t) \to 0$ as $t \to \infty$, such that for every $y > 0$,

$$\lim_{t\to\infty} \frac{1}{A(t)} \left[\frac{U(ty)}{U(t)} - y^\gamma\right] \;=\; y^\gamma\, \frac{y^\rho - 1}{\rho},$$

where the right-hand side is read as $y^\gamma \log y$ when $\rho = 0$.

The function $|A|$ is itself regularly varying with index ρ (Haan and Ferreira 2006, Theorem 2.3.9); smaller $|\rho|$ means slower convergence and therefore larger second-order bias of the tail-index estimators we shall encounter in Section 2.2.

Condition 2.4 (Lower-tail moment condition)**.** For some $\delta > 0$, the negative part $X^- = \max(-X, 0)$ satisfies

$$\mathbb{E}(X^-)^{2+\delta} < \infty.$$

Daouia, Girard, et al. (2018) write the same assumption as $\mathbb{E}\,|Y_-|^{2+\delta} < \infty$ with $Y_- = \min(Y, 0)$; here X^- denotes the nonnegative negative part.

For the expectile CLTs used below we shall work in the range $\gamma \in (0, 1/2)$ together with an explicit lower-tail moment condition. The restriction $\gamma < 1/2$ controls the positive tail and ensures $\mathbb{E}(X^+)^2 < \infty$; since X is real-valued, it does not by itself control the negative part. Condition 2.4 is the corresponding left-tail assumption used in the extreme-expectile limit theorems.

2.2 Tail-index and extreme-quantile estimation

Under the first-order condition, the extreme-value index $\gamma > 0$ is the primary parameter determining the behaviour of F in the tail. We collect here the two classical estimators that the thesis builds on: the Hill estimator of γ and the Weissman extrapolation for extreme quantiles. Both are standard; the textbook reference is Haan and Ferreira (2006, Ch. 3–4).

2.2.1 The Hill estimator

Given an iid sample $X_1, \ldots, X_n$ from F and an integer $k = k_n \in \{1, \ldots, n-1\}$, the *Hill estimator* (Hill 1975) of γ based on the $k+1$ top order statistics is

$$\hat{\gamma}_n(k) \;=\; \frac{1}{k}\sum_{i=1}^{k} \log\left(\frac{X_{n-i+1\,:\,n}}{X_{n-k\,:\,n}}\right). \tag{2.1}$$

Since $U(t) \to \infty$ under Condition 2.2, the event $X_{n-k:n} > 0$ has probability tending to one for intermediate k. The log-based tail estimators in this section are understood on that event; equivalently one may assume that the upper tail is eventually supported on $(0, \infty)$, or apply a deterministic shift, which leaves the extreme-value index unchanged. The interpretation is direct: if the upper tail of F were exactly Pareto with index $1/\gamma$, the logarithms of excess ratios above a fixed tail threshold would be exponential with mean γ; in Hill's order-statistic derivation, this appears through the scaled consecutive log-spacings $i\{\log X_{n-i+1:n} - \log X_{n-i:n}\}$, which are independent exponential variables with mean γ in the exact Pareto case and whose average is (2.1). Condition 2.2 asserts that the upper tail is *approximately* Pareto. The consistency statement below is Haan and Ferreira (2006, Theorem 3.2.2); the normal limit is Haan and Ferreira (2006, Theorem 3.2.5) when $\rho < 0$, with the $\rho = 0$ endpoint in the same $\mathcal{C}_2$ convention covered by the $m = 1$ specialisation of Daouia, Padoan, et al. (2024, Theorem 1).

Theorem 2.5. *Assume $\mathcal{C}_2(\gamma, \rho, A)$ with $\gamma > 0$. Let $k = k_n \to \infty$ and $k/n \to 0$ as $n \to \infty$. Then $\hat{\gamma}_n(k) \xrightarrow{\mathbb{P}} \gamma$. If, in addition, $\sqrt{k}\,A(n/k) \to \lambda \in \mathbb{R}$, then*

$$\sqrt{k}\,(\hat{\gamma}_n(k) - \gamma) \;\xrightarrow{d}\; \mathcal{N}\left(\frac{\lambda}{1-\rho},\, \gamma^2\right).$$

The choice of k controls the well-known bias–variance trade-off: too small a k inflates the variance, while too large a k injects second-order bias through the term $\lambda/(1-\rho)$. Adaptive choice of k is a research topic in its own right; we do not pursue it in this thesis and treat k as fixed by the data analyst.

2.2.2 Weissman extrapolation to extreme quantiles

The *Weissman extrapolation* (Weissman 1978) extrapolates from the intermediate-order statistic $X_{n-k:n}$ — which estimates the quantile $Q_X(1 - k/n)$ at the intermediate level $1 - k/n$ — to a quantile at a very-extreme level $1 - p$ with $p = p_n \to 0$ as $n \to \infty$. Plugging the Hill estimator into the first-order identity $U(ty)/U(t) \to y^\gamma$ gives the Weissman estimator

$$\widehat{q}^{\star}_n(1-p \mid k) \;=\; \left(\frac{k}{np}\right)^{\hat\gamma_n(k)} X_{n-k:n}. \tag{2.2}$$

For $p = c/n$ and fixed k, (2.2) reduces to the large-quantile estimator proposed by Weissman (1978, §4) in the Fréchet case, written there with tail parameter $\alpha = 1/\gamma$ and descending order statistics. The theorem below covers the far-tail regime $np = o(k)$. The especially severe very-extreme case has $np = O(1)$, where the expected number of observations beyond $Q_X(1-p)$ remains bounded; when $p < 1/n$, the target is beyond the usual sample-maximum scale. It is the Hill-estimator specialisation of Haan and Ferreira (2006, Theorem 4.3.8); since $\log k = o(\sqrt{k})$, the source condition $\log(np_n) = o(\sqrt{k})$ is equivalently written here as $\log(k/(np_n)) = o(\sqrt{k})$.

Theorem 2.6. *Assume $\mathcal{C}_2(\gamma, \rho, A)$ with $\gamma > 0$ and $\rho < 0$. Let $k = k_n \to \infty$, $k/n \to 0$, $p = p_n \to 0$ with $k/(np) \to \infty$ and $\sqrt{k}/\log(k/(np)) \to \infty$. Suppose $\sqrt{k}\,A(n/k) \to \lambda \in \mathbb{R}$. Then*

$$\frac{\sqrt{k}}{\log(k/(np))}\left(\frac{\widehat{q}^{\star}_n(1-p \mid k)}{Q_X(1-p)} - 1\right) \xrightarrow{d} \mathcal{N}\!\left(\frac{\lambda}{1-\rho},\, \gamma^2\right).$$

The rate $\sqrt{k}/\log(k/(np))$ reflects the cost of extrapolation: the further into the tail one extrapolates, the larger the log-denominator and the slower the rate. The asymptotic variance is identical to that of the Hill estimator (Theorem 2.5); the extrapolation contributes only through the multiplicative log-scale factor. This is the structural fact that Daouia, Padoan, et al. (2024, §2.3) exploits to motivate the weighted-*geometric* pooling of Weissman estimators — rather than the arithmetic pooling natural for centred quantities — and that we shall in turn exploit in Chapter 4 for the extrapolation of pooled expectiles.

2.3 Expectiles

This section collects the definition and the basic properties of expectiles that we shall use in the rest of the thesis. Expectiles were introduced by Newey and Powell (1987) as an asymmetric-least-squares analogue of quantiles. Their role as risk measures, in particular their coherence properties, was clarified by Bellini et al. (2014) and Ziegel (2016); their elicitability and scoring-function representations are treated in Gneiting (2011) and Ehm et al. (2016).

2.3.1 Definition

Let X be a real-valued random variable with $\mathbb{E}\,|X| < \infty$, and fix a level $\tau \in (0,1)$. The τ-*expectile* of X is defined as

$$\xi_\tau(X) \;:=\; \operatorname*{arg\,min}_{\theta \in \mathbb{R}} \; \mathbb{E}[\,\eta_\tau(X-\theta) - \eta_\tau(X)\,], \qquad \eta_\tau(u) := |\tau - \mathbb{1}\{u \le 0\}|\; u^2. \tag{2.3}$$

The asymmetric squared check function η_τ penalises positive and negative deviations from θ with the asymmetric weights τ and $1-\tau$, respectively. Recovering the mean in the symmetric case is immediate: $\eta_{1/2}(u) = u^2/2$, so $\xi_{1/2}(X) = \mathbb{E}\,X$. The subtraction of $\eta_\tau(X)$ in (2.3) is a centering device that makes the criterion finite under the first-moment condition $\mathbb{E}|X| < \infty$; it does not affect the minimiser in θ.

The convex objective (2.3) has a unique minimiser, and its first-order condition reduces to the implicit equation

$$\tau\,\mathbb{E}[(X - \xi_\tau(X))_+] \;=\; (1-\tau)\,\mathbb{E}[(\xi_\tau(X) - X)_+], \tag{2.4}$$

where $u_+ = \max(u, 0)$. Thus $\xi_\tau(X)$ is the unique point where the upper and lower first partial moments, with asymmetric weights, balance. In the continuous case, this mirrors the implicit quantile equation

$$\tau\,\mathbb{P}(X > Q_X(\tau)) = (1-\tau)\,\mathbb{P}(X < Q_X(\tau)),$$

with first partial moments replacing exceedance probabilities.

2.3.2 Basic properties

The following properties are immediate from (2.3) or (2.4); we collect them without proof and refer to Newey and Powell (1987, §2) and Bellini et al. (2014, Sections 2–3) for details.

Proposition 2.7. *Let X and Y be real-valued random variables with $\mathbb{E}|X|, \mathbb{E}|Y| < \infty$. For each $\tau \in (0,1)$:*

(i) ***Existence and uniqueness.*** *$\xi_\tau(X)$ exists and is uniquely defined.*

(ii) ***Law invariance.*** *If X and Y have the same distribution, then $\xi_\tau(X) = \xi_\tau(Y)$.*

(iii) ***Mean recovery.*** *$\xi_{1/2}(X) = \mathbb{E}\,X$.*

(iv) ***Location equivariance.*** *$\xi_\tau(X + c) = \xi_\tau(X) + c$ for $c \in \mathbb{R}$.*

(v) ***Positive homogeneity.*** *$\xi_\tau(\lambda X) = \lambda\,\xi_\tau(X)$ for $\lambda \geq 0$.*

(vi) ***Sign reflection.*** *$\xi_\tau(-X) = -\xi_{1-\tau}(X)$.*

(vii) ***Monotonicity in τ.*** *The map $\tau \mapsto \xi_\tau(X)$ is nondecreasing on $(0,1)$, and is strictly increasing when X is nondegenerate, with the boundary limits $\xi_\tau(X) \to \operatorname{ess\,inf}(X)$ as $\tau \to 0$ and $\xi_\tau(X) \to \operatorname{ess\,sup}(X)$ as $\tau \to 1$ (infinite limits allowed).*

(viii) ***Monotonicity in the argument.*** *If $X \leq Y$ almost surely, then $\xi_\tau(X) \leq \xi_\tau(Y)$.*

Location equivariance, positive homogeneity, and monotonicity in the argument give three of the four axioms in Artzner et al. (1999) that define a coherent risk measure. Sign reflection is useful for switching between loss and position conventions. The fourth axiom is subadditivity, to which we turn next.

2.3.3 Coherence and elicitability

In Sections 2.1, 2.2 and 2.4 and in the contribution chapters, X denotes a loss and high risk is represented by the upper expectile $\xi_\tau(X)$ with $\tau \uparrow 1$. The coherence literature is often written for financial positions, where positive values are gains and larger values are better. In this subsection only, write Y for such a position and define the capital requirement

$$\rho_\tau(Y) \;=\; \xi_\tau(-Y).$$

Subadditivity of ρ_τ requires $\rho_\tau(Y+Z) \le \rho_\tau(Y) + \rho_\tau(Z)$ for all positions Y, Z.

Theorem 2.8 (Bellini et al. (2014))**.** *The functional $\rho_\tau(Y) = \xi_\tau(-Y)$ is a coherent risk measure if and only if $\tau \ge 1/2$.*

Recall that a real-valued functional T on a class of distributions is *elicitable* (Gneiting 2011) if there exists a strictly consistent scoring function $S : \mathbb{R} \times \mathbb{R} \to [0, \infty)$ whose expected score is uniquely minimised by reporting $T(P)$ for every P in the class. Gneiting (2011, §3.3) shows that the τ-expectile is elicitable on classes with finite first moment; the familiar asymmetric squared score $S_\tau(\theta, x) = \eta_\tau(x - \theta)$ is strictly consistent on classes with finite second moment, while the general characterisation uses Bregman-type scores with the corresponding integrability conditions. Thus competing expectile forecasts can be compared by empirical averages of strictly consistent scores. Taken together with Theorem 2.8, this places expectiles in a privileged position among risk measures. On suitable convex classes of distributions and under the usual monetary-risk-measure convention, Bellini et al. show that, within the class of M-quantiles, the coherent risk measures are exactly the expectiles, while Ziegel (2016) characterises elicitable law-invariant coherent risk measures as certain expectiles, with the expectation appearing as the special case.

2.3.4 Choquet representation

Ehm et al. (2016) place expectiles within a broader theory of consistent scoring functions. They show that regular scoring functions consistent for expectiles admit a mixture representation over elementary scores, analogous in spirit to the corresponding representation for quantile scores. After a normalisation that removes trivial dilations of the scoring class, these elementary scores are extreme points, yielding the advertised Choquet-type representation. This is relevant for forecast comparison and backtesting, but it is not used in the asymptotic theory below; all subsequent arguments rely only on the expectile first-order condition and the extreme expectile–quantile equivalence.

2.4 Extreme expectile inference: the iid heavy-tailed baseline

We now restrict to a heavy-tailed loss X satisfying Condition 2.2 with extreme-value index $\gamma \in (0, 1)$, and consider inference for ξ_τ at levels τ approaching one. We distinguish, throughout, between two asymptotic regimes:

- *intermediate-level*: $\tau_n \to 1$ with $n(1 - \tau_n) \to \infty$. There are still many observations above $Q_X(\tau_n)$, and central limit theorems hold at the rate $\sqrt{n(1 - \tau_n)}$.
- *extreme-level*: $\tau'_n \to 1$ with $n(1 - \tau'_n) \to c \in [0, \infty)$. There are at most finitely many

observations above $Q_X(\tau_n')$, and direct estimation is infeasible; extrapolation is required.

The results in this section are due, in the iid setting, to Daouia, Girard, et al. (2018), Daouia, Girard, et al. (2019), and Daouia, Girard, et al. (2020); the time-series extension that we have used to organise the exposition is Davison et al. (2023), and we shall borrow Proposition C.6 of that paper's supplement to formulate the abstract plug-in lemma that the contribution chapters use.

2.4.1 The expectile–quantile asymptotic equivalence

The single most important fact for extreme expectile inference is that, in the heavy-tailed regime, the expectile is asymptotically proportional to the quantile of the same level.

Proposition 2.9 (Asymptotic equivalence; Bellini et al. (2014), Daouia, Girard, et al. (2019))**.** *Suppose X satisfies Condition 2.2 with index $\gamma \in (0,1)$ and $\mathbb{E}\, X^- < \infty$. Then*

$$\frac{\xi_\tau(X)}{Q_X(\tau)} \xrightarrow{\tau\uparrow 1} \psi(\gamma) \;:=\; (\gamma^{-1}-1)^{-\gamma}. \tag{2.5}$$

The factor $\psi(\gamma)$ is the linchpin of all extreme-expectile inference: it converts asymptotic statements about extreme quantiles into asymptotic statements about extreme expectiles, and vice versa. The first-order equivalence in Proposition 2.9 requires $\gamma < 1$ together with the integrability of the lower tail; the stronger range $\gamma < 1/2$ enters the CLT-level inference below. The function ψ is continuous on $(0,1)$, with $\psi(\gamma) \to 1$ as $\gamma \downarrow 0$ and $\psi(\gamma) \to \infty$ as $\gamma \uparrow 1$. It is *not* monotone: its logarithmic derivative

$$m(\gamma) \;:=\; \frac{d}{d\gamma}\log\psi(\gamma) \;=\; \frac{1}{1-\gamma} - \log\left(\gamma^{-1}-1\right) \tag{2.6}$$

changes sign at the value $\gamma^\star \approx 0.2178$ solving $m(\gamma^\star) = 0$, so ψ decreases on $(0,\gamma^\star)$ to a minimum $\psi(\gamma^\star) \approx 0.757$ and increases on $(\gamma^\star, 1)$; in particular it returns to $\psi = 1$ at $\gamma = 1/2$. Only for tail indices beyond $\gamma^\star$ does a heavier tail place the expectile further above the same-level quantile.

Under the second-order condition $\mathcal{C}_2(\gamma,\rho,A)$ with $\gamma \in (0,1/2)$ and Condition 2.4, the first-order equivalence (2.5) sharpens to the second-order expansion (Daouia, Girard, et al. 2018)

$$\begin{aligned} \frac{\xi_\tau(X)}{Q_X(\tau)} &= \psi(\gamma)\Big[1 + c(\gamma,\rho)\,A((1-\tau)^{-1}) + O(Q_X(\tau)^{-1})\Big],\\ c(\gamma,\rho) &:= \frac{(\gamma^{-1}-1)^{-\rho}}{1-\gamma-\rho} + \frac{(\gamma^{-1}-1)^{-\rho}-1}{\rho}. \end{aligned} \tag{2.7}$$

The displayed form is the version used in the thesis: it combines the second-order tail expansion with the lower-tail moment condition sufficient to control the non-tail contribution to the expectile equation. As $\tau \uparrow 1$, the second term in $c(\gamma,\rho)$ is read by continuity when $\rho = 0$, so that $c(\gamma,0) = m(\gamma)$. The constant $c(\gamma,\rho)$ measures the deterministic gap between the extreme expectile and the same-level quantile beyond the leading factor $\psi(\gamma)$. The remaining second-order term $O(Q_X(\tau)^{-1})$ is governed by the lower moments of X; under the moment condition

$\sqrt{n(1-\tau_n)}\,/Q_X(\tau_n) \to 0$ adopted below it is $\sqrt{n(1-\tau_n)}$-negligible, so the A-driven term alone governs the asymptotic bias of the quantile-based estimator.

2.4.2 Two families of estimators at intermediate levels

Two natural estimators of ξ_{τ_n} at an intermediate level τ_n arise from the asymmetric-LS definition (2.3) on the one hand, and from the asymptotic equivalence (2.5) on the other.

LAWS direct estimator. The *asymmetric-least-squares* (LAWS) estimator replaces the population minimisation (2.3) with its empirical analogue:

$$\widetilde{\xi}_{\tau_n} \;=\; \arg\min_{\theta\in\mathbb{R}} \sum_{i=1}^{n} \eta_{\tau_n}(X_i-\theta). \tag{2.8}$$

The estimator $\widetilde{\xi}_{\tau_n}$ is computed by an iteratively-reweighted least-squares scheme that converges in a finite number of iterations; see Daouia, Girard, et al. (2018, §3) for implementation details.

Theorem 2.10 (Intermediate LAWS CLT; Daouia, Girard, et al. (2018))**.** *Assume Condition 2.2 with $\gamma \in (0, 1/(2+\delta))$, and assume Condition 2.4 for the same $\delta > 0$. Let $\tau_n \to 1$ with $n(1-\tau_n) \to \infty$. Then*

$$\sqrt{n(1-\tau_n)}\left(\frac{\widetilde{\xi}_{\tau_n}}{\xi_{\tau_n}} - 1\right) \xrightarrow{d} \mathcal{N}\left(0,\; \frac{2\gamma^3}{1-2\gamma}\right).$$

Quantile-based plug-in estimator. A second family substitutes estimators of γ and of the intermediate quantile $Q_X(\tau_n)$ into the asymptotic-equivalence formula (2.5):

$$\widehat{\xi}_{\tau_n} \;=\; \psi(\hat{\gamma}_n)\,\widehat{Q}(\tau_n), \tag{2.9}$$

where $\widehat{Q}(\tau_n) = X_{n-\lfloor n(1-\tau_n)\rfloor:n}$ is the empirical quantile (an intermediate-order statistic) and $\hat{\gamma}_n$ is any $\sqrt{n(1-\tau_n)}$-consistent tail-index estimator. The natural choice is the Hill estimator of (2.1) with effective sample size $k = \lfloor n(1-\tau_n)\rfloor$.

Theorem 2.11 (Intermediate QB CLT; Daouia, Girard, et al. (2018))**.** *Assume $\mathcal{C}_2(\gamma,\rho,A)$ with $\gamma \in (0, 1/2)$ and Condition 2.4. Let $\tau_n \to 1$ with $n(1-\tau_n) \to \infty$ and suppose*

$$\sqrt{n(1-\tau_n)}\,A\big((1-\tau_n)^{-1}\big) \to \lambda \in \mathbb{R}, \qquad \frac{\sqrt{n(1-\tau_n)}}{Q_X(\tau_n)} \to 0.$$

With $\hat{\gamma}_n$ the Hill estimator at effective sample size $k = \lfloor n(1-\tau_n)\rfloor$,

$$\sqrt{n(1-\tau_n)}\left(\frac{\widehat{\xi}_{\tau_n}}{\xi_{\tau_n}} - 1\right) \xrightarrow{d} \mathcal{N}\left(\Big(\frac{m(\gamma)}{1-\rho} - c(\gamma,\rho)\Big)\lambda,\; \gamma^2\big(1+m(\gamma)^2\big)\right),$$

with $m(\gamma)$ as in (2.6) and $c(\gamma,\rho)$ the second-order expectile–quantile constant of (2.7). The asymptotic bias has two pieces: the Hill bias $\lambda/(1-\rho)$ propagated through the logarithmic derivative $m(\gamma)$ of ψ, and the deterministic expectile–quantile gap $-c(\gamma,\rho)\,\lambda$. The empirical *quantile $\widehat{Q}(\tau_n)$ contributes no asymptotic mean term at the $\sqrt{n(1-\tau_n)}$ scale when centered at $Q_X(\tau_n)$; the condition $\sqrt{n(1-\tau_n)}\,/Q_X(\tau_n) \to 0$ inherited from the theorem ensures that the moment-term contribution to the expectile–quantile gap is negligible.*

This is the $\lambda_2 = 0$ specialisation of the more general intermediate QB CLT, where

$$\lambda_2 = \lim_{n\to\infty} \frac{\sqrt{n(1-\tau_n)}}{Q_X(\tau_n)}.$$

In the general statement, the asymptotic mean contains an additional term of order λ_2 depending on the first moment of X. The present thesis imposes $\sqrt{n(1-\tau_n)}/Q_X(\tau_n) \to 0$ because this is the regime needed later to isolate the second-order tail bias.

Comparison. The LAWS estimator has asymptotic variance $2\gamma^3/(1-2\gamma)$, which is independent of the choice of tail-index estimator. The QB estimator's variance $\gamma^2(1+m(\gamma)^2)$ depends on the tail-index estimator used and reflects the contribution of $\hat{\gamma}_n$ via the term $m(\gamma)^2$. Numerical comparison over $\gamma \in (0, 1/2)$ reveals that no single estimator dominates the other across the whole range: the two variances cross once, at $\gamma \approx 0.2623$. For lighter heavy tails, i.e. small γ, the LAWS variance is smaller. For sufficiently heavy tails, and especially as $\gamma \uparrow 1/2$, the QB variance is smaller because the LAWS variance diverges. The thesis works with the QB construction throughout not because it uniformly dominates LAWS, but because it composes directly with the pooled $(\hat{\gamma}, \widehat{Q})$ inputs of the pooling framework.

2.4.3 Weissman extrapolation to very-extreme expectile levels

For levels τ_n' so close to one that $n(1-\tau_n') \to c \in [0, \infty)$, neither $\widetilde{\xi}_{\tau_n'}$ nor $\widehat{\xi}_{\tau_n'}$ is defined consistently. Extrapolation from an intermediate level τ_n to the extreme level τ_n' is required.

The asymptotic equivalence (2.5) and the first-order identity $U(ty)/U(t) \to y^\gamma$ together yield the heuristic

$$\frac{\xi_{\tau_n'}}{\xi_{\tau_n}} \sim \frac{Q_X(\tau_n')}{Q_X(\tau_n)} \sim \left(\frac{1-\tau_n'}{1-\tau_n}\right)^{-\gamma},$$

which suggests the *Weissman-type* extrapolated estimator

$$\overline{\xi}^{\star}_{\tau_n'} = \left(\frac{1-\tau_n'}{1-\tau_n}\right)^{-\hat{\gamma}_n} \overline{\xi}_{\tau_n}, \tag{2.10}$$

where $\overline{\xi}_{\tau_n}$ is any consistent intermediate-level estimator satisfying a $\sqrt{n(1-\tau_n)}$-CLT, such as $\widetilde{\xi}_{\tau_n}$ or $\widehat{\xi}_{\tau_n}$ under the conditions above, and $\hat{\gamma}_n$ is a $\sqrt{n(1-\tau_n)}$-consistent tail-index estimator. We write $\tilde{\xi}^\star$ and $\hat{\xi}^\star$ for the LAWS-anchored and QB-anchored extrapolations, respectively.

Theorem 2.12 (Extreme-level CLT; Daouia, Girard, et al. (2018))**.** *Assume $\mathcal{C}_2(\gamma, \rho, A)$ with $\gamma \in (0, 1/2)$, $\rho < 0$, and Condition 2.4. Let $\tau_n \to 1$ satisfy $n(1-\tau_n) \to \infty$ and let $\tau_n' \to 1$ satisfy $n(1-\tau_n') \to c \in [0, \infty)$, $(1-\tau_n')/(1-\tau_n) \to 0$ and $\sqrt{n(1-\tau_n)}\,/\log((1-\tau_n)/(1-\tau_n')) \to \infty$.*

Suppose moreover that

$$\sqrt{n(1-\tau_n)}\,A((1-\tau_n)^{-1}) \to \lambda \in \mathbb{R}, \qquad \frac{\sqrt{n(1-\tau_n)}}{Q_X(\tau_n)} \to 0.$$

If $\hat\gamma_n$ is the Hill estimator and $\overline{\xi}_{\tau_n}$ is either the LAWS or QB intermediate estimator above, then

$$\frac{\sqrt{n(1-\tau_n)}}{\log((1-\tau_n)/(1-\tau_n'))}\left(\frac{\overline{\xi}^{\star}_{\tau_n'}}{\xi_{\tau_n'}}-1\right) \xrightarrow{d} \mathcal{N}\left(\frac{\lambda}{1-\rho},\ \gamma^2\right).$$

Two remarks are in order. First, the rate $\sqrt{n(1-\tau_n)}\,/\log((1-\tau_n)/(1-\tau_n'))$ is the classical Weissman rate of Theorem 2.6: the further into the tail one extrapolates, the larger the log-denominator and the slower the rate of convergence. Second, the asymptotic variance is γ^2 *regardless of whether the anchor estimator is LAWS or QB*: the contribution of $\overline{\xi}_{\tau_n}$ is dominated, at the extreme level, by that of $\hat\gamma_n$ entering the multiplicative power. This is the central structural feature that makes extreme-expectile inference tractable; at the extreme level, asymptotic inference for $\xi_{\tau_n'}$ reduces to asymptotic inference for the tail index.

2.4.4 The plug-in central limit theorem

Theorems 2.10 to 2.12 together specify a complete asymptotic theory for extreme expectiles in the iid heavy-tailed setting. For the pooled inference developed in Chapters 3 and 4, however, we require a more abstract formulation that decouples the asymptotic distribution of the inputs $(\hat\gamma_n, \overline{\xi}_{\tau_n})$ from the way in which they are obtained. The required abstraction is essentially *Proposition C.6* in the supplement to Davison et al. (2023), restated here in a form adapted to the iid setting.

Proposition 2.13 (Plug-in CLT for extrapolated expectiles; cf. Davison et al. (2023), Prop. C.6)**.** *Assume $\mathcal{C}_2(\gamma,\rho,A)$ with $\gamma \in (0,1/2)$ and $\rho<0$, and assume Condition 2.4. Let $a_n \to \infty$, let $\hat\gamma_n$ be an estimator of γ, and let $\widehat{\xi}^{I}_{\tau_n}$ be an intermediate-level estimator of ξ_{τ_n} satisfying the joint asymptotic representation*

$$a_n \begin{pmatrix} \hat\gamma_n - \gamma \\ \widehat{\xi}^{I}_{\tau_n}/\xi_{\tau_n} - 1 \end{pmatrix} \xrightarrow{d} \begin{pmatrix} \Gamma \\ \Delta \end{pmatrix}, \tag{2.11}$$

for some random vector $(\Gamma,\Delta)^\top$. Let $\tau_n, \tau_n' \to 1$ with $n(1-\tau_n)\to\infty$, $n(1-\tau_n') \to c \in [0,\infty)$, $(1-\tau_n')/(1-\tau_n) \to 0$, put

$$\ell_n \;:=\; \log\left(\frac{1-\tau_n}{1-\tau_n'}\right),$$

and suppose

$$\frac{a_n}{\ell_n} \to \infty, \qquad a_n A((1-\tau_n)^{-1}) \to \lambda_1 \in \mathbb{R}, \qquad \frac{a_n}{Q_X(\tau_n)} \to \lambda_2 \in \mathbb{R}.$$

Then the Weissman-extrapolated expectile

$$\widehat{\xi}^{\star}_{\tau_n'} = \left(\frac{1-\tau_n'}{1-\tau_n}\right)^{-\hat\gamma_n} \widehat{\xi}^{I}_{\tau_n}$$

satisfies

$$\frac{a_n}{\log((1-\tau_n)/(1-\tau_n'))}\left(\frac{\widehat{\xi}^{\star}_{\tau_n'}}{\xi_{\tau_n'}}-1\right) \xrightarrow{d} \Gamma.$$

Remark 2.14 (Deterministic ratio control). The proof of Proposition 2.13 uses the deterministic log-ratio negligibility

$$\frac{a_n}{\ell_n}\left\{\log \xi_{\tau_n'} - \log \xi_{\tau_n} - \gamma \ell_n\right\} \longrightarrow 0.$$

In the original supplement of Davison et al. (2023), this control is part of the high-level condition C^*. In the iid heavy-tailed setting of this thesis, it follows from the second-order expansion (2.7): the A-term contributes $O(a_n A((1-\tau_n)^{-1})/\ell_n) = o(1)$ and the lower-tail remainder contributes $O(a_n/\{Q_X(\tau_n)\ell_n\}) = o(1)$.

Proposition 2.13 is the technical workhorse of the thesis. The original supplement of Davison et al. (2023) states the result under broader high-level assumptions; the version above is deliberately restricted to the heavy-tailed iid regime used throughout this thesis; Remark 2.14 records the deterministic step that links this concrete regime to the abstract formulation. It says that the asymptotic distribution of the extrapolated expectile at very-extreme levels is governed *only* by the asymptotic distribution of the tail-index estimator: the contribution of the intermediate-level expectile estimator $\widehat{\xi}^{I}_{\tau_n}$ washes out through the log-rate. In Chapter 4 we shall feed the pooled inputs $(\hat{\gamma}_n(\boldsymbol{\omega}), \overline{\xi}^{\text{pool}}_{\tau_n})$ of Chapter 3 through this proposition to obtain the CLT for the pooled extrapolated expectile in essentially one step.

2.5 The optimal pooling framework

We now turn to the pooling framework of Daouia, Padoan, et al. (2024), which the rest of the thesis extends from quantiles to expectiles. The framework concerns inference for the common tail index and the common extreme quantile of m samples that may have heterogeneous sizes, heterogeneous second-order properties, and non-trivial cross-sample tail dependence.

2.5.1 Setup

Throughout Section 2.5, unless explicitly stated otherwise, m is fixed while $n = \sum_{j=1}^m n_j \to \infty$. This is the finite-$m$ regime imported into Chapters 3 and 4. We have m samples, each obtained by iid sampling of one component of an m-dimensional random vector $\boldsymbol{X} = (X_1, \ldots, X_m)^\top$. The jth sample comprises n_j observations $X_{1,j}, X_{2,j}, \ldots, X_{n_j,j}$ with marginal distribution function F_j, tail quantile function $U_j = (1/\overline{F}_j)^{\leftarrow}$, and extreme-value index $\gamma_j > 0$. The order statistics of the jth sample are denoted $X_{1:n_j,j} \leq \cdots \leq X_{n_j:n_j,j}$. The total sample size is $n = \sum_{j=1}^m n_j$.

Tail inference in sample j uses its top $k_j + 1$ order statistics, where the effective sample size $k_j = k_j(n)$ satisfies $k_j \to \infty$ and $k_j/n_j \to 0$ as $n \to \infty$. The total effective sample size is $k = \sum_{j=1}^m k_j$. Order the samples so that $n_1 \leq \cdots \leq n_m$, and assume the asymptotic-proportionality conditions

$$\frac{n_1}{n_j} \to b_j \in (0,1], \qquad \frac{k_1}{k_j} \to c_j \in (0,\infty), \tag{2.12}$$

with $b_1 = c_1 = 1$. The marginal F_j satisfies $\mathcal{C}_2(\gamma_j, \rho_j, A_j)$ (Condition 2.3). The thesis works under the *common-tail-index* assumption

$$(H_\gamma) \qquad \gamma_1 \;=\; \cdots \;=\; \gamma_m \;=\; \gamma,$$

which is the natural setup when the m samples are believed to come from the same underlying tail behaviour.

The cross-sample asymptotic dependence is captured by a pairwise *tail-copula* structure (Daouia, Padoan, et al. 2024, §2.1).

Condition $\mathcal{J}(\boldsymbol{R})$ (tail-copula condition). For each pair (j,ℓ) with $j \neq \ell$, there exists a function $R_{j,\ell} : [0,\infty]^2 \setminus \{(\infty,\infty)\} \to [0,\infty)$ such that, for every $(x_j, x_\ell) \in [0,\infty)^2$,

$$\lim_{s\to\infty} s\,\mathbb{P}\big(1 - F_j(X_j) \le x_j/s,\; 1 - F_\ell(X_\ell) \le x_\ell/s\big) \;=\; R_{j,\ell}(x_j, x_\ell).$$

The function $R_{j,\ell}$ is the bivariate tail copula of (X_j, X_ℓ) (Haan and Ferreira 2006, §8.2); it measures the rate at which the joint upper-tail probability of (X_j, X_ℓ) decays in comparison with the marginals. The case $R_{j,\ell} \equiv 0$ corresponds to *asymptotic tail independence* of X_j and X_ℓ, which is in particular implied by independence of the samples themselves — the setting of this thesis. This formulation covers the general pooling framework in which cross-sample extremal dependence may be represented through a tail copula. In the distributed-inference case used later in the thesis, the samples are independent marginal subsamples and all pairwise tail-copula terms vanish.

2.5.2 The pooled Hill estimator

Each sample contributes a Hill estimator $\hat\gamma_j(k_j)$ as in (2.1); we collect these into the random vector

$$\hat{\boldsymbol{\gamma}}_n \;=\; \big(\hat\gamma_1(k_1), \ldots, \hat\gamma_m(k_m)\big)^\top.$$

Given a weight vector $\boldsymbol{\omega} \in \mathbb{R}^m$ with $\boldsymbol{\omega}^\top \mathbf{1} = 1$, the *pooled Hill estimator* is the weighted arithmetic combination

$$\hat\gamma_n(\boldsymbol{\omega}) \;:=\; \boldsymbol{\omega}^\top \hat{\boldsymbol{\gamma}}_n \;=\; \sum_{j=1}^m \omega_j\, \hat\gamma_j(k_j). \tag{2.13}$$

The constraint $\boldsymbol{\omega}^\top \mathbf{1} = 1$ ensures consistency of $\hat\gamma_n(\boldsymbol{\omega})$ for the common γ under (H_γ). The naive choice $\boldsymbol{\omega} = m^{-1}\mathbf{1}$ recovers a simple arithmetic average; the choice $\boldsymbol{\omega} \propto (k_1, \ldots, k_m)$ recovers effective-sample-size weighting. The goal of the optimal-pooling literature is to identify weight choices that improve on these heuristics in terms of asymptotic variance or AMSE.

The fundamental asymptotic result is the joint CLT for the marginal Hill estimators.

Theorem 2.15 (Daouia, Padoan, et al. (2024), Theorem 1)**.** *Assume $\mathcal{C}_2(\gamma_j, \rho_j, A_j)$ for every j, the tail-copula condition $\mathcal{J}(\boldsymbol{R})$, the proportionality conditions (2.12), and $\sqrt{k_j}\, A_j(n_j/k_j) \to \lambda_j \in \mathbb{R}$ for every j. Then*

$$\left(\sqrt{k_1}(\hat{\gamma}_1(k_1) - \gamma_1), \ldots, \sqrt{k_m}(\hat{\gamma}_m(k_m) - \gamma_m)\right)^\top \xrightarrow{d} \mathcal{N}(\boldsymbol{B}, \boldsymbol{V}),$$

with $\boldsymbol{B} = \left(\lambda_j/(1-\rho_j)\right)_{j=1}^m$ and $\boldsymbol{V}$ the symmetric matrix whose entries are

$$V_{j,\ell} \;=\; \begin{cases} \gamma_j^2, & j = \ell, \\ \gamma_j\,\gamma_\ell\,\sqrt{c_j/c_\ell}\,R_{j,\ell}(c_\ell/c_j,\, b_\ell/b_j), & j < \ell. \end{cases}$$

The matrix is symmetric, so $V_{\ell,j} = V_{j,\ell}$ for $\ell > j$. Suppose, in addition, the common-γ condition (H_γ). Then the same marginal estimators, now viewed on the common $\sqrt{k}$ scale, satisfy

$$\sqrt{k}\,(\hat{\boldsymbol{\gamma}}_n - \gamma\mathbf{1}) \;\xrightarrow{d}\; \mathcal{N}(\boldsymbol{B_c}, \boldsymbol{V_c}),$$

where

$$(\boldsymbol{B_c})_j = \sqrt{c_j \sum_{i=1}^m c_i^{-1}}\,\frac{\lambda_j}{1-\rho_j} \qquad j = 1, \ldots, m.$$

The covariance matrix $\boldsymbol{V_c}$ is given by

$$(\boldsymbol{V_c})_{j,\ell} = \left(\sum_{i=1}^m c_i^{-1}\right) \begin{cases} \gamma^2 c_j, & j = \ell, \\ \gamma^2 c_j\, R_{j,\ell}(c_\ell/c_j, b_\ell/b_j), & j < \ell, \end{cases}$$

with $(\boldsymbol{V_c})_{\ell,j} = (\boldsymbol{V_c})_{j,\ell}$ for $\ell > j$. Consequently, for any $\boldsymbol{\omega}$ with $\boldsymbol{\omega}^\top\mathbf{1} = 1$,

$$\sqrt{k}\,(\hat{\gamma}_n(\boldsymbol{\omega}) - \gamma) \;\xrightarrow{d}\; \mathcal{N}(\boldsymbol{\omega}^\top\boldsymbol{B_c},\; \boldsymbol{\omega}^\top\boldsymbol{V_c}\boldsymbol{\omega}).$$

In the case of *independent* samples ($R_{j,\ell} \equiv 0$ for every $j \neq \ell$), the off-diagonal entries of $\boldsymbol{V_c}$ vanish identically and $\boldsymbol{V_c}$ becomes the diagonal matrix

$$\boldsymbol{V_c} \;=\; \gamma^2\left(\sum_{i=1}^m c_i^{-1}\right)\mathrm{diag}(c_1, \ldots, c_m). \tag{2.14}$$

Equation (2.14) is the form of $\boldsymbol{V_c}$ that the thesis will exploit in Chapters 3 and 4.

2.5.3 Variance- and AMSE-optimal weights

Given Theorem 2.15, two natural choices of $\boldsymbol{\omega}$ emerge as solutions to convex quadratic programmes. Unless explicitly stated otherwise, weights are allowed to be real and are constrained only by $\boldsymbol{\omega}^\top\mathbf{1} = 1$; the optimal weights therefore need not be convex weights.

Proposition 2.16 (Optimal weights; Daouia, Padoan, et al. (2024), §2.2)**.** *Suppose $\boldsymbol{V_c}$ is positive definite. Under the common-γ condition (H_γ):*

(i) The variance-optimal weights, minimising $\boldsymbol{\omega}^\top \boldsymbol{V_c}\boldsymbol{\omega}$ subject to $\boldsymbol{\omega}^\top \mathbf{1} = 1$, are

$$\boldsymbol{\omega}^{\mathrm{Var}} = \frac{\boldsymbol{V_c}^{-1}\mathbf{1}}{\mathbf{1}^\top \boldsymbol{V_c}^{-1}\mathbf{1}},$$

with corresponding asymptotic variance $\boldsymbol{\omega}^{\mathrm{Var},\top}\boldsymbol{V_c}\boldsymbol{\omega}^{\mathrm{Var}} = (\mathbf{1}^\top \boldsymbol{V_c}^{-1}\mathbf{1})^{-1}$.

(ii) The AMSE-optimal weights, minimising $(\boldsymbol{\omega}^\top \boldsymbol{B_c})^2 + \boldsymbol{\omega}^\top \boldsymbol{V_c}\boldsymbol{\omega}$ subject to $\boldsymbol{\omega}^\top \mathbf{1} = 1$, are

$$\boldsymbol{\omega}^{\mathrm{AMSE}} = \frac{(1 + \boldsymbol{B_c}^\top \boldsymbol{V_c}^{-1}\boldsymbol{B_c})\,\boldsymbol{V_c}^{-1}\mathbf{1} - (\mathbf{1}^\top \boldsymbol{V_c}^{-1}\boldsymbol{B_c})\,\boldsymbol{V_c}^{-1}\boldsymbol{B_c}}{(1 + \boldsymbol{B_c}^\top \boldsymbol{V_c}^{-1}\boldsymbol{B_c})(\mathbf{1}^\top \boldsymbol{V_c}^{-1}\mathbf{1}) - (\mathbf{1}^\top \boldsymbol{V_c}^{-1}\boldsymbol{B_c})^2}.$$

In the independent-samples case, $\boldsymbol{V_c}$ is diagonal (Equation (2.14)) and the variance-optimal weights reduce to

$$\omega_j^{\mathrm{Var}} = \left(\sum_{i=1}^m c_i^{-1}\right)^{-1} c_j^{-1} \sim \frac{k_j}{k}, \tag{2.15}$$

where the equivalence uses $c_i^{-1} = \lim k_i/k_1$. Thus the samples contribute to the pooled tail index in proportion to their effective sample size, the same rule that one would expect from the unfeasible Hill estimator built from the entire combined sample.

In practice $\boldsymbol{V_c}$ and $\boldsymbol{B_c}$ must be replaced by consistent estimators $\widehat{\boldsymbol{V_c}}$ and $\widehat{\boldsymbol{B_c}}$ to render the optimal weights data-driven. The estimators are built from the marginal $\hat\gamma_j$, estimators of the second-order quantities entering A_j and ρ_j, and a plug-in estimator $\widehat{R_{j,\ell}}$ of the bivariate tail copula (Daouia, Padoan, et al. 2024, §2.2). The composite estimator $\hat\gamma_n(\hat{\boldsymbol{\omega}})$ is then $\sqrt{k}$-asymptotically equivalent to $\hat\gamma_n(\boldsymbol{\omega})$ provided $\hat{\boldsymbol{\omega}}^\top \mathbf{1} = 1$ exactly and $\hat{\boldsymbol{\omega}} \xrightarrow{\mathbb{P}} \boldsymbol{\omega}$. If exact normalisation is not imposed, one must instead require $\sqrt{k}\,\left|\hat{\boldsymbol{\omega}}^\top \mathbf{1} - 1\right| \xrightarrow{\mathbb{P}} 0$.

2.5.4 Weighted geometric pooling of extreme quantiles

For estimation of an extreme quantile $q(1-p)$ with $p = p(n) \to 0$, each sample contributes a Weissman estimator

$$\hat q_j^\star(1 - p \mid k_j) = \left(\frac{k_j}{n_j\, p}\right)^{\hat\gamma_j(k_j)} X_{n_j - k_j : n_j,\, j}.$$

The aggregation of these per-sample Weissman estimators must be adapted to their multiplicative-power structure (Equation (2.2)), and is done by *weighted geometric* — rather than arithmetic — averaging. Under the additional assumption that the samples have asymptotically equivalent extreme quantiles, formalised as the asymptotic-proportionality condition

Condition 2.17 (Tail homoskedasticity ($\mathcal{H}$))**.** For every $j \neq \ell$, $U_j(t)/U_\ell(t) \to 1$ as $t \to \infty$.

The terminology reflects the fact that the marginal tail scales are asymptotically equal. Equivalently, $Q_j(1-p)/Q_\ell(1-p) \to 1$ as $p \downarrow 0$. The notation in Daouia, Padoan, et al. (2024) writes this population extreme quantile as $q_j(1-p)$; here we keep the thesis-wide notation $Q_j(1-p)$.

The pooled extreme-quantile estimator of Daouia, Padoan, et al. (2024, §2.3) is

$$\widehat{q}_n^{\star}(1-p \mid \boldsymbol{\omega}) \;\coloneqq\; \prod_{j=1}^{m} \left[\hat{q}_j^{\star}(1-p \mid k_j)\right]^{\omega_j}. \tag{2.16}$$

The reason for the geometric form is visible from the log-scale identity $\log \hat{q}_j^{\star}(1-p \mid k_j) = \log(k_j/(n_j p))\,\hat{\gamma}_j(k_j) + \log X_{n_j-k_j:n_j,j}$: a weighted sum on the log scale puts the per-sample $\hat{\gamma}_j(k_j)$ on the standard linear scale, which is what the joint CLT of Theorem 2.15 addresses. Under the proportionality conditions (2.12), $\log(k_j/(n_j p)) = \log(k/(np)) + O(1)$, so the per-sample logarithmic normalisers are asymptotically equivalent to the common normaliser in the diverging-log regime.

Theorem 2.18 (Daouia, Padoan, et al. (2024), Theorem 2)**.** *Assume the hypotheses of Theorem 2.15, the common-γ condition (H_γ), the tail-homoskedasticity condition $(\mathcal{H})$, and, in addition, $\rho_j < 0$ for every $j = 1, \ldots, m$. Let $p = p(n) \to 0$ with $k/(np) \to \infty$ and $\sqrt{k}/\log(k/(np)) \to \infty$. Then for any $\boldsymbol{\omega}$ with $\boldsymbol{\omega}^\top \mathbf{1} = 1$ and any $j \in \{1, \ldots, m\}$,*

$$\frac{\sqrt{k}}{\log(k/(np))}\left(\frac{\widehat{q}_n^{\star}(1-p \mid \boldsymbol{\omega})}{Q_j(1-p)} - 1\right) \xrightarrow{d} \mathcal{N}(\boldsymbol{\omega}^\top \boldsymbol{B_c},\; \boldsymbol{\omega}^\top \boldsymbol{V_c} \boldsymbol{\omega}).$$

The strict condition $\rho_j < 0$ is part of the Weissman extrapolation theorem and is stronger than the $\rho_j \le 0$ condition needed for the marginal Hill CLT. Condition $(\mathcal{H})$ itself is intentionally only the first-order tail-homoskedasticity condition of Daouia, Padoan, et al. (2024). The recentering by any marginal $Q_j(1-p)$ in Theorem 2.18 is valid because the strict second-order assumption $\rho_j < 0$ supplies the missing rate: by Daouia, Padoan, et al. (2024, Supplement, Lemma B.2), for any j, ℓ,

$$\log Q_j(1-p) - \log Q_\ell(1-p) = O(\,|A_j(1/p)|\,) + O(\,|A_\ell(1/p)|\,) = o(k^{-1/2}),$$

along the Weissman sequence. After multiplication by $\sqrt{k}/\log(k/(np))$, this deterministic centering difference is negligible.

The asymptotic distribution of the pooled Weissman estimator is *identical*, up to the log-rate factor, to that of the pooled Hill estimator. Two structural consequences follow. First, the variance- and AMSE-optimal weights of Proposition 2.16 are also optimal for the pooled extreme-quantile estimator. Second, the entire information needed for inference at very-extreme quantile levels is contained in the joint distribution of the per-sample Hill estimators; the extreme-quantile inputs contribute only through the deterministic log-scale factor. This structural fact is the analogue, in the multi-sample setting, of the dominance of the tail-index estimator at the extreme level that we have already encountered in Theorem 2.12 and Proposition 2.13. In the geometric quantile pool, real weights remain algebraically well-defined because the Weissman estimators are positive with probability tending to one under the positivity convention of Section 2.2.1.

2.5.5 Tail-homogeneity testing and confidence intervals

The covariance structure $\boldsymbol{V_c}$ also yields a Wald-type chi-square test for the common-γ hypothesis (H_γ), sometimes called *tail homogeneity* across samples.

Theorem 2.19 (Tail-homogeneity test; Daouia, Padoan, et al. (2024), Corollary 3)**.** *Assume the hypotheses of Theorem 2.15, the null $H_0 : \gamma_1 = \cdots = \gamma_m$, and the bias-free calibration $\lambda_j = 0$ for every j. The statistic*

$$\Lambda_n \;=\; k\,(\hat{\boldsymbol{\gamma}}_n - \bar{\mu}_n\mathbf{1})^\top \widehat{\boldsymbol{V}}_{\boldsymbol{c}}^{-1}(\hat{\boldsymbol{\gamma}}_n - \bar{\mu}_n\mathbf{1}), \qquad \bar{\mu}_n \;=\; \frac{\mathbf{1}^\top \widehat{\boldsymbol{V}}_{\boldsymbol{c}}^{-1}\hat{\boldsymbol{\gamma}}_n}{\mathbf{1}^\top \widehat{\boldsymbol{V}}_{\boldsymbol{c}}^{-1}\mathbf{1}},$$

satisfies $\Lambda_n \xrightarrow{d} \chi^2_{m-1}$ under H_0 and $\Lambda_n \xrightarrow{\mathbb{P}} \infty$ under fixed alternatives. A test of asymptotic size α rejects H_0 when $\Lambda_n > \chi^2_{m-1,\,1-\alpha}$.

If the asymptotic bias is retained rather than removed by undersmoothing or bias correction, the same GLS deviance has the noncentral limit

$$\Lambda_n \xrightarrow{d} \chi^2_{m-1}(\delta_\gamma), \qquad \delta_\gamma = \boldsymbol{B_c}^\top \boldsymbol{V_c}^{-1}\boldsymbol{B_c} - \frac{(\mathbf{1}^\top \boldsymbol{V_c}^{-1}\boldsymbol{B_c})^2}{\mathbf{1}^\top \boldsymbol{V_c}^{-1}\mathbf{1}}.$$

Thus the central chi-square limit holds whenever $\boldsymbol{B_c} \in \operatorname{span}\{\mathbf{1}\}$, in particular under $\lambda_j = 0$ for every j. Under contiguous alternatives of order $k^{-1/2}$, the limit is likewise noncentral rather than divergent.

Under tail homogeneity, Proposition 2.16 furnishes asymptotic confidence intervals for the common γ on the $\sqrt{k}$ scale; log-scale confidence intervals for the extrapolated extreme quantile follow from Theorem 2.18. These variance-only intervals are centered correctly under undersmoothing or after valid bias correction; otherwise the asymptotic bias term must be included in the centering.

2.5.6 Distributed inference as a special case

The distributed-inference setting is the special case in which the m samples are obtained as iid subdivisions of one common underlying iid sample: data points are iid both within and across samples, with a common marginal distribution. The framework simplifies considerably and matches the assumptions of this thesis.

Under distributed-inference assumptions:

- the common-γ condition (H_γ) and the tail-homoskedasticity condition $(\mathcal{H})$ are trivially satisfied;
- the tail copula $R_{j,\ell}$ between any pair of distinct samples vanishes identically (independence implies tail independence);
- $\boldsymbol{V_c}$ in Theorem 2.15 becomes the diagonal matrix (2.14);
- the variance-optimal weights of Proposition 2.16 reduce to $\omega_j^{\mathrm{Var}} \propto k_j$ as in (2.15).

Corollary 2.20 (Daouia, Padoan, et al. (2024), Corollary 6 and Theorem 3)**.** *Under the distributed-inference assumptions, and provided that the sample fractions are asymptotically equal,* $k_j/n_j = (k/n)(1 + O(k^{-1/2}))$ *for every* j*, the variance-optimally-pooled Hill estimator* $\hat{\gamma}_n(\boldsymbol{\omega}^{\mathrm{Var}})$ *is* $\sqrt{k}$*-asymptotically equivalent to the unfeasible benchmark Hill estimator* $\hat{\gamma}_n^{(\mathrm{Hill})}(k)$ *built from the entire combined sample. Analogously, the variance-optimally-pooled Weissman estimator* $\widehat{q}_n^{\star}(1 - p \mid \boldsymbol{\omega}^{\mathrm{Var}})$ *is asymptotically equivalent to the unfeasible benchmark Weissman estimator built from the combined sample.*

Corollary 2.20 is the central efficiency result of the optimal pooling framework: under the distributed-inference assumptions, equal effective sample fractions, and variance-optimal weighting, the pooled estimators pay no first-order asymptotic efficiency price relative to the infeasible pooled-sample benchmark. It is this property that we shall seek to preserve in the pooled-expectile estimators of Chapters 3 and 4.

Chapter 3

Pooled inference for extreme expectiles at intermediate levels

Overview

This chapter introduces the pooled extreme-expectile estimator at intermediate levels and develops its asymptotic theory. The construction plugs the pooled tail-index estimator $\hat{\gamma}_n(\boldsymbol{\omega}_\gamma)$ and the pooled weighted-geometric Weissman extreme-quantile estimator $\widehat{q}_n^\star(\,\cdot\,;\boldsymbol{\omega}_q)$ of Daouia, Padoan, et al. (2024) into the expectile–quantile asymptotic equivalence $\xi_\tau \approx \psi(\gamma)\,Q_X(\tau)$. The delta method applied through ψ yields a joint Gaussian limit, from which we read off the variance- and AMSE-optimal pooling weights tailored to the pooled expectile target.

3.1 Setup and notation

The contribution of this chapter rests on the pooling framework of Section 2.5, specialised to independent samples with a common marginal distribution — the tail-homogeneous distributed-inference case of Section 2.5.6. The equal effective sample fraction condition $k_j/n_j = (k/n)(1+o(1))$ of Corollary 2.20 is *not* part of the standing setup; it is invoked only in Sections 3.5 and 3.6 where it simplifies the variance-optimal weights. We restate below only those assumptions used throughout; the general pooling machinery and its supporting joint central limit theorems (Theorems 2.15 and 2.18) are imported without modification.

The m samples and the pooling primitives. We have m *independent* samples $\{X_{i,j}\}_{i=1,\dots,n_j}$, $j=1,\dots,m$, with common marginal distribution function F and common extreme-value index γ. The number of samples m is fixed as $n\to\infty$ throughout the chapter; all CLTs below are finite-m statements. Independence of the samples implies asymptotic tail independence, hence the pairwise tail-copula functions in Condition $\mathcal{J}(\boldsymbol{R})$ vanish:

$$R_{j,\ell} \;\equiv\; 0 \qquad \text{for every } j\neq\ell. \tag{3.1}$$

The proportionality conditions (2.12) are in force, with $k_j \to \infty$ and $k_j/n_j \to 0$, and the total effective sample size is $k = \sum_{j=1}^{m} k_j$. Each sample j supplies the two pooling primitives of Section 2.5: a local Hill estimator $\hat{\gamma}_j(k_j)$ as in (2.1), and a local Weissman extreme-quantile estimator $\hat{q}_j^\star(\,\cdot \mid k_j)$ as in (2.2). These $2m$ scalars, together with their joint asymptotic covariance, constitute the entire information shared between samples; no raw observation and no expectile-specific summary statistic crosses between machines.

Heavy-tail regime. The common marginal F satisfies the second-order condition $\mathcal{C}_2(\gamma, \rho, A)$ of Condition 2.3 with

$$\gamma \in (0, 1/2), \qquad \mathbb{E}\,\big|\min(X, 0)\big|^{2+\delta} < \infty \quad \text{for some } \delta > 0, \tag{3.2}$$

the per-sample bias-rate condition $\sqrt{k_j}\, A(n_j/k_j) \to \lambda_j \in \mathbb{R}$ holds for every j, and the moment-negligibility condition

$$\frac{\sqrt{k}}{Q_X(\tau_n)} \;\to\; 0 \tag{3.3}$$

is in force. The restriction $\gamma < 1/2$ is the moment regime in which expectile inference is statistically meaningful and in which the expectile–quantile asymptotic equivalence (2.5) composes with the iid intermediate central limit theorems Theorems 2.10 and 2.11. Condition (3.3) is the multi-sample counterpart of the $\sqrt{n(1-\tau_n)}\,/Q_X(\tau_n) \to 0$ hypothesis of those theorems: it ensures that the moment term in the expectile–quantile expansion (2.7) is asymptotically negligible, so that the deterministic bias of the pooled estimator is governed by the single second-order rate carried by A. It is mild in the heavy-tailed regime, where $Q_X(\tau_n)$ diverges at the regularly varying rate $(1-\tau_n)^{-\gamma}$.

The target. We target the common population expectile ξ_{τ_n} at an *intermediate* level $\tau_n \to 1$ with $n(1-\tau_n) \to \infty$, in the regime sense of Section 2.4. In addition, the per-sample log-rate factor

$$L_j(\tau_n) \;:=\; \log \frac{k_j}{n_j(1-\tau_n)}$$

is assumed to admit a finite limit $L_j^\star \in \mathbb{R}$ for every j; equivalently, $1 - \tau_n$ has the same order as k_j/n_j for each sample. This condition controls the Hill–Weissman cross-covariance in Proposition 3.2 and is the chapter's link between the chosen target level and the local effective sample fractions. The very-extreme regime $n(1-\tau_n') \to c \in [0, \infty)$ is deferred to Chapter 4, where the pooled intermediate-level result proved here serves as the anchor for Weissman-style extrapolation through Proposition 2.13.

Per-sample QB plug-in: a forced consequence of the pooling protocol. Because each sample shares the pooling primitives $(\hat{\gamma}_j(k_j), \hat{q}_j^\star(\,\cdot \mid k_j))$ and no expectile-specific summary statistic, the only expectile estimator that the chapter can construct *locally* on each sample, before pooling, is the QB plug-in (2.9) obtained by factoring through the expectile–quantile equivalence of Proposition 2.9:

$$\widehat{\xi}_{\tau, j} \;=\; \psi(\hat{\gamma}_j(k_j))\, \hat{q}_j^\star(1 - \tau \mid k_j).$$

The LAWS direct estimator (2.8) would require sharing $\widetilde{\xi}_{\tau_n,j}$ across samples, which lies outside the pooling protocol of Section 2.5. The comparison of Theorems 2.10 and 2.11 shows that neither LAWS nor QB uniformly dominates the other over $\gamma \in (0, 1/2)$; the QB construction is therefore adopted not because it is uniformly more efficient, but because it is the only intermediate-expectile estimator that can be assembled from the communicated pooling primitives. The choice is a derived consequence of the pooling protocol, not an independent methodological preference.

Roadmap. Section 3.2 defines the pooled QB intermediate expectile estimator from the pooled Hill and pooled Weissman inputs. Section 3.3 derives the joint $\sqrt{k}$-CLT for the pooled Hill and log-Weissman inputs, combining the marginal pooled limits of Section 2.5 (Theorems 2.15 and 2.18) with the local Hill / intermediate-threshold joint CLT, specialised to the independent-samples case (3.1). The output is the Gaussian input to which the delta method is then applied. Section 3.4 applies the delta method through ψ and states the main asymptotic-normality theorem of the chapter. Section 3.5 derives the variance- and AMSE-optimal pooling weights for the expectile target and compares them with the corresponding optima for the quantile target. Section 3.6 provides feasible confidence intervals built from estimated weights, and Section 3.7 adapts the tail-homogeneity test of Theorem 2.19 to the expectile-tail-homogeneity null hypothesis.

3.2 The pooled QB expectile estimator

The shared primitives $\big(\hat{\gamma}_j(k_j),\, \hat{q}_j^{\star}(1-\tau_n \mid k_j)\big)_{j=1,\dots,m}$ admit two natural *single-weight* constructions of a pooled QB intermediate-expectile estimator. The first, denoted (A), applies the QB plug-in once, at the centre, after pooling the Hill and Weissman primitives separately; the second, denoted (B), computes a per-sample QB expectile on each machine and then geometric-pools the m local estimates. At matched single weights — the only setting in which the two constructions are directly comparable — Proposition 3.1 below shows they share the same $\sqrt{k}$-asymptotic distribution, so on this common domain the choice between them is asymptotically inert. We adopt (A) by parsimony, since it is the direct lift of the iid QB pipeline of Theorem 2.11 — one tail-index input, one quantile input, one application of ψ — to the pooled setting.

Crucially, (A) admits an intrinsic *two-weight* extension that (B) does not: the pooled Hill and pooled Weissman primitives can in principle receive distinct weight vectors $(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q)$. Once we relax to two weights, construction (B) has no analogue — it is built on a single per-sample QB and a single geometric pool — so the two-weight machinery developed in the remainder of the chapter is necessarily (A)-only. Section 3.5 will return to the question of whether this two-weight freedom is ever genuinely exploited, and show that under the distributed-inference setting of Section 3.1 the variance-optimal pair lies on the matched-weight diagonal.

Pooled Hill and pooled Weissman primitives. For weight vectors $\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q \in \mathbb{R}^m$ satisfying $\boldsymbol{\omega}_\gamma^\top \mathbf{1} = \boldsymbol{\omega}_q^\top \mathbf{1} = 1$, the linear pooled Hill estimator (2.13) and the weighted-geometric pooled

Weissman estimator (2.16) read

$$\hat{\gamma}_n(\boldsymbol{\omega}_\gamma) \;=\; \boldsymbol{\omega}_\gamma^\top \hat{\boldsymbol{\gamma}}_n \;=\; \sum_{j=1}^{m} \omega_{\gamma,j}\, \hat{\gamma}_j(k_j), \tag{3.4}$$

$$\widehat{q}_n^\star(1-\tau_n \mid \boldsymbol{\omega}_q) \;=\; \prod_{j=1}^{m} \left[\widehat{q}_j^\star(1-\tau_n \mid k_j)\right]^{\omega_{q,j}}, \tag{3.5}$$

both evaluated at the chapter's intermediate level τ_n.

The two candidate estimators. The first construction, which we adopt as canonical, pools the primitives separately and applies the QB plug-in once at the centre,

$$\widehat{\xi}^{(A)}_{\tau_n}(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q) \;:=\; \psi(\hat{\gamma}_n(\boldsymbol{\omega}_\gamma))\, \widehat{q}_n^\star(1-\tau_n \mid \boldsymbol{\omega}_q). \tag{3.6}$$

The second computes a per-sample QB expectile on each machine and geometric-pools the resulting m local estimates,

$$\widehat{\xi}^{(B)}_{\tau_n}(\boldsymbol{\omega}) \;:=\; \prod_{j=1}^{m} \left[\psi(\hat{\gamma}_j(k_j))\, \widehat{q}_j^\star(1-\tau_n \mid k_j)\right]^{\omega_j} \;=\; \Big(\prod_{j=1}^{m} \psi(\hat{\gamma}_j(k_j))^{\omega_j}\Big) \cdot \widehat{q}_n^\star(1-\tau_n \mid \boldsymbol{\omega}). \tag{3.7}$$

The Weissman factor is identical in the two formulae; the constructions differ only in whether the nonlinearity ψ acts once on the pooled $\hat{\gamma}_n$ (construction (A)) or componentwise on the m per-sample $\hat{\gamma}_j$'s followed by weighted-geometric averaging (construction (B)). The iid QB plug-in of Theorem 2.11 ($m = 1$) reduces both formulae to $\psi(\hat{\gamma})\,\hat{q}^\star$; construction (A) is the direct $m \geq 2$ generalisation of that structure, whereas (B) introduces m independent applications of ψ that have no counterpart in the iid baseline.

Proposition 3.1 ($\sqrt{k}$-asymptotic equivalence of (A) and (B))**.** *Under the assumptions of Section 3.1, for fixed $m \in \mathbb{N}$ and every deterministic weight vector $\boldsymbol{\omega} \in \mathbb{R}^m$ with $\boldsymbol{\omega}^\top \mathbf{1} = 1$,*

$$\sqrt{k}\, \log \frac{\widehat{\xi}^{(A)}_{\tau_n}(\boldsymbol{\omega}, \boldsymbol{\omega})}{\widehat{\xi}^{(B)}_{\tau_n}(\boldsymbol{\omega})} \;\xrightarrow{\mathbb{P}}\; 0.$$

In fact the unscaled log-ratio is itself $O_p(k^{-1})$, which is strictly stronger than the $o_p(k^{-1/2})$ statement required by the $\sqrt{k}$-CLT below. Consequently, when construction (A) is evaluated at matched weights $(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q) = (\boldsymbol{\omega}, \boldsymbol{\omega})$, the two estimators share the same $\sqrt{k}$-asymptotic distribution and Theorem 3.3 applies to either without modification. The same conclusion extends to estimated weights $\hat{\boldsymbol{\omega}}$ satisfying $\hat{\boldsymbol{\omega}} \xrightarrow{\mathbb{P}} \boldsymbol{\omega}$: since $\boldsymbol{\omega}$ is a fixed finite vector, the components of $\hat{\boldsymbol{\omega}}$ are automatically $O_p(1)$ (cf. Corollary 3.7).

Sketch. A first-order Taylor expansion of $\log \psi$ around γ gives $\log \psi(\hat{\gamma}_j) - \log \psi(\gamma) = m(\gamma)(\hat{\gamma}_j - \gamma) + O_p((\hat{\gamma}_j - \gamma)^2)$, and analogously for $\hat{\gamma}_n(\boldsymbol{\omega})$. The linear terms in $\log \widehat{\xi}^{(A)}$ and $\log \widehat{\xi}^{(B)}$ coincide because $\sum_j \omega_j (\hat{\gamma}_j - \gamma) = \hat{\gamma}_n(\boldsymbol{\omega}) - \gamma$, so the discrepancy is the difference of quadratic-and-higher remainders, which is $O_p(1/k)$. On the $\sqrt{k}$-scale this is $O_p(k^{-1/2}) = o_p(1)$. Full proof in Appendix A.1. $\square$

Canonical choice. We adopt

$$\widehat{\xi}^{\mathrm{pool}}_{\tau_n}(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q) \;:=\; \psi(\hat{\gamma}_n(\boldsymbol{\omega}_\gamma))\, \widehat{q}^{\star}_n(1-\tau_n \mid \boldsymbol{\omega}_q), \tag{3.8}$$

identical to (3.6), as the canonical pooled QB intermediate-expectile estimator for the remainder of this chapter. The choice is justified by parsimony: (3.6) reproduces the iid QB pipeline of Theorem 2.11 — one tail-index input, one quantile input, one application of ψ — directly at the pooled level, whereas (3.7) introduces m applications of ψ which, by Proposition 3.1, contribute no additional asymptotic information.[1]

Two weight vectors, not one. Definition (3.8) carries two free weight vectors rather than a single one. In the quantile-only analysis of Theorem 2.18 the joint asymptotic distribution of the pooled Hill and pooled Weissman estimators is governed by a single $\boldsymbol{\omega}$, and the variance- and AMSE-optima of Proposition 2.16 apply uniformly across the two primitives. The expectile target combines these two primitives through the nonlinear map ψ, and the asymptotic variance of $\widehat{\xi}^{\mathrm{pool}}_{\tau_n}$ acquires a contribution from the within-sample cross-covariance of $\hat{\gamma}_j$ and $\hat{q}^{\star}_j$ that is absent in the quantile-only analysis. This contribution can in principle pull the variance-optimal $\boldsymbol{\omega}_\gamma$ apart from the variance-optimal $\boldsymbol{\omega}_q$; Section 3.5 identifies the exact discrepancy and the sufficient conditions under which the two optima coincide. Keeping the weights separate in the definition leaves that section free to specify their relationship rather than imposing it by hand.

Why the pooled Weissman at the intermediate level. At the intermediate level τ_n, the per-sample Weissman estimator $\hat{q}^{\star}_j(1-\tau_n \mid k_j) = (k_j/(n_j(1-\tau_n)))^{\hat{\gamma}_j(k_j)}\, X_{n_j-k_j:n_j,j}$ plays a dual role. When τ_n matches the sample-specific intermediate level $\tau_{n_j} := 1-k_j/n_j$, the multiplicative power factor collapses to one and $\hat{q}^{\star}_j$ reduces to the bare (k_j+1)-th upper order statistic of sample j — the empirical intermediate quantile. Otherwise, $\hat{q}^{\star}_j$ interpolates or mildly extrapolates across intermediate levels, and remains $\sqrt{k_j}$-consistent by the within-sample joint CLT for the Hill estimator and the intermediate log-order statistic invoked in Proposition 3.2 below — this is the finite-log-rate analogue of the diverging-log-rate Weissman CLT Theorem 2.6, not a direct consequence of it. Adopting the Weissman primitive uniformly — in lieu of switching to bare order statistics at intermediate levels — has two virtues: (i) it is the primitive that the pooling protocol of Section 2.5 already shares between samples, so no additional summary statistic is required; (ii) it carries the same form to the extreme regime treated in Chapter 4, keeping the two chapters on a single notational footing.

3.3 Joint asymptotic distribution of the pooled inputs

This section establishes the joint $\sqrt{k}$-Gaussian limit of $(\hat{\gamma}_n(\boldsymbol{\omega}_\gamma), \log \widehat{q}^{\star}_n(1-\tau_n \mid \boldsymbol{\omega}_q))$ at intermediate level τ_n. The result is the input that Section 3.4 feeds through the delta method to obtain the asymptotic distribution of the pooled QB expectile estimator (3.8).

[1] At the second order, the two constructions differ through a Jensen-type term of order k^{-1} involving $m'(\gamma)$ and the dispersion of the local Hill estimators across samples; its sign depends on the weight vector and need not favour either construction in general. This term sits below the order of the $\sqrt{k}$-asymptotic theory developed here and below the standard EVI bias control $\sqrt{k_j}\, A(n_j/k_j) \to \lambda_j$ of Theorem 2.5, so we do not use it.

Within-sample decomposition. For each sample j, the log of the Weissman estimator decomposes exactly as

$$\log \hat{q}_j^{\star}(1-\tau_n \mid k_j) \;=\; \log X_{n_j-k_j:n_j,\,j} \;+\; L_j(\tau_n)\,\hat{\gamma}_j(k_j), \qquad L_j(\tau_n) := \log\frac{k_j}{n_j(1-\tau_n)}. \tag{3.9}$$

The log-rate factor $L_j(\tau_n)$ records the extent to which sample j extrapolates from its own intermediate level $\tau_{n_j} := 1 - k_j/n_j$ to the chapter target τ_n: $L_j = 0$ when the two coincide, $L_j > 0$ for extrapolation, $L_j < 0$ for interpolation. The only non-algebraic input to the entire proof of Proposition 3.2 is the within-sample joint asymptotic behaviour of the Hill estimator and its companion intermediate log-order statistic at effective sample size k_j. By the tail-empirical-process representation (Haan and Ferreira 2006, Theorem 2.4.8 and §5.1), the two statistics are jointly Gaussian on the $\sqrt{k_j}$-scale, *asymptotically independent*, and each has asymptotic variance γ^2. The zero off-diagonal entry is structural: the Hill estimator is a smooth functional of the centred log-spacings above $X_{n_j-k_j:n_j,\,j}$, while the intermediate-threshold fluctuation is the centred (k_j+1)-th upper log-order statistic itself, and the two decouple in the tail-empirical-process limit. This same joint behaviour underlies the iid QB plug-in CLT of Daouia, Girard, et al. (2018) reproduced in Theorem 2.11. Combining it with (3.9) yields the within-sample joint central limit theorem

$$\sqrt{k_j}\begin{pmatrix} \hat{\gamma}_j(k_j) - \gamma \\ \log \hat{q}_j^{\star}(1-\tau_n \mid k_j) - \log Q_X(\tau_n) \end{pmatrix} \xrightarrow{d} \mathcal{N}\left(\boldsymbol{\mu}_j(\tau_n),\, \gamma^2 \begin{pmatrix} 1 & L_j^{\star} \\ L_j^{\star} & 1 + (L_j^{\star})^2 \end{pmatrix}\right), \tag{3.10}$$

under the common-marginal assumption of Section 3.1 and the condition $L_j(\tau_n) \to L_j^{\star} \in \mathbb{R}$. The bias vector $\boldsymbol{\mu}_j(\tau_n)$ is determined by the second-order condition $\mathcal{C}_2(\gamma, \rho, A)$; its explicit form is deferred to the proof appendix. The 2×2 covariance reveals the within-sample Hill–Weissman correlation induced by the log-rate factor: cross-covariance $\gamma^2 L_j^{\star}$, and a Weissman variance $\gamma^2(1 + (L_j^{\star})^2)$ obtained from the asymptotically independent order-statistic and rescaled-Hill contributions.

Pooling under independence. Independence of the m samples implies that the cross-sample covariance blocks of the $2m$-vector $\big(\hat{\gamma}_j - \gamma,\, \log \hat{q}_j^{\star} - \log Q_X(\tau_n)\big)_{j=1,\dots,m}$ vanish identically. Linear pooling with weight vectors $(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q)$ then collapses this $2m$-vector onto a 2-vector whose covariance is assembled block-by-block from the within-sample blocks of (3.10), each rescaled to the common $\sqrt{k}$-scale via the proportionality conditions (2.12).

Proposition 3.2 (Joint pooled CLT at intermediate level)**.** *Under the assumptions of Section 3.1 and the convergence $L_j(\tau_n) \to L_j^{\star} \in \mathbb{R}$ for every j, for any weight vectors $\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q \in \mathbb{R}^m$ satisfying $\boldsymbol{\omega}_\gamma^\top \mathbf{1} = \boldsymbol{\omega}_q^\top \mathbf{1} = 1$,*

$$\sqrt{k}\begin{pmatrix} \hat{\gamma}_n(\boldsymbol{\omega}_\gamma) - \gamma \\ \log \widehat{q}_n^{\star}(1-\tau_n \mid \boldsymbol{\omega}_q) - \log Q_X(\tau_n) \end{pmatrix} \xrightarrow{d} \mathcal{N}(\boldsymbol{\mu}(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q),\, \boldsymbol{\Sigma}(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q)), \tag{3.11}$$

with covariance

$$\boldsymbol{\Sigma}(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q) \;=\; \begin{pmatrix} \boldsymbol{\omega}_\gamma^\top \boldsymbol{V_c} \boldsymbol{\omega}_\gamma & \boldsymbol{\omega}_\gamma^\top \operatorname{diag}(\boldsymbol{L}^\star)\, \boldsymbol{V_c}\, \boldsymbol{\omega}_q \\ \boldsymbol{\omega}_q^\top \operatorname{diag}(\boldsymbol{L}^\star)\, \boldsymbol{V_c}\, \boldsymbol{\omega}_\gamma & \boldsymbol{\omega}_q^\top (\boldsymbol{V_c} + \operatorname{diag}(\boldsymbol{L}^\star)^2 \boldsymbol{V_c})\, \boldsymbol{\omega}_q \end{pmatrix}, \tag{3.12}$$

where $\boldsymbol{V_c}$ *is the diagonal matrix* (2.14), $\boldsymbol{L}^\star = (L_1^\star, \ldots, L_m^\star)^\top$, *and mean*

$$\boldsymbol{\mu}(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q) \;=\; \begin{pmatrix} \boldsymbol{\omega}_\gamma^\top \boldsymbol{B_c} \\ \boldsymbol{\omega}_q^\top \boldsymbol{B_c^\star} \end{pmatrix}, \tag{3.13}$$

where $\boldsymbol{B_c}$ *is the pooled-Hill bias vector of Theorem 2.15 and the limiting pooled-Weissman bias vector at the intermediate level has entries*

$$(\boldsymbol{B_c^\star})_j \;=\; \sqrt{c_j \textstyle\sum_{i=1}^m c_i^{-1}}\; \lambda_j \Big[\frac{L_j^\star}{1-\rho} \;-\; \Psi_\rho(e^{L_j^\star})\Big], \qquad \Psi_\rho(y) \;=\; \frac{y^\rho - 1}{\rho}, \tag{3.14}$$

combining the Hill bias $b_j^H = \lambda_j/(1-\rho)$, *weighted by the log-rate* $L_j^\star$, *with the limiting second-order Weissman remainder* $-\lambda_j \Psi_\rho(e^{L_j^\star})$. *The intermediate log-order statistic contributes no bias term of its own: centred at* $\log Q_X(\tau_{n_j})$ *it is* $\sqrt{k_j}$*-unbiased (Daouia, Girard, et al. 2018; Haan and Ferreira 2006), as established in Appendix A.2. The covariance* (3.12) *does not depend on* $\boldsymbol{B_c^\star}$. *The proposition is stated for deterministic weights; the extension to estimated weights* $\hat{\boldsymbol{\omega}}_\gamma, \hat{\boldsymbol{\omega}}_q \xrightarrow{\mathbb{P}} \boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q$ *is a Slutsky corollary, deferred to Section 3.6.*

Proof. See Appendix A.2. □

Proposition 3.2 is the joint counterpart, at the intermediate finite-log-rate level $L_j(\tau_n) \to L_j^\star \in \mathbb{R}$, of the marginal pooled Hill and pooled Weissman limits recalled in Section 2.5 (Theorems 2.15 and 2.18). The extreme-level, diverging-log-rate joint analogue is the subject of Chapter 4. The new feature for the expectile target is the cross-covariance $\boldsymbol{\omega}_\gamma^\top \operatorname{diag}(\boldsymbol{L}^\star)\boldsymbol{V_c}\boldsymbol{\omega}_q$: it is the structural reason why the variance-optimal weights $(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q)$ for the pooled expectile estimator need not coincide, an observation that Section 3.5 develops quantitatively.

3.4 Asymptotic normality of the pooled expectile estimator

The main result of this chapter is the central limit theorem for the pooled QB intermediate expectile estimator (3.8). The result is obtained by feeding the joint pooled CLT of Proposition 3.2 through the smooth log-linear map $(g, x) \mapsto \log\psi(g) + x$ via a first-order Taylor expansion, combined with the second-order expectile–quantile expansion that controls the deterministic gap between $\log \xi_{\tau_n}$ and $\log\psi(\gamma) + \log Q_X(\tau_n)$.

Derivative of $\log\psi$. A short calculation gives

$$\frac{d}{dg}\log\psi(g) \;=\; \frac{1}{1-g} - \log(g^{-1} - 1) \;=:\; m(g), \tag{3.15}$$

which is the same $m(\gamma)$ that governs the variance of the QB intermediate-level estimator in the iid baseline Theorem 2.11. The function $m(\,\cdot\,)$ is continuous on $(0,1)$, vanishes at the value $\gamma^\star \approx 0.2178$ solving $(1-\gamma)^{-1} = \log(\gamma^{-1}-1)$, and is positive (resp. negative) for γ above (resp. below) $\gamma^\star$.

The deterministic expectile–quantile gap. The first-order asymptotic equivalence $\xi_\tau/Q_X(\tau) \to \psi(\gamma)$ of Proposition 2.9 sharpens, under Condition 2.3, to the logarithm of the second-order expansion (2.7) (Daouia, Girard, et al. 2018),

$$\log \xi_\tau \;-\; \log \psi(\gamma) \;-\; \log Q_X(\tau) \;=\; c(\gamma,\rho)\, A\big((1-\tau)^{-1}\big) \;+\; o\big(A((1-\tau)^{-1})\big), \qquad \tau \uparrow 1, \quad (3.16)$$

where the constant $c(\gamma,\rho)$ is given explicitly in (2.7) and the moment-rate term has been absorbed into the remainder by (3.3); only the finiteness of $c(\gamma,\rho)$ is used below. The chapter setup of Section 3.1 already *implies* a finite limit

$$\Lambda^\bullet \;\coloneqq\; \lim_{n\to\infty} \sqrt{k}\, A\big((1-\tau_n)^{-1}\big) \;\in\; \mathbb{R}, \qquad (3.17)$$

without any additional hypothesis: starting from the per-sample bias rate $\sqrt{k_j}\,A(n_j/k_j) \to \lambda_j$, the log-rate convergence $L_j(\tau_n) \to L_j^\star$, and the regular variation $|A| \in \mathrm{RV}_\rho$ at infinity (Haan and Ferreira 2006, Theorem 2.3.9), one obtains $\sqrt{k_j}\,A((1-\tau_n)^{-1}) \to \lambda_j\,(e^{L_j^\star})^\rho$, and a further rescaling by $\sqrt{k/k_j} \to \sqrt{c_j \sum_i c_i^{-1}}$ delivers (3.17) (the per-j expressions for $\Lambda^\bullet$ agree by joint compatibility of $\{\lambda_j\}, \{L_j^\star\}, \{c_j\}$ under a single global A; see Appendix A.3 for the explicit verification). The finite asymptotic constant

$$b^{\xi/Q} \;\coloneqq\; \lim_{n\to\infty} \sqrt{k}\,\big[\log\psi(\gamma) + \log Q_X(\tau_n) - \log \xi_{\tau_n}\big] \;=\; -\,c(\gamma,\rho)\,\Lambda^\bullet \qquad (3.18)$$

follows. The sign convention in (3.18) is chosen so that $b^{\xi/Q}$ enters the asymptotic mean of the pooled estimator with a $+$ sign in (3.21) below; that is, $b^{\xi/Q}$ is the deterministic *bias contribution* of the gap, not the gap itself. The contribution $b^{\xi/Q}$ is *absent* from the analogue quantile-target theorem Theorem 2.18 of Daouia, Padoan, et al. (2024): it is the deterministic correction that arises when the quantile target is replaced by the expectile target through the QB plug-in.

Theorem 3.3 (Pooled QB intermediate expectile CLT)**.** *Under the assumptions of Proposition 3.2 (in particular Condition 2.3), for any weight vectors $\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q \in \mathbb{R}^m$ satisfying $\boldsymbol{\omega}_\gamma^\top \mathbf{1} = \boldsymbol{\omega}_q^\top \mathbf{1} = 1$,*

$$\sqrt{k}\left(\frac{\widehat{\xi}^{\mathrm{pool}}_{\tau_n}(\boldsymbol{\omega}_\gamma,\boldsymbol{\omega}_q)}{\xi_{\tau_n}} - 1\right) \;\xrightarrow{d}\; \mathcal{N}\big(\mu^\xi(\boldsymbol{\omega}_\gamma,\boldsymbol{\omega}_q),\, \sigma^\xi(\boldsymbol{\omega}_\gamma,\boldsymbol{\omega}_q)^2\big), \qquad (3.19)$$

with asymptotic variance

$$\sigma^\xi(\boldsymbol{\omega}_\gamma,\boldsymbol{\omega}_q)^2 \;=\; m(\gamma)^2\,\boldsymbol{\omega}_\gamma^\top \boldsymbol{V_c}\boldsymbol{\omega}_\gamma \;+\; 2\,m(\gamma)\,\boldsymbol{\omega}_\gamma^\top \mathrm{diag}(\boldsymbol{L}^\star)\boldsymbol{V_c}\boldsymbol{\omega}_q \;+\; \boldsymbol{\omega}_q^\top\big(\boldsymbol{V_c} + \mathrm{diag}(\boldsymbol{L}^\star)^2\boldsymbol{V_c}\big)\boldsymbol{\omega}_q, \quad (3.20)$$

and asymptotic bias

$$\mu^\xi(\boldsymbol{\omega}_\gamma,\boldsymbol{\omega}_q) \;=\; m(\gamma)\,\boldsymbol{\omega}_\gamma^\top \boldsymbol{B_c} \;+\; \boldsymbol{\omega}_q^\top \boldsymbol{B}_{\boldsymbol{c}}^\star \;+\; b^{\xi/Q}, \qquad (3.21)$$

where $\boldsymbol{B_c}$ and $\boldsymbol{B}_{\boldsymbol{c}}^\star$ are as in Proposition 3.2 (the pooled-Hill and limiting intermediate-level *pooled-Weissman bias vectors) and $b^{\xi/Q}$ is the asymptotic expectile–quantile bias contribution of* (3.18)*.*

Proof. See Appendix A.3. □

Reduction to the iid baseline. Theorem 3.3 contains Theorem 2.11 as the special case $m = 1$ with the chapter target aligned to the single sample's intermediate level. The weight vectors collapse to scalars $\omega_\gamma = \omega_q = 1$; the alignment forces $L_1^\star = 0$, $\Psi_\rho(e^{L_1^\star}) = 0$, and $\boldsymbol{V_c} \to \gamma^2$. The asymptotic variance (3.20) collapses to $\gamma^2(1 + m(\gamma)^2)$, in exact agreement with Theorem 2.11. The bias (3.21) collapses likewise. The pooled-Weissman bias vanishes, $\boldsymbol{B}_{\boldsymbol{c}}^\star = 0$: at $L_1^\star = 0$ the bracket in (3.14) reduces to $L_1^\star/(1-\rho) - \Psi_\rho(1) = 0$, since the intermediate order statistic carries no bias of its own. Hence

$$\mu^\xi \;=\; m(\gamma)\,\boldsymbol{B_c} + \boldsymbol{B}_{\boldsymbol{c}}^\star + b^{\xi/Q} \;=\; \frac{m(\gamma)}{1-\rho}\lambda \;-\; c(\gamma,\rho)\,\lambda \;=\; \Big(\frac{m(\gamma)}{1-\rho} - c(\gamma,\rho)\Big)\lambda, \tag{3.22}$$

the Hill bias $\lambda/(1-\rho)$ propagated through $m(\gamma)$, minus the deterministic expectile–quantile gap $c(\gamma,\rho)\lambda$ of (2.7). This is precisely the bias of Theorem 2.11: there is no cancellation between an order-statistic bias and the gap, because — as Appendix A.2 establishes — the intermediate order statistic is $\sqrt{k}$-unbiased and contributes no bias term at all.

Distributed-inference simplification. Under the distributed-inference assumption that the sample fractions are asymptotically equal, $k_j/n_j = (k/n)(1 + o(1))$ for every j (Corollary 2.20), the log-rate factor $L_j(\tau_n) = \log(k/(n(1-\tau_n))) + o(1)$ becomes a sample-free quantity. Writing $L^\star$ for its common limit, (3.20) simplifies to

$$\sigma^\xi(\boldsymbol{\omega}_\gamma,\boldsymbol{\omega}_q)^2 \;=\; m(\gamma)^2\,\boldsymbol{\omega}_\gamma^\top\boldsymbol{V_c}\boldsymbol{\omega}_\gamma \;+\; 2\,m(\gamma)\,L^\star\,\boldsymbol{\omega}_\gamma^\top\boldsymbol{V_c}\boldsymbol{\omega}_q \;+\; (1 + (L^\star)^2)\,\boldsymbol{\omega}_q^\top\boldsymbol{V_c}\boldsymbol{\omega}_q. \tag{3.23}$$

With matched weights $\boldsymbol{\omega}_\gamma = \boldsymbol{\omega}_q = \boldsymbol{\omega}$ and the chapter target aligned to the common per-sample intermediate level ($L^\star = 0$), (3.23) collapses to $(1 + m(\gamma)^2)\,\boldsymbol{\omega}^\top\boldsymbol{V_c}\boldsymbol{\omega}$ — the natural pooled analogue of the iid intermediate QB variance $\gamma^2(1 + m(\gamma)^2)$ of Theorem 2.11.

Why two weight vectors. The separate weight pair $(\boldsymbol{\omega}_\gamma,\boldsymbol{\omega}_q)$ is needed to expose the full variance structure of the pooled QB expectile estimator: the cross term in (3.20) couples the two pooled inputs through $\mathrm{diag}(\boldsymbol{L}^\star)$ and is a genuine structural feature of the expectile target. As Section 3.5 shows, however, the additional freedom is not exploited by the variance optimum in the equal-fraction distributed-inference case: under Corollary 2.20 the variance-optimal pair collapses to a single Paper 3 quantile-style weight vector $\boldsymbol{\omega}_\gamma^{\xi,\mathrm{Var}} = \boldsymbol{\omega}_q^{\xi,\mathrm{Var}} = \boldsymbol{\omega}^{\mathrm{Var}}$ (Proposition 3.4).

3.5 Variance- and AMSE-optimal weights

We now identify the weight vectors $(\boldsymbol{\omega}_\gamma,\boldsymbol{\omega}_q)$ that minimise the asymptotic variance (3.20) and the AMSE $(\mu^\xi)^2 + \sigma^{\xi,2}$ of the pooled QB expectile estimator, subject to the simplex constraints $\boldsymbol{\omega}_\gamma^\top\mathbf{1} = \boldsymbol{\omega}_q^\top\mathbf{1} = 1$. The qualitative question driving this section is whether the expectile-optimal weights coincide with those for the quantile target of Proposition 2.16, or whether the expectile target requires its own weight scheme.

Variance-optimal weights

The variance (3.20) admits the identity

$$\sigma^{\xi}(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q)^2 \;=\; \boldsymbol{u}^\top \boldsymbol{V_c} \boldsymbol{u} \;+\; \boldsymbol{\omega}_q^\top \boldsymbol{V_c} \boldsymbol{\omega}_q, \qquad \boldsymbol{u} \;:=\; m(\gamma)\,\boldsymbol{\omega}_\gamma + \mathrm{diag}(\boldsymbol{L}^\star)\,\boldsymbol{\omega}_q, \tag{3.24}$$

which one verifies by direct expansion using the diagonality of $\boldsymbol{V_c}$ (and hence its commutativity with $\mathrm{diag}(\boldsymbol{L}^\star)$). Both quadratic forms in (3.24) are individually minimised by aligning their argument with $\boldsymbol{V_c}^{-1}\mathbf{1}$; whether this can be done by a single choice of $(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q)$ depends on whether the linear constraint induced on $\boldsymbol{u}$ by the simplex constraints is sample-free. It is, precisely under the equal-fraction distributed-inference assumption $k_j/n_j = (k/n)(1 + o(1))$ of Corollary 2.20, which renders $L_j(\tau_n)$ sample-free and collapses $\mathrm{diag}(\boldsymbol{L}^\star)$ to a scalar multiple of the identity.

Proposition 3.4 (Variance-optimal weights, equal-fraction distributed inference)**.** *Under the assumptions of Theorem 3.3 and the equal-sample-fraction condition $k_j/n_j = (k/n)(1 + o(1))$ of Corollary 2.20, the variance-optimal weights for the pooled QB expectile estimator satisfy*

$$\boldsymbol{\omega}_q^{\xi,\mathrm{Var}} \;=\; \boldsymbol{\omega}^{\mathrm{Var}} \;=\; \frac{\boldsymbol{V_c}^{-1}\mathbf{1}}{\mathbf{1}^\top \boldsymbol{V_c}^{-1}\mathbf{1}}, \tag{3.25}$$

with corresponding asymptotic variance

$$\sigma^{\xi}\big(\boldsymbol{\omega}_\gamma^{\xi,\mathrm{Var}}, \boldsymbol{\omega}_q^{\xi,\mathrm{Var}}\big)^2 \;=\; \big[(m(\gamma) + L^\star)^2 + 1\big]\,\big(\mathbf{1}^\top \boldsymbol{V_c}^{-1}\mathbf{1}\big)^{-1}. \tag{3.26}$$

If $m(\gamma) \neq 0$, the variance-optimal pair is unique and $\boldsymbol{\omega}_\gamma^{\xi,\mathrm{Var}} = \boldsymbol{\omega}^{\mathrm{Var}}$ as well, so both pooled inputs receive the Paper 3 quantile-pooling weights. If $m(\gamma) = 0$, the asymptotic variance does not depend on $\boldsymbol{\omega}_\gamma$: any admissible vector is then variance-optimal, and the matched representative $\boldsymbol{\omega}_\gamma^{\xi,\mathrm{Var}} = \boldsymbol{\omega}^{\mathrm{Var}}$ is one canonical choice.

Proof. The proposition is the specialisation $\boldsymbol{L}^\star = L^\star\mathbf{1}$ of Proposition 3.5; see Appendix A.4, which discharges both the $m(\gamma) \neq 0$ closed form and the $m(\gamma) = 0$ degenerate case. □

Proposition 3.4 is the central variance-efficiency statement of this chapter: under equal-fraction distributed inference, *the variance-optimal pooling weights are invariant under the change of target from extreme quantile to extreme expectile.* The two targets share the same data-fusion recipe; the only change in the asymptotic variance is the multiplicative factor $(m(\gamma) + L^\star)^2 + 1$ replacing the quantile-target factor of 1. The distributed-inference oracle equivalence (Corollary 2.20) thus extends to the expectile target at the variance criterion without modification of the pooling rule. The AMSE-optimum, by contrast, is genuinely target-dependent even under equal-fraction distributed inference, as the second subsection below makes precise.

Beyond the equal-fraction setting. When the sample fractions k_j/n_j vary across j, $\mathrm{diag}(\boldsymbol{L}^\star)$ ceases to be a scalar multiple of the identity, the constraint $\boldsymbol{u}^\top\mathbf{1} = m(\gamma) + \boldsymbol{\omega}_q^\top \boldsymbol{L}^\star$ acquires explicit dependence on $\boldsymbol{\omega}_q$, and the decoupling argument used in Proposition 3.4 breaks. The variance-optimal pair then lifts off the matched-weight diagonal and admits the explicit closed form recorded below. Set $\boldsymbol{D} \coloneqq \operatorname{diag}(\boldsymbol{L}^\star)$ and abbreviate the three scalar invariants of $(\boldsymbol{V_c}, \boldsymbol{L}^\star)$:

$$a \;\coloneqq\; \mathbf{1}^\top \boldsymbol{V_c}^{-1}\mathbf{1}, \qquad b \;\coloneqq\; \mathbf{1}^\top \boldsymbol{V_c}^{-1}\boldsymbol{L}^\star, \qquad c \;\coloneqq\; (\boldsymbol{L}^\star)^\top \boldsymbol{V_c}^{-1}\boldsymbol{L}^\star, \qquad \Delta \;\coloneqq\; a(a+c) - b^2. \tag{3.27}$$

Proposition 3.5 (Variance-optimal weights, general case)**.** *Under the assumptions of Theorem 3.3 and $m(\gamma) \neq 0$, the Cauchy–Schwarz inequality in the $\boldsymbol{V_c}^{-1}$-inner product gives $ac - b^2 \geq 0$, with equality iff $\boldsymbol{L}^\star$ is a scalar multiple of $\mathbf{1}$ — precisely the equal-fraction case in which $L_j^\star$ is sample-free across j. Consequently*

$$\Delta \;=\; a^2 + (ac - b^2) \;\geq\; a^2 \;>\; 0,$$

with $\Delta = a^2$ exactly under the equal-fraction distributed-inference setting of Proposition 3.4. The variance-optimal pooling weights for the pooled QB intermediate expectile estimator are unique and given by

$$\boldsymbol{\omega}_q^{\xi,\mathrm{Var}} \;=\; \frac{(a + c + m(\gamma)\,b)\,\boldsymbol{V_c}^{-1}\mathbf{1} \;-\; (m(\gamma)\,a + b)\,\boldsymbol{V_c}^{-1}\boldsymbol{L}^\star}{\Delta}, \tag{3.28}$$

$$\boldsymbol{\omega}_\gamma^{\xi,\mathrm{Var}} \;=\; \frac{(m(\gamma)\,a + b)\,\boldsymbol{V_c}^{-1}(\mathbf{1} + \boldsymbol{D}^2\mathbf{1}) \;-\; (a + c + m(\gamma)\,b)\,\boldsymbol{V_c}^{-1}\boldsymbol{L}^\star}{m(\gamma)\,\Delta}, \tag{3.29}$$

and the associated minimum asymptotic variance is

$$\sigma^\xi\big(\boldsymbol{\omega}_\gamma^{\xi,\mathrm{Var}}, \boldsymbol{\omega}_q^{\xi,\mathrm{Var}}\big)^2 \;=\; \frac{(m(\gamma)^2 + 1)\,a \;+\; 2\,m(\gamma)\,b \;+\; c}{\Delta}. \tag{3.30}$$

In the distributed-inference setting $\boldsymbol{L}^\star = L^\star\mathbf{1}$ one has $b = L^\star a$, $c = (L^\star)^2 a$, $\Delta = a^2$; formulas (3.28)–(3.30) then collapse to (3.25) and (3.26), recovering Proposition 3.4.

Proof. See Appendix A.4, where the change of variables $(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q) \mapsto (\boldsymbol{u}, \boldsymbol{\omega}_q)$ with $\boldsymbol{u} = m(\gamma)\boldsymbol{\omega}_\gamma + \boldsymbol{D}\boldsymbol{\omega}_q$, positivity of Δ from Cauchy–Schwarz, and the explicit Lagrangian dual system are spelled out step-by-step. $\square$

The assumption $m(\gamma) \neq 0$ is essential for the two-vector closed form: it is the condition that makes the change of variables $(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q) \mapsto (\boldsymbol{u}, \boldsymbol{\omega}_q)$ a bijection, and it appears in the denominator of (3.29). When $m(\gamma) = 0$ the asymptotic variance does not identify $\boldsymbol{\omega}_\gamma$, and the optimisation collapses to the single-vector problem in $\boldsymbol{\omega}_q$ recorded in Appendix A.4.

Departure from the matched-weight diagonal. The discrepancy between $(\boldsymbol{\omega}_\gamma^{\xi,\mathrm{Var}}, \boldsymbol{\omega}_q^{\xi,\mathrm{Var}})$ and the Paper 3 matched-weight pair $(\boldsymbol{\omega}^{\mathrm{Var}}, \boldsymbol{\omega}^{\mathrm{Var}})$ is controlled by the $\boldsymbol{V_c}^{-1}$-Cauchy–Schwarz defect $ac - b^2 \geq 0$, which vanishes exactly when $\boldsymbol{L}^\star$ is a scalar multiple of $\mathbf{1}$. Writing $\bar{L}^\star \coloneqq b/a$ for the $\boldsymbol{V_c}^{-1}\mathbf{1}$-weighted mean of $\boldsymbol{L}^\star$, a direct expansion of (3.28) delivers the closed-form discrepancy

$$\boldsymbol{\omega}_q^{\xi,\mathrm{Var}} - \boldsymbol{\omega}^{\mathrm{Var}} \;=\; -\,\frac{m(\gamma)\,a + b}{\Delta}\,\boldsymbol{V_c}^{-1}(\boldsymbol{L}^\star - \bar{L}^\star\mathbf{1}), \tag{3.31}$$

verified in Appendix A.4 (Step (v)), with an analogous identity for $\boldsymbol{\omega}_\gamma^{\xi,\mathrm{Var}}$ obtained by inversion of the $\boldsymbol{u}$-definition. The correction lies entirely along the $\boldsymbol{V_c}^{-1}$-centred direction $\boldsymbol{V_c}^{-1}(\boldsymbol{L}^\star - \bar{L}^\star\mathbf{1})$. Writing $\Delta = a^2 + (ac - b^2)$ exhibits its scalar prefactor as

$$\frac{m(\gamma)\,a + b}{\Delta} \;=\; \frac{m(\gamma) + \bar{L}^\star}{a}\cdot\left[1 \;+\; O\!\left(\frac{ac-b^2}{a^2}\right)\right]$$

in the small Cauchy–Schwarz defect regime $ac - b^2 \ll a^2$. The Paper 3 weights therefore remain near-optimal whenever (i) the log-rate vector is nearly sample-free ($ac - b^2 \approx 0$, i.e. $\boldsymbol{L}^\star$ nearly constant in the $\boldsymbol{V_c}^{-1}\mathbf{1}$-weighted sense), or (ii) $m(\gamma) + \bar{L}^\star \approx 0$, which combines the expectile-specific zero $\gamma^\star \approx 0.218$ of $m(\,\cdot\,)$ with the log-rate weighted mean. The degenerate case $m(\gamma) = 0$ leaves $\boldsymbol{\omega}_\gamma$ indeterminate and reduces the optimisation to the single-vector problem $\min \boldsymbol{\omega}_q^\top \boldsymbol{V_c}(\boldsymbol{I} + \boldsymbol{D}^2)\boldsymbol{\omega}_q$ subject to $\boldsymbol{\omega}_q^\top\mathbf{1} = 1$, whose solution is recorded in Appendix A.4; it coincides with $\boldsymbol{\omega}^{\mathrm{Var}}$ in the equal-fraction setting and deviates from it otherwise.

AMSE-optimal weights

The AMSE-optimal pair $(\boldsymbol{\omega}_\gamma^{\xi,\mathrm{AMSE}}, \boldsymbol{\omega}_q^{\xi,\mathrm{AMSE}})$ minimises the asymptotic mean squared error

$$\mathrm{AMSE}^\xi(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q) \;:=\; \big(\mu^\xi(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q)\big)^2 + \sigma^\xi(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q)^2 \tag{3.32}$$

subject to the simplex constraints $\boldsymbol{\omega}_\gamma^\top\mathbf{1} = \boldsymbol{\omega}_q^\top\mathbf{1} = 1$, with μ^ξ and σ^ξ as in (3.21) and (3.20). The objective parallels Proposition 2.16(ii), but the expectile target reshapes the bias landscape on two structural counts. First, in the two-weight problem the bias is a linear functional of the stacked vector $(\boldsymbol{\omega}_\gamma^\top, \boldsymbol{\omega}_q^\top)^\top$ with *stacked gradient* $(m(\gamma)\boldsymbol{B_c}, \boldsymbol{B_c}^\star)^\top$, in which the pooled-Hill bias enters rescaled by the QB Jacobian $m(\gamma)$ and the pooled-Weissman bias enters unrescaled; the composite direction $m(\gamma)\boldsymbol{B_c} + \boldsymbol{B_c}^\star$ that replaces the Paper 3 vector $\boldsymbol{B_c}^\star$ arises only when restricted to the matched-weight diagonal $\boldsymbol{\omega}_\gamma = \boldsymbol{\omega}_q$. Second, μ^ξ carries the additive offset $b^{\xi/Q}$ of (3.18), which translates the AMSE minimum away from the Paper 3 variance-optimal direction.

Stacked notation. To state the optimum compactly, stack the two pooling weight vectors into $\boldsymbol{w} := (\boldsymbol{\omega}_\gamma^\top, \boldsymbol{\omega}_q^\top)^\top \in \mathbb{R}^{2m}$ and assemble the bias gradient and variance Hessian implied by (3.21)–(3.20):

$$\boldsymbol{h} \;:=\; \begin{pmatrix} m(\gamma)\,\boldsymbol{B_c} \\ \boldsymbol{B_c}^\star \end{pmatrix} \in \mathbb{R}^{2m}, \qquad \boldsymbol{Q} \;:=\; \begin{pmatrix} m(\gamma)^2\,\boldsymbol{V_c} & m(\gamma)\,\boldsymbol{D}\,\boldsymbol{V_c} \\ m(\gamma)\,\boldsymbol{D}\,\boldsymbol{V_c} & \boldsymbol{V_c} + \boldsymbol{D}^2\boldsymbol{V_c} \end{pmatrix} \in \mathbb{R}^{2m\times 2m}, \tag{3.33}$$

so that $\mu^\xi(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q) = \boldsymbol{h}^\top\boldsymbol{w} + b^{\xi/Q}$ and $\sigma^\xi(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q)^2 = \boldsymbol{w}^\top\boldsymbol{Q}\boldsymbol{w}$. Write $\boldsymbol{A} \in \mathbb{R}^{2m\times 2}$ for the matrix with columns $\mathbf{1}_g := (\mathbf{1}^\top, \mathbf{0}^\top)^\top$ and $\mathbf{1}_q := (\mathbf{0}^\top, \mathbf{1}^\top)^\top$, so that the simplex constraints read $\boldsymbol{A}^\top\boldsymbol{w} = \mathbf{1}_2$.

Proposition 3.6 (AMSE-optimal weights)**.** *Under the assumptions of Theorem 3.3 and $m(\gamma) \neq 0$, the matrix $\boldsymbol{M} := \boldsymbol{Q} + \boldsymbol{h}\boldsymbol{h}^\top$ is positive definite, and the AMSE objective (3.32) is strictly convex on the affine constraint set $\{\boldsymbol{w} \in \mathbb{R}^{2m} : \boldsymbol{A}^\top\boldsymbol{w} = \mathbf{1}_2\}$. The unique minimiser admits the* *closed form*

$$\boldsymbol{w}^{\xi,\mathrm{AMSE}} \;=\; \boldsymbol{M}^{-1}\Big[\boldsymbol{A}\,(\boldsymbol{A}^\top\boldsymbol{M}^{-1}\boldsymbol{A})^{-1}(\mathbf{1}_2 \;+\; b^{\xi/Q}\,\boldsymbol{A}^\top\boldsymbol{M}^{-1}\boldsymbol{h}) \;-\; b^{\xi/Q}\,\boldsymbol{h}\Big], \tag{3.34}$$

and is equivalently characterised by the first-order conditions

$$\begin{aligned} m(\gamma)^2\,\boldsymbol{V_c}\,\boldsymbol{\omega}_\gamma + m(\gamma)\,\boldsymbol{D}\,\boldsymbol{V_c}\,\boldsymbol{\omega}_q \;&=\; \tfrac{1}{2}\lambda_\gamma\mathbf{1} \;-\; m(\gamma)\,\mu^\xi\,\boldsymbol{B_c}, \\ m(\gamma)\,\boldsymbol{D}\,\boldsymbol{V_c}\,\boldsymbol{\omega}_\gamma + (\boldsymbol{V_c}+\boldsymbol{D}^2\boldsymbol{V_c})\,\boldsymbol{\omega}_q \;&=\; \tfrac{1}{2}\lambda_q\mathbf{1} \;-\; \mu^\xi\,\boldsymbol{B}^\star_{\boldsymbol{c}}, \end{aligned} \tag{3.35}$$

together with the simplex constraints $\boldsymbol{\omega}_\gamma^\top\mathbf{1} = \boldsymbol{\omega}_q^\top\mathbf{1} = 1$ *and the self-referential identity* $\mu^\xi = m(\gamma)\,\boldsymbol{\omega}_\gamma^\top\boldsymbol{B_c} + \boldsymbol{\omega}_q^\top\boldsymbol{B}^\star_{\boldsymbol{c}} + b^{\xi/Q}$. *The Lagrange multipliers* $\lambda_\gamma, \lambda_q \in \mathbb{R}$ *are uniquely determined by these constraints.*

Proof. See Appendix A.5, where positive-definiteness of $\boldsymbol{Q}$ (and hence of $\boldsymbol{M} = \boldsymbol{Q} + \boldsymbol{h}\boldsymbol{h}^\top$) is verified in Step (ii) via the variance factorisation, the Lagrangian and explicit FOC normalisation are recorded in Step (iii), and the block decomposition matching (3.35) is in Step (iv). □

Comparison with the Paper 3 AMSE optimum. The expectile AMSE-optimum does *not* reduce to the Paper 3 quantile AMSE-optimum of Proposition 2.16(ii), even under the equal-fraction distributed-inference simplification $\boldsymbol{D} = L^\star\boldsymbol{I}$ that aligned the two variance-optima in Proposition 3.4. Three structural differences persist:

(i) the bias gradient is the stacked vector $\boldsymbol{h} = (m(\gamma)\boldsymbol{B_c}, \boldsymbol{B}^\star_{\boldsymbol{c}})^\top$ rather than the single-block $\boldsymbol{B}^\star_{\boldsymbol{c}}$ of Proposition 2.16;

(ii) μ^ξ carries the additive offset $b^{\xi/Q}$, which shifts the AMSE minimum away from the variance-optimal direction $\boldsymbol{V_c}^{-1}\mathbf{1}$;

(iii) the expectile minimiser is genuinely a pair $(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q)$, and the diagonal $\boldsymbol{\omega}_\gamma = \boldsymbol{\omega}_q$ is generically not a solution to the FOC (3.35) because the Hill and Weissman blocks of $\boldsymbol{h}$ enter asymmetrically through $m(\gamma)\boldsymbol{B_c}$ and $\boldsymbol{B}^\star_{\boldsymbol{c}}$.

The two AMSE-optima coincide only under restrictive alignment conditions; a transparent sufficient case is $b^{\xi/Q} = 0$ together with the structural alignment $\boldsymbol{B}^\star_{\boldsymbol{c}} \propto m(\gamma)\,\boldsymbol{B_c}$ — the deterministic expectile–quantile gap must vanish at the chapter rate, and the pooled-Hill and pooled-Weissman biases must reinforce each other in proportion to the QB Jacobian. We do not characterise the exact necessary condition: it would require inverting the closed-form (3.34) against the Paper 3 expression and isolating the locus of coincidence in $(\boldsymbol{V_c}, \boldsymbol{D}, \boldsymbol{B_c}, \boldsymbol{B}^\star_{\boldsymbol{c}}, b^{\xi/Q})$-space, a calculation that is not generically a single algebraic constraint. Generically, however, the AMSE-optimum is target-dependent. This asymmetry between the variance-optimum, which is target-invariant under equal-fraction distributed inference (Proposition 3.4), and the AMSE-optimum, which is target-dependent even there, is the qualitative signature of carrying the expectile target through the QB plug-in.

3.6 Estimated weights and confidence intervals

The optimal weights of Propositions 3.4 to 3.6 are unknown in practice: they depend on the population covariance $\boldsymbol{V_c}$, on the log-rate vector $\boldsymbol{L}^\star$, on $m(\gamma)$, and — for the AMSE optimum — on the bias quantities $\boldsymbol{B_c}, \boldsymbol{B_c^\star}$ and the expectile–quantile offset $b^{\xi/Q}$. We follow the plug-in scheme of Daouia, Padoan, et al. (2024, §2.2) recalled in Section 2.5: each population quantity is replaced by a consistent empirical analogue, and the resulting plug-in weights are substituted into the pooled estimator (3.8). This section organises the plug-in inputs by what each optimum needs, states the resulting confidence interval under undersmoothing (Corollary 3.7) with the variance-optimal plug-in as the default implementation, and contrasts it with the bias-corrected variant required in the AMSE-optimal regime (Remark 3.8).

Feasible plug-in weights

The three optima of Section 3.5 have different plug-in demands. Listing them in order of increasing complexity:

(a) *Equal-fraction variance-optimal weights* (Proposition 3.4). Under $k_j/n_j = (k/n)(1+o(1))$ the optimum collapses to $\boldsymbol{\omega}^{\mathrm{Var}} = \boldsymbol{V_c}^{-1}\mathbf{1}/(\mathbf{1}^\top \boldsymbol{V_c}^{-1}\mathbf{1})$, which depends on $\boldsymbol{V_c}$ alone. The plug-in

$$\hat{\boldsymbol{\omega}}^{\mathrm{Var}} \;=\; \widehat{\boldsymbol{V}}_{\boldsymbol{c}}^{-1}\mathbf{1} \,/\, (\mathbf{1}^\top \widehat{\boldsymbol{V}}_{\boldsymbol{c}}^{-1}\mathbf{1}) \tag{3.36}$$

therefore requires only a consistent estimator of $\boldsymbol{V_c}$, a first-order quantity built from the per-sample $\hat{\gamma}_j(k_j)$ via the diagonal formula (2.14).

(b) *General variance-optimal weights* (Proposition 3.5). When k_j/n_j varies across j, the closed forms (3.28)–(3.29) also depend on $m(\gamma)$ and on the log-rate vector $\boldsymbol{L}^\star$. The plug-in therefore additionally requires $\hat{m} \xrightarrow{\mathbb{P}} m(\gamma)$ and $\hat{L}_j^\star \to L_j^\star$; the latter is deterministic ($\hat{L}_j^\star := L_j(\tau_n) = \log(k_j/(n_j(1-\tau_n)))$) converges by the chapter setup of Section 3.1), so the only random plug-in beyond $\widehat{\boldsymbol{V}}_{\boldsymbol{c}}$ is $\hat{m}$. To avoid circularity — the general formula for $\hat{\boldsymbol{\omega}}_\gamma$ would otherwise depend on $\hat{m}$, which would itself depend on $\hat{\boldsymbol{\omega}}_\gamma$ — we seed $\hat{m}$ from a preliminary consistent pooled Hill estimator that needs no weight optimisation, for instance the effective-sample-size pool

$$\hat{m}^{(0)} \;:=\; m\Big(\sum_{j=1}^{m}(k_j/k)\,\hat{\gamma}_j(k_j)\Big), \qquad \hat{m}^{(0)} \xrightarrow{\mathbb{P}} m(\gamma), \tag{3.37}$$

which is consistent under Theorem 2.5 and avoids the self-reference. One may optionally iterate after computing the weights — updating $\hat{m} \leftarrow m(\hat{\gamma}_n(\hat{\boldsymbol{\omega}}_\gamma))$ once the weights have been formed — but this is asymptotically immaterial.

(c) *AMSE-optimal weights* (Proposition 3.6). The closed form (3.34) depends on all of $\boldsymbol{V_c}, \boldsymbol{L}^\star, m(\gamma), \boldsymbol{B_c}, \boldsymbol{B_c^\star}, b^{\xi/Q}$ through the stacked vector $\boldsymbol{h}$ and Hessian $\boldsymbol{Q}$ of (3.33). The plug-in therefore requires, on top of $\widehat{\boldsymbol{V}}_{\boldsymbol{c}}$ and $\hat{m}$, consistent estimators of the second-order quantities

$$\widehat{\boldsymbol{B}}_{\boldsymbol{c}} \xrightarrow{\mathbb{P}} \boldsymbol{B_c}, \qquad \widehat{\boldsymbol{B}}_{\boldsymbol{c}}^\star \xrightarrow{\mathbb{P}} \boldsymbol{B_c^\star}, \qquad \hat{b}^{\xi/Q} \xrightarrow{\mathbb{P}} b^{\xi/Q}. \tag{3.38}$$

These are built from estimators $\hat{\beta}_j, \hat{\rho}_j$ of the second-order parameters of $\mathcal{C}_2(\gamma, \rho, A)$ via (3.14)–(3.18) and $\boldsymbol{B_c}$ from Theorem 2.15.

In each case the population optimum map is a continuous function of its arguments on the parameter region where the denominators $\mathbf{1}^\top \boldsymbol{V_c}^{-1}\mathbf{1}, \Delta, \det(\boldsymbol{A}^\top \boldsymbol{M}^{-1}\boldsymbol{A})$ are bounded away from zero (which holds whenever $\boldsymbol{V_c}$ and $\boldsymbol{M}$ are positive definite, and, for the general variance-optimal and AMSE-optimal closed forms, $m(\gamma) \neq 0$). The continuous-mapping theorem then transfers the plug-in convergences above into consistency $\hat{\boldsymbol{\omega}}_\gamma \xrightarrow{\mathbb{P}} \boldsymbol{\omega}_\gamma, \hat{\boldsymbol{\omega}}_q \xrightarrow{\mathbb{P}} \boldsymbol{\omega}_q$ of the optimal-weight plug-ins, and from there into consistency of the plug-in variance $\hat{\sigma}^\xi(\hat{\boldsymbol{\omega}}_\gamma, \hat{\boldsymbol{\omega}}_q) \xrightarrow{\mathbb{P}} \sigma^\xi(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q)$ via (3.20). By construction the closed forms (3.36), (3.28)–(3.29) and (3.34) are normalised against the simplex constraint, so $\hat{\boldsymbol{\omega}}_\gamma^\top \mathbf{1} = \hat{\boldsymbol{\omega}}_q^\top \mathbf{1} = 1$ holds *exactly*, not just in probability — a property that the estimated-weight lift in Corollary 3.7 below relies on.

Practical recommendation. The complexity gradient (a) → (b) → (c) above is also a finite-sample stability gradient. Estimators of the second-order parameter ρ and the rate function $A(\cdot)$ that (3.38) relies on are notoriously delicate in the heavy-tailed regime (Haan and Ferreira 2006, §5.2), and the resulting $\widehat{\boldsymbol{B}}_{\boldsymbol{c}}, \widehat{\boldsymbol{B}}^\star_{\boldsymbol{c}}, \hat{b}^{\xi/Q}$ converge slowly enough that the AMSE-optimal plug-in (c) is substantially less stable in finite samples than the variance-optimal plug-ins (a)–(b). The variance-optimal plug-in is therefore the robust default in practice; the AMSE-optimal plug-in is best reserved for settings where reliable estimators of ρ and A are available.

Confidence interval under undersmoothing

The asymptotic bias (3.21) is the linear combination

$$\mu^\xi(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q) \;=\; m(\gamma)\, \boldsymbol{\omega}_\gamma^\top \boldsymbol{B_c} \;+\; \boldsymbol{\omega}_q^\top \boldsymbol{B}^\star_{\boldsymbol{c}} \;+\; b^{\xi/Q}.$$

Inspection of the bias formulas (3.14)–(3.18) together with $\boldsymbol{B_c}$ from Theorem 2.15 shows that every term is a continuous linear combination of the per-sample bias rates $\lambda_j := \lim \sqrt{k_j}\, A(n_j/k_j)$. The *undersmoothing calibration*

$$\sqrt{k_j}\, A(n_j/k_j) \;\to\; 0 \qquad \text{for every } j = 1, \ldots, m, \tag{3.39}$$

therefore forces $\boldsymbol{B_c} = \mathbf{0}$, $\boldsymbol{B}^\star_{\boldsymbol{c}} = \mathbf{0}$, $\Lambda^\bullet = 0$, $b^{\xi/Q} = 0$, and consequently $\mu^\xi(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q) = 0$ for every admissible pair $(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q)$. This is the hypothesis under which the variance-only confidence interval below is asymptotically valid. Without (3.39) the limit-bias is generically non-zero and the interval must be re-centred on a bias-corrected point estimator (Remark 3.8).

Corollary 3.7 (Asymptotic confidence interval for ξ_{τ_n})**.** *Assume the setting of Theorem 3.3 together with the undersmoothing condition (3.39), and let $(\hat{\boldsymbol{\omega}}_\gamma, \hat{\boldsymbol{\omega}}_q)$ be estimated weight pairs satisfying the simplex constraints $\hat{\boldsymbol{\omega}}_\gamma^\top \mathbf{1} = \hat{\boldsymbol{\omega}}_q^\top \mathbf{1} = 1$ exactly and the consistency conditions*

$$\hat{\boldsymbol{\omega}}_\gamma \xrightarrow{\mathbb{P}} \boldsymbol{\omega}_\gamma, \qquad \hat{\boldsymbol{\omega}}_q \xrightarrow{\mathbb{P}} \boldsymbol{\omega}_q, \qquad \widehat{\boldsymbol{V}}_{\boldsymbol{c}} \xrightarrow{\mathbb{P}} \boldsymbol{V_c}, \qquad \hat{m} \xrightarrow{\mathbb{P}} m(\gamma), \tag{3.40}$$

for some deterministic limit pair $(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q)$ on the simplex. Then, for every $\alpha \in (0,1)$, the random *log-scale interval*

$$I_n^{\log} \;:=\; \log \widehat{\xi}^{\text{pool}}_{\tau_n}(\hat{\boldsymbol{\omega}}_\gamma, \hat{\boldsymbol{\omega}}_q) \;\pm\; z_{1-\alpha/2}\,\frac{\hat{\sigma}^{\xi}(\hat{\boldsymbol{\omega}}_\gamma, \hat{\boldsymbol{\omega}}_q)}{\sqrt{k}} \tag{3.41}$$

and the associated original-scale interval $I_n := \exp(I_n^{\log})$ *are asymptotic* $(1-\alpha)$ *confidence intervals for* $\log \xi_{\tau_n}$ *and* ξ_{τ_n} *respectively:*

$$\mathbb{P}\big(\log \xi_{\tau_n} \in I_n^{\log}\big) \;\to\; 1-\alpha, \qquad \mathbb{P}\big(\xi_{\tau_n} \in I_n\big) \;\to\; 1-\alpha,$$

as $n \to \infty$*, where* $z_{1-\alpha/2}$ *is the standard-normal quantile of order* $1-\alpha/2$ *and* $\hat{\sigma}^{\xi}$ *is the plug-in version of* (3.20)*.*

Proof. See Appendix A.6, where the four-step argument is spelled out: (i) the log-scale recasting of Theorem 3.3; (ii) the lift to estimated weights via the exact simplex constraint and Slutsky; (iii) self-normalisation through the plug-in variance under undersmoothing; (iv) transport of coverage to the original scale by strict monotonicity of exp. □

Practical remarks. First, the log-scale interval (3.41) is symmetric in $\log \xi_{\tau_n}$ while its exponential is not; the latter is the natural form for applications in which ξ_{τ_n} is interpreted as a monetary risk measure. Second, under the equal-fraction distributed-inference setting of Proposition 3.4 the variance-optimal weights are identical across the two pooled inputs, so the single estimated $\hat{\boldsymbol{\omega}}^{\text{Var}}$ of (3.36) suffices and the plug-in variance simplifies to

$$\hat{\sigma}^{\xi}(\hat{\boldsymbol{\omega}}^{\text{Var}}, \hat{\boldsymbol{\omega}}^{\text{Var}})^2 \;=\; \big[(\hat{m} + \hat{L}^{\star})^2 + 1\big]/(\mathbf{1}^{\top}\widehat{\boldsymbol{V}}_{\boldsymbol{c}}^{-1}\mathbf{1}), \qquad \hat{L}^{\star} \;=\; \log\big(k/(n(1-\tau_n))\big),$$

which is the natural plug-in image of the limit (3.26).

CI validity at $m(\gamma) = 0$. The hypotheses of Corollary 3.7 do *not* require $m(\gamma) \neq 0$: the proof of Appendix A.6 uses only (i) consistency of the estimated weights toward a deterministic simplex pair, (ii) consistency of the plug-in variance, and (iii) positivity of the limit variance $\sigma^{\xi}(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q)^2$. The restriction $m(\gamma) \neq 0$ enters elsewhere: it is needed for the variance-optimal closed form of Proposition 3.5 (because the change of variables $(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q) \mapsto (\boldsymbol{u}, \boldsymbol{\omega}_q)$ ceases to be a bijection at $m(\gamma) = 0$) and for the AMSE positive-definiteness of Proposition 3.6. At the boundary $\gamma = \gamma^{\star} \approx 0.2178$, the equal-fraction variance-optimal $\hat{\boldsymbol{\omega}}^{\text{Var}}$ of (3.36) remains well defined and consistent, and the CI (3.41) retains its $(1-\alpha)$ asymptotic coverage; the variance simply degenerates to $\hat{\sigma}^{\xi,2} = [(\hat{L}^{\star})^2 + 1]/(\mathbf{1}^{\top}\widehat{\boldsymbol{V}}_{\boldsymbol{c}}^{-1}\mathbf{1})$.

Remark 3.8 (Bias-corrected confidence interval)*.* The undersmoothing condition (3.39) is the natural companion of the variance-only interval (3.41) but is incompatible with the AMSE-optimal calibration of Proposition 3.6: the AMSE-optimal choice of k is intended for the regime in which at least one λ_j may be non-zero, so that the limit-bias $\mu^{\xi}(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q)$ need not vanish (if all $\lambda_j = 0$ then $\boldsymbol{B}_{\boldsymbol{c}} = \boldsymbol{B}_{\boldsymbol{c}}^{\star} = b^{\xi/Q} = 0$ and the AMSE-optimum collapses to the variance-optimum, so the distinction between the two criteria has no content). When $\mu^{\xi} \neq 0$, (3.41) does not have nominal asymptotic coverage. The standard remedy is to centre the interval on the bias-corrected point estimator $\widehat{\xi}^{\text{pool}}_{\tau_n}\exp\big(-\hat{\mu}^{\xi}(\hat{\boldsymbol{\omega}}_\gamma, \hat{\boldsymbol{\omega}}_q)/\sqrt{k}\big)$, where $\hat{\mu}^{\xi}$ substitutes $\widehat{\boldsymbol{B}}_{\boldsymbol{c}}, \widehat{\boldsymbol{B}}_{\boldsymbol{c}}^{\star}, \hat{b}^{\xi/Q}, \hat{m}$

into (3.21). The corresponding bias-corrected log-scale interval inherits the $(1-\alpha)$ asymptotic coverage of Corollary 3.7 provided $\hat{\mu}^{\xi}(\hat{\boldsymbol{\omega}}_\gamma, \hat{\boldsymbol{\omega}}_q) \xrightarrow{\mathbb{P}} \mu^{\xi}(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q)$ — *i.e.* the second-order consistency conditions (3.38) hold — a hypothesis substantially more demanding than (3.40) because it requires consistent estimation of the second-order parameter ρ and the rate function A (Haan and Ferreira 2006, §5.2).

Standard practice in the extreme-value literature accordingly favours the variance-only interval (3.41) under (3.39) over the bias-corrected variant of Remark 3.8: undersmoothing trades a small amount of asymptotic efficiency for the absence of a hard-to-estimate correction, and operates the pooled estimator away from the AMSE-optimal k of Proposition 3.6. This trade-off is the operational reason why Proposition 3.4 — not Proposition 3.6 — is the natural default companion to Corollary 3.7 in applications.

3.7 A test of expectile-tail-homogeneity

The variance-optimal pooling of Proposition 3.4 rests on the common-γ assumption (H_γ) across the m samples. Although the chapter assumes a common marginal F throughout, in practice this assumption ought to be validated before pooling. The chi-square test of Theorem 2.19 provides such a diagnostic at the tail-index level; we adapt it here to a target formulated directly in terms of the expectile, which is the quantity actually being inferred.

Let

$$\widehat{\xi}_{\tau_n,j} \;:=\; \psi(\hat{\gamma}_j(k_j))\, \hat{q}_j^{\star}(1-\tau_n \mid k_j), \qquad j = 1, \ldots, m, \tag{3.42}$$

denote the per-sample QB intermediate-expectile estimators of Section 3.1 — one per machine, each assembled from that machine's own pooling primitives $(\hat{\gamma}_j(k_j),\, \hat{q}_j^{\star})$ — and collect their logarithms into the vector

$$\hat{\boldsymbol{\Xi}}_n \;:=\; \big(\log \widehat{\xi}_{\tau_n,1}, \ldots, \log \widehat{\xi}_{\tau_n,m}\big)^{\top}.$$

Joint limit of the per-sample log-expectiles. Applying the expectile-specific Jacobian row $(m(\gamma),\, 1)$ — the same linearisation of $\log\psi$ that produced Theorem 3.3 — to the within-sample 2-vector (3.10) of each sample, and aggregating across the m independent samples on the common $\sqrt{k}$ scale, yields the joint central limit theorem

$$\sqrt{k}\,\big(\hat{\boldsymbol{\Xi}}_n - \log \xi_{\tau_n}\mathbf{1}\big) \;\xrightarrow{d}\; \mathcal{N}(\boldsymbol{\beta},\, \boldsymbol{\Sigma}), \tag{3.43}$$

with diagonal covariance matrix

$$\boldsymbol{\Sigma} \;=\; \operatorname{diag}\!\big((m(\gamma) + L_j^{\star})^2 + 1\big)_{j=1}^{m} \boldsymbol{V_c}, \qquad (\boldsymbol{\Sigma})_{j,j} \;=\; \big[(m(\gamma) + L_j^{\star})^2 + 1\big]\,(\boldsymbol{V_c})_{j,j}, \tag{3.44}$$

and asymptotic mean vector

$$\boldsymbol{\beta} \;=\; m(\gamma)\,\boldsymbol{B_c} \;+\; \boldsymbol{B_c^{\star}} \;+\; b^{\xi/Q}\,\mathbf{1}, \tag{3.45}$$

where $\boldsymbol{V_c}$, $\boldsymbol{B_c}$, $\boldsymbol{B_c^\star}$, $b^{\xi/Q}$ and the log-rate limits $L_j^\star$ are exactly the quantities of Proposition 3.2 and Theorem 3.3. The covariance (3.44) is diagonal because the m samples are independent (3.1); its jth diagonal entry is the $\sqrt{k}$-rescaled per-sample log-expectile variance $\gamma^2\left[(m(\gamma)+L_j^\star)^2+1\right](k/k_j)$, read off from (3.10) through the row $(m(\gamma),1)$ and recognising $\gamma^2\,(k/k_j)\to(\boldsymbol{V_c})_{j,j}$. The mean (3.45) is the per-sample analogue of the pooled bias (3.21), the deterministic expectile–quantile offset $b^{\xi/Q}$ of (3.18) entering every coordinate equally (it is sample-free, hence collinear with $\mathbf{1}$). The derivation of (3.43) is recorded in Appendix A.7.

The GLS deviance statistic. Under the common-marginal assumption all $\widehat{\xi}_{\tau_n,j}$ estimate the same population ξ_{τ_n}, and their log-scale dispersion about the generalised-least-squares (GLS) weighted mean is captured by the deviance statistic

$$\Lambda_n^\xi \;:=\; k\,\big(\hat{\boldsymbol{\Xi}}_n-\hat{\nu}_n\mathbf{1}\big)^\top\hat{\boldsymbol{\Sigma}}_n^{-1}\big(\hat{\boldsymbol{\Xi}}_n-\hat{\nu}_n\mathbf{1}\big),\qquad \hat{\nu}_n\;=\;\frac{\mathbf{1}^\top\hat{\boldsymbol{\Sigma}}_n^{-1}\hat{\boldsymbol{\Xi}}_n}{\mathbf{1}^\top\hat{\boldsymbol{\Sigma}}_n^{-1}\mathbf{1}},\tag{3.46}$$

in which $\hat{\nu}_n$ is the GLS estimator of the common log-expectile under (3.43) and $\hat{\boldsymbol{\Sigma}}_n$ is the plug-in image of (3.44) assembled from the consistent estimators of Section 3.6,

$$\hat{\boldsymbol{\Sigma}}_n\;=\;\mathrm{diag}\big((\hat{m}+\hat{L}_j^\star)^2+1\big)_{j=1}^m\,\widehat{\boldsymbol{V}}_{\boldsymbol{c}},\qquad \hat{L}_j^\star\;=\;\log\big(k_j/(n_j(1-\tau_n))\big),\tag{3.47}$$

with $\hat{m}=\hat{m}^{(0)}$ the preliminary effective-sample-size seed (3.37) and $\widehat{\boldsymbol{V}}_{\boldsymbol{c}}$ the diagonal covariance plug-in of (3.36). The residual $\hat{\boldsymbol{\Xi}}_n-\hat{\nu}_n\mathbf{1}$ is the vector of GLS residuals about $\hat{\nu}_n$; by construction it annihilates the direction $\mathbf{1}$, which is the structural reason the statistic is insensitive both to the common centring $\log\xi_{\tau_n}$ and — as the theorem makes precise — to any asymptotic bias collinear with $\mathbf{1}$.

Theorem 3.9 (Expectile-tail-homogeneity test)**.** *Adopt the assumptions of Theorem 3.3 and suppose $\hat{\boldsymbol{\Sigma}}_n\xrightarrow{\mathbb{P}}\boldsymbol{\Sigma}$ with $\boldsymbol{\Sigma}$ positive definite. Under the null hypothesis*

$$H_0^\xi:\quad \frac{\xi_{\tau_n,j}}{\xi_{\tau_n,\ell}}\;\to\;1\qquad\text{for every } j\neq\ell,$$

the statistic (3.46) converges to a noncentral chi-square,

$$\Lambda_n^\xi\;\xrightarrow{d}\;\chi^2_{m-1}(\delta),\qquad \delta\;=\;\boldsymbol{\beta}^\top\boldsymbol{\Sigma}^{-1}\boldsymbol{\beta}\;-\;\frac{\big(\mathbf{1}^\top\boldsymbol{\Sigma}^{-1}\boldsymbol{\beta}\big)^2}{\mathbf{1}^\top\boldsymbol{\Sigma}^{-1}\mathbf{1}}\;\geq\;0,\tag{3.48}$$

with $\boldsymbol{\beta}$, $\boldsymbol{\Sigma}$ as in (3.44)–(3.45). The noncentrality δ is the squared $\boldsymbol{\Sigma}^{-1}$-norm of $\boldsymbol{\beta}$ projected off the direction $\mathbf{1}$, and vanishes — so that the limit is the central χ^2_{m-1} *— exactly when $\boldsymbol{\beta}\in\mathrm{span}\{\mathbf{1}\}$. Two standing regimes of the chapter guarantee $\delta=0$:*

(i) the undersmoothing calibration (3.39), under which $\boldsymbol{B_c}=\boldsymbol{B_c^\star}=\mathbf{0}$ and $b^{\xi/Q}=0$, hence $\boldsymbol{\beta}=\mathbf{0}$;

(ii) the equal-fraction distributed-inference setting $k_j/n_j=(k/n)(1+o(1))$ of Corollary 2.20, under which every coordinate of $\boldsymbol{\beta}$ collapses to one common value, so that $\boldsymbol{\beta}\in\mathrm{span}\{\mathbf{1}\}$.

Under any alternative for which the population log-expectiles are separated faster than $k^{-1/2}$ —

formally, $\sqrt{k}\,\min_{c\in\mathbb{R}}\big\|(\log\xi_{\tau_n,j})_{j=1}^{m}-c\,\mathbf{1}\big\|\to\infty$ *— the statistic diverges,* $\Lambda_n^{\xi}\xrightarrow{\mathbb{P}}\infty$*. A test of asymptotic size* α *rejects* H_0^{ξ} *when* $\Lambda_n^{\xi}>\chi^2_{m-1,\,1-\alpha}$*, the* $(1-\alpha)$*-quantile of the* central χ^2_{m-1} *law.*

Proof. See Appendix A.7. □

Discussion. Three comments situate the test. First, under the chapter's standing assumption of a common marginal F the null H_0^{ξ} holds automatically, so the test is a specification check on the pooling hypothesis rather than a test of a substantive scientific null: a non-rejection licenses the variance-optimal pooling recipes of Sections 3.5 and 3.6, while a rejection warns that the samples are not tail-homogeneous and that pooling would import bias. Second, the central-limit calibration of the test is the same one that underlies the confidence interval of Corollary 3.7: case (i) of Theorem 3.9 is exactly the undersmoothing condition (3.39) adopted there, and case (ii) is the equal-fraction distributed-inference regime in which the variance-optimal weights of Proposition 3.4 are used — so whenever the pooling recipe of this chapter is operated, one of the two regimes that delivers the central χ^2_{m-1} is in force. Outside both regimes the asymptotic bias $\boldsymbol{\beta}$ is generically not collinear with $\mathbf{1}$, the limit is the noncentral $\chi^2_{m-1}(\delta)$ of (3.48), and the nominal-size test is over- or under-sized according to the magnitude of δ; this is the expectile-level counterpart of the identical bias caveat carried, but left implicit, by the tail-index test of Theorem 2.19.

Third, the expectile null H_0^{ξ} and the tail-index null $H_0\colon \gamma_1=\cdots=\gamma_m$ of Theorem 2.19 are related but not identical. Per-sample expectile–quantile equivalence (Proposition 2.9) gives $\xi_{\tau_n,j}\sim\psi(\gamma_j)\,Q_{X_j}(\tau_n)$, so H_0^{ξ} requires the products $\psi(\gamma_j)\,Q_{X_j}(\tau_n)$ to agree across samples in the limit. Under the tail-homoskedasticity condition $(\mathcal{H})$ of Condition 2.17 the quantile factors $Q_{X_j}(\tau_n)$ already coincide, and H_0^{ξ} collapses to homogeneity of the scalars $\psi(\gamma_j)$ — a consequence of, but strictly weaker than, H_0, because ψ is not injective on $(0,1/2)$: it tends to 1 at *both* endpoints $\gamma\downarrow 0$ and $\gamma\uparrow 1/2$, so distinct tail indices can share a common ψ value. Against such ψ-preserving tail-index alternatives the expectile test is asymptotically blind, whereas the tail-index test of Theorem 2.19 detects them. The complementary strength of Λ_n^{ξ} surfaces when $(\mathcal{H})$ *fails*: scale (extreme-quantile) heterogeneity across samples can leave H_0 intact while violating H_0^{ξ}, so the expectile test registers departures in the actual inferential target — the expectile — to which a test of the tail index alone is by construction blind. The two diagnostics are therefore complementary, carrying *different* blind spots rather than one uniformly dominating the other.

Chapter 4

Extrapolation to very-extreme expectile levels

Overview

This chapter extrapolates the pooled intermediate-level expectile estimator of Chapter 3 to very-extreme levels $\tau_n' \uparrow 1$ with $n(1-\tau_n') \to c \in [0,\infty)$. The extrapolation is the analogue of the Weissman extrapolation for extreme quantiles (Weissman 1978) and follows the same multiplicative-power structure that motivates the weighted-geometric pooling of Daouia, Padoan, et al. (2024, §2.3). As in the single-sample case (Davison et al. 2023, Theorem 3.5), the asymptotic distribution at the extreme level is dominated by the tail-index estimator: the contribution of the intermediate-level expectile estimator washes out through the log-rate.

4.1 Motivation

[**TODO:** Risk-management applications routinely require expectiles at levels τ_n' so close to one that there are essentially no observations in the corresponding tail region (typical regime: $n(1-\tau_n') \to c \in [0,\infty)$, equivalently $1-\tau_n' \leq 1/n$). Direct estimation by either LAWS or QB at τ_n' is not feasible; one must extrapolate from an intermediate-level estimator using a model for the tail.]

4.2 The pooled extrapolated expectile estimator

[**TODO:** Define the pooled extrapolated estimator by Weissman-style lifting of the pooled intermediate-level estimator of Chapter 3:

$$\widehat{\xi}^{\text{pool},\star}_{\tau_n'} := \left(\frac{1-\tau_n'}{1-\tau_n}\right)^{-\hat{\gamma}_n(\boldsymbol{\omega}_\gamma)} \widehat{\xi}^{\text{pool}}_{\tau_n}(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q).$$

Discuss two natural alternatives that turn out to be asymptotically equivalent: (i) plug the pooled *extrapolated* quantile $\widehat{q}^{\star}_n(1-(1-\tau_n')/(\gamma^{-1}-1) \mid \boldsymbol{\omega}_q)$ into $\psi(\hat{\gamma}_n)$; (ii) the geometric pooling of marginal extrapolated expectiles. Show via the multiplicative-power identity that all three constructions agree at the asymptotic level.]

4.3 Asymptotic distribution

[**TODO:** State and prove the central limit theorem

$$\frac{\sqrt{k}}{\log((1-\tau_n)/(1-\tau_n'))}\left(\frac{\widehat{\xi}^{\text{pool},\star}_{\tau_n'}}{\xi_{\tau_n'}}-1\right) \xrightarrow{d} \mathcal{N}\Big(\mu^{\xi,\star}(\boldsymbol{\omega}_\gamma),\ \sigma^{\xi,\star}(\boldsymbol{\omega}_\gamma)^2\Big),$$

where the limit law depends *only* on the pooled tail-index estimator (the contribution of $\widehat{\xi}^{\text{pool}}_{\tau_n}$ enters through a log-multiplicative term that vanishes at the displayed rate). Explicitly, $\sigma^{\xi,\star}(\boldsymbol{\omega}_\gamma)^2 = \boldsymbol{\omega}_\gamma^\top \boldsymbol{V_c}\boldsymbol{\omega}_\gamma$. Cite Davison et al. (2023, Proposition C.6) as the plug-in lemma and follow the proof template of Davison et al. (2023, Theorem 3.5).]

4.4 Optimal weights for extrapolation

[**TODO:** Because the asymptotic variance is governed solely by $\hat{\gamma}_n(\boldsymbol{\omega}_\gamma)$, the variance-optimal weights for extrapolation coincide with $\boldsymbol{\omega}^{\text{Var}}$ of Daouia, Padoan, et al. (2024): $\boldsymbol{\omega}^{\text{Var},\star} = \boldsymbol{V_c}^{-1}\mathbf{1}/(\mathbf{1}^\top\boldsymbol{V_c}^{-1}\mathbf{1})$. In the distributed-inference case ($R_{j,\ell}\equiv 0$), these reduce to $\omega_j \propto k_j$ (effective-sample-size weighting), which is the same conclusion as for extreme-quantile extrapolation. Independently, $\boldsymbol{\omega}_q$ becomes irrelevant at the extreme level.]

4.5 Bias-reduction

[**TODO:** Bias-reduced versions of the pooled tail index lift to bias-reduced versions of the pooled extrapolated expectile. Following the bias-reduction scheme of Daouia, Padoan, et al. (2024, §2.2), define $\overline{\gamma}_n(\boldsymbol{\omega}) = \hat{\gamma}_n(\boldsymbol{\omega}) - k^{-1/2}\boldsymbol{\omega}^\top\widehat{\boldsymbol{B}_{\boldsymbol{c}}}$, and propagate this into $\widehat{\xi}^{\text{pool},\star}_{\tau_n'}$. State the resulting unbiased CLT.]

4.6 Asymptotic confidence intervals

[**TODO:** Log-scale confidence intervals for $\xi_{\tau_n'}$ in the spirit of Daouia, Padoan, et al. (2024, Corollary 4):

$$\mathbb{P}\left(\xi_{\tau_n'} \in \widehat{\xi}^{\text{pool},\star}_{\tau_n'} \exp\left[\pm z_{1-\alpha/2}\log\Big(\frac{1-\tau_n}{1-\tau_n'}\Big)\sqrt{\frac{\boldsymbol{\omega}_\gamma^\top\widehat{\boldsymbol{V_c}}\boldsymbol{\omega}_\gamma}{k}}\right]\right) \to 1-\alpha.$$

Discuss finite-sample performance heuristically.]

Chapter 5

Discussion and future work

5.1 Summary of contributions

[**TODO:** Restate the contributions of Chapters 3 and 4 in the language of Section 1.5. Emphasise the modular architecture: the pooled inputs $(\hat{\gamma}_n, \widehat{q}_n^\star)$ of Daouia, Padoan, et al. (2024) compose cleanly with the expectile–quantile asymptotic equivalence via the plug-in CLT of Davison et al. (2023, Prop. C.6), so the only expectile-specific computation is the delta-method calculus around $\psi(\gamma) = (\gamma^{-1} - 1)^{-\gamma}$.]

5.2 Connections to Expected Shortfall

[**TODO:** For a heavy-tailed loss with index $\gamma \in (0,1)$ there is an asymptotic relation $\xi_\tau \sim \mathrm{ES}_{\tau'}$ for a specific matching level $\tau' = \tau'(\tau, \gamma)$ that can be expressed in closed form (cf. Daouia, Girard, et al. (2018)). Thus pooled expectile inference yields, as an immediate corollary, pooled inference on the Expected Shortfall, sidestepping the non-elicitability of ES. Sketch the level-matching identity and the corresponding plug-in estimator.]

5.3 Limitations

[**TODO:** (i) Tail-homogeneity across samples is assumed when pooling extreme quantiles geometrically (cf. assumption $(\mathcal{H})$ in Daouia, Padoan, et al. (2024, §2.3)) and is also needed for pooling expectiles. The test of Section 3.7 should be applied first.
(ii) The choice of effective sample sizes k_j is data-driven and introduces variability that the asymptotic theory does not account for.
(iii) The asymptotic equivalence (1.1) requires $\gamma < 1/2$ for a finite second moment; very heavy tails ($\gamma \geq 1/2$) fall outside the scope.
(iv) The framework is iid within and across samples. Time-series dependence within a sample and dependence across samples are excluded.]

5.4 Future work

[**TODO:** (i) *Time-series dependence within each sample.* Extending the framework to β-mixing observations within a sample, in the spirit of Davison et al. (2023), would require inflating the within-sample asymptotic variance by a serial-dependence factor. The pooled inference architecture itself is preserved.

(ii) *Multivariate observations.* Each sample could be d-dimensional, with the goal of pooling marginal-component-wise expectiles. The joint expectile machinery of Padoan and Stupfler (2022) provides the within-sample tools; the cross-sample step is identical to the univariate case.

(iii) *Tail-heteroscedasticity.* A simple parametric model for $q_j(\tau)/q_\ell(\tau) \to \kappa_{j,\ell}$ would allow pooling even when assumption $(\mathcal{H})$ fails.

(iv) *Empirical applications.* Distributed-inference settings in operational risk (cross-bank loss data), insurance (multi-line claims), and climate (multi-station extremes) are natural use cases.

(v) *Bias-robust pooled estimators.* Use bias-reduced tail-index estimators (e.g. corrected Hill) in place of the plain Hill estimator in the pooled construction.]

Appendix A

Proofs

This appendix collects the detailed proofs of the named results of Chapters 3 and 4 whose proof sketches in the main text omit routine bookkeeping. Each section is self-contained and re-uses only the notation and named results of the body of the thesis.

A.1 Proof of Proposition 3.1

We prove the $\sqrt{k}$-asymptotic equivalence of the two pooled QB intermediate-expectile estimators $\widehat{\xi}^{(A)}_{\tau_n}$ and $\widehat{\xi}^{(B)}_{\tau_n}$ introduced in Section 3.2, applied to a common, *deterministic* weight vector $\boldsymbol{\omega} \in \mathbb{R}^m$ with $\boldsymbol{\omega}^\top \mathbf{1} = 1$, the number of samples m being fixed as $n \to \infty$.

Reduction to a ψ-difference. From the definitions (3.6) and (3.7),

$$\begin{aligned}\widehat{\xi}^{(A)}_{\tau_n}(\boldsymbol{\omega},\boldsymbol{\omega}) &= \psi(\hat{\gamma}_n(\boldsymbol{\omega}))\, \widehat{q}^{\star}_n(1-\tau_n \mid \boldsymbol{\omega}),\\ \widehat{\xi}^{(B)}_{\tau_n}(\boldsymbol{\omega}) &= \Big(\prod_{j=1}^{m} \psi(\hat{\gamma}_j(k_j))^{\omega_j}\Big)\, \widehat{q}^{\star}_n(1-\tau_n \mid \boldsymbol{\omega}).\end{aligned}$$

The Weissman factor is identical in the two estimators, so the ratio reduces to a function of the Hill estimators only:

$$\log \frac{\widehat{\xi}^{(A)}_{\tau_n}(\boldsymbol{\omega},\boldsymbol{\omega})}{\widehat{\xi}^{(B)}_{\tau_n}(\boldsymbol{\omega})} = \log\psi(\hat{\gamma}_n(\boldsymbol{\omega})) - \sum_{j=1}^{m} \omega_j \log\psi(\hat{\gamma}_j(k_j)). \tag{A.1}$$

The proposition therefore reduces to

$$\sqrt{k}\left[\log\psi(\hat{\gamma}_n(\boldsymbol{\omega})) - \sum_j \omega_j \log\psi(\hat{\gamma}_j(k_j))\right] \xrightarrow{\mathbb{P}} 0. \tag{A.2}$$

Good event supporting the Taylor expansion. The map $g \mapsto \log\psi(g) = -g\log(g^{-1}-1)$ is defined on the open interval $(0,1)$ and singular at its endpoints, so the Taylor expansion below requires $\hat{\gamma}_j(k_j)$ and $\hat{\gamma}_n(\boldsymbol{\omega})$ to lie in $(0,1)$. By Theorem 2.5, $\hat{\gamma}_j(k_j) \xrightarrow{\mathbb{P}} \gamma$ for every j, and

$\gamma \in (0, 1/2)$ by (3.2); hence

$$\mathbb{P}\big(\,\hat{\gamma}_j(k_j) \in (0,1) \text{ for every } j = 1, \dots, m\,\big) \;\to\; 1.$$

The pooled estimator inherits the same property: $\hat{\gamma}_n(\boldsymbol{\omega}) = \sum_j \omega_j\, \hat{\gamma}_j(k_j) \xrightarrow{\mathbb{P}} \gamma$ (a finite weighted sum of consistent terms with deterministic weights summing to one), so $\mathbb{P}(\hat{\gamma}_n(\boldsymbol{\omega}) \in (0,1)) \to 1$. Define the event

$$E_n \;\coloneqq\; \{\hat{\gamma}_1(k_1), \dots, \hat{\gamma}_m(k_m),\, \hat{\gamma}_n(\boldsymbol{\omega}) \in (0,1)\}, \qquad \mathbb{P}(E_n) \to 1.$$

It suffices to establish the claimed $O_p(k^{-1})$ bound on E_n: for any $\epsilon > 0$ and $M > 0$, $\mathbb{P}(|X_n| > Ma_n) \le \mathbb{P}(|X_n|\mathbf{1}_{E_n} > Ma_n) + \mathbb{P}(E_n^c)$, and the second term tends to zero, so modifying a sequence on an event of vanishing probability does not affect convergence in probability or $O_p(\cdot)$ bounds. Throughout the remainder of the proof we therefore work on E_n without further comment.

Taylor expansion with Lagrange remainder. The map $g \mapsto \log\psi(g) = -g\log(g^{-1} - 1)$ is C^∞ on the open interval $(0,1)$, with first derivative $m(g) = (1-g)^{-1} - \log(g^{-1} - 1)$ (the function introduced in (3.15)) and second derivative

$$m'(g) \;=\; \frac{1}{(1-g)^2} + \frac{1}{1-g} + \frac{1}{g} \;>\; 0, \qquad g \in (0,1).$$

By Taylor's theorem with Lagrange remainder, for each $j \in \{1, \dots, m\}$ there exists a random $\tilde{\gamma}_j$ lying between $\hat{\gamma}_j(k_j)$ and γ such that

$$\log\psi(\hat{\gamma}_j(k_j)) - \log\psi(\gamma) \;=\; m(\gamma)\,(\hat{\gamma}_j(k_j) - \gamma) + \tfrac{1}{2}\, m'(\tilde{\gamma}_j)\,(\hat{\gamma}_j(k_j) - \gamma)^2, \tag{A.3}$$

and similarly there exists $\tilde{\gamma}_\star$ between $\hat{\gamma}_n(\boldsymbol{\omega})$ and γ with

$$\log\psi(\hat{\gamma}_n(\boldsymbol{\omega})) - \log\psi(\gamma) \;=\; m(\gamma)\,(\hat{\gamma}_n(\boldsymbol{\omega}) - \gamma) + \tfrac{1}{2}\, m'(\tilde{\gamma}_\star)\,(\hat{\gamma}_n(\boldsymbol{\omega}) - \gamma)^2. \tag{A.4}$$

Subtract the $\boldsymbol{\omega}$-weighted sum of (A.3) across j from (A.4). The constant terms $\log\psi(\gamma)$ cancel because $\sum_j \omega_j = 1$, and the linear terms cancel because the pooled Hill estimator is the weighted linear combination $\hat{\gamma}_n(\boldsymbol{\omega}) = \sum_j \omega_j \hat{\gamma}_j(k_j)$, which implies $\sum_j \omega_j(\hat{\gamma}_j - \gamma) = \hat{\gamma}_n(\boldsymbol{\omega}) - \gamma$. We obtain

$$\log\psi(\hat{\gamma}_n(\boldsymbol{\omega})) - \sum_j \omega_j \log\psi(\hat{\gamma}_j(k_j)) \;=\; \tfrac{1}{2}\, m'(\tilde{\gamma}_\star)\,(\hat{\gamma}_n(\boldsymbol{\omega}) - \gamma)^2 - \tfrac{1}{2}\sum_j \omega_j\, m'(\tilde{\gamma}_j)\,(\hat{\gamma}_j(k_j) - \gamma)^2. \tag{A.5}$$

Order of the remainder terms. By Theorem 2.5, $\sqrt{k_j}(\hat{\gamma}_j(k_j) - \gamma) \xrightarrow{d} \mathcal{N}(\lambda_j/(1-\rho), \gamma^2)$ for every j, hence $\hat{\gamma}_j(k_j) - \gamma = O_p(k_j^{-1/2})$. The proportionality condition (2.12), $k_1/k_j \to c_j \in (0,\infty)$, gives $k_j \asymp k$ uniformly in j, so

$$\hat{\gamma}_j(k_j) - \gamma \;=\; O_p(k^{-1/2}), \qquad (\hat{\gamma}_j(k_j) - \gamma)^2 \;=\; O_p(k^{-1}), \qquad j = 1, \dots, m. \tag{A.6}$$

The pooled Hill estimator inherits the rate as a finite weighted combination of $\sqrt{k}$-consistent terms:

$$\hat{\gamma}_n(\boldsymbol{\omega}) - \gamma \;=\; O_p(k^{-1/2}), \qquad \big(\hat{\gamma}_n(\boldsymbol{\omega}) - \gamma\big)^2 \;=\; O_p(k^{-1}). \tag{A.7}$$

For the Lagrange-remainder pre-factors, the consistency $\hat{\gamma}_j(k_j) \xrightarrow{\mathbb{P}} \gamma$ and the squeeze $\tilde{\gamma}_j$ between $\hat{\gamma}_j$ and γ give $\tilde{\gamma}_j \xrightarrow{\mathbb{P}} \gamma$; continuity of m' on $(0,1)$ then yields, by the continuous mapping theorem, $m'(\tilde{\gamma}_j) \xrightarrow{\mathbb{P}} m'(\gamma) = O_p(1)$. The same argument gives $m'(\tilde{\gamma}_\star) \xrightarrow{\mathbb{P}} m'(\gamma)$.

Inserting these orders into (A.5): each summand on the right-hand side is the product of an $O_p(1)$ factor $(m'(\tilde{\gamma}_\cdot))$ and an $O_p(k^{-1})$ factor (a squared deviation), so each is $O_p(k^{-1})$. Since the sum on the right has $m+1$ terms and m is fixed, the right-hand side of (A.5) is itself $O_p(k^{-1})$.

Conclusion. The right-hand side of (A.5) is a finite sum of $m+1$ terms, each a product of an $O_p(1)$ Lagrange factor and an $O_p(k^{-1})$ squared deviation. With m fixed,

$$\log \psi(\hat{\gamma}_n(\boldsymbol{\omega})) - \sum_j \omega_j \log \psi(\hat{\gamma}_j(k_j)) \;=\; O_p(k^{-1}).$$

Combining this with the reduction (A.1) gives

$$\log \frac{\widehat{\xi}^{(A)}_{\tau_n}(\boldsymbol{\omega}, \boldsymbol{\omega})}{\widehat{\xi}^{(B)}_{\tau_n}(\boldsymbol{\omega})} \;=\; O_p(k^{-1}),$$

hence

$$\sqrt{k}\, \log \frac{\widehat{\xi}^{(A)}_{\tau_n}(\boldsymbol{\omega}, \boldsymbol{\omega})}{\widehat{\xi}^{(B)}_{\tau_n}(\boldsymbol{\omega})} \;=\; O_p(k^{-1/2}) \;=\; o_p(1),$$

which proves the convergence in probability claimed by Proposition 3.1. Since $e^x - 1 \sim x$ as $x \to 0$ and our log-ratio is $O_p(k^{-1})$, the continuous mapping theorem applied to exp further yields

$$\frac{\widehat{\xi}^{(A)}_{\tau_n}(\boldsymbol{\omega}, \boldsymbol{\omega})}{\widehat{\xi}^{(B)}_{\tau_n}(\boldsymbol{\omega})} - 1 \;=\; O_p(k^{-1}).$$

The k^{-1} rate is one full order *stronger* than the $o_p(k^{-1/2})$ statement that the $\sqrt{k}$-CLT Theorem 3.3 requires: the two constructions agree asymptotically not only at the CLT scale but at a strictly finer scale, so the two estimators are interchangeable to leading order in every quantity downstream of the joint Gaussian limit. □

A.2 Proof of Proposition 3.2

We establish the joint $\sqrt{k}$-Gaussian limit (3.11) of the pair $\big(\hat{\gamma}_n(\boldsymbol{\omega}_\gamma), \log \widehat{q}^{\star}_n(1-\tau_n \mid \boldsymbol{\omega}_q)\big)$ under the standing assumptions of Section 3.1: m independent samples sharing the common marginal F and the common tail index $\gamma \in (0, 1/2)$, the proportionality conditions (2.12), the second-order condition $\mathcal{C}_2(\gamma, \rho, A)$ with the per-sample bias rate $\sqrt{k_j}\, A(n_j/k_j) \to \lambda_j \in \mathbb{R}$, an intermediate target level $\tau_n \to 1$ with $n(1-\tau_n) \to \infty$, and the chapter's log-rate convergence $L_j(\tau_n) \to L_j^\star \in \mathbb{R}$ for every j. The number of samples m is fixed.

The proof composes four steps: (i) the within-sample joint CLT for the Hill estimator and the intermediate log-order statistic; (ii) the linear lift of (i) to the (Hill, log-Weissman) pair via the exact decomposition (3.9); (iii) cross-sample independence, which renders the $2m$-vector of all per-sample inputs jointly Gaussian with block-diagonal covariance on the common $\sqrt{k}$ scale; (iv) the linear pooling map, which collapses the $2m$-vector onto the 2-vector (3.11) and yields the covariance (3.12).

Step (i): within-sample joint CLT for Hill and intermediate log-order statistic. Fix $j \in \{1, \dots, m\}$ and write $\tau_{n_j} \coloneqq 1 - k_j/n_j$ for the sample-j intermediate level. The joint $\sqrt{k_j}$-asymptotics of the Hill estimator and its companion intermediate log-order statistic is a standard tail-empirical-process result (Haan and Ferreira 2006, Theorem 2.4.8 and §5.1); in the form we use it is the input to the iid QB plug-in CLT of Daouia, Girard, et al. (2018) reproduced as Theorem 2.11. Under $\mathcal{C}_2(\gamma, \rho, A)$ with $k_j \to \infty$, $k_j/n_j \to 0$, and $\sqrt{k_j}\, A(n_j/k_j) \to \lambda_j \in \mathbb{R}$,

$$\sqrt{k_j}\begin{pmatrix} \hat{\gamma}_j(k_j) - \gamma \\ \log X_{n_j - k_j:n_j,\, j} - \log Q_X(\tau_{n_j}) \end{pmatrix} \xrightarrow{d} \mathcal{N}\left(\begin{pmatrix} b_j^H \\ 0 \end{pmatrix},\ \gamma^2\, \boldsymbol{I}_2 \right), \tag{A.8}$$

where $b_j^H = \lambda_j/(1-\rho)$ is the Hill bias and $\boldsymbol{I}_2$ is the 2×2 identity matrix. The *zero* second component is the asymptotic unbiasedness of the intermediate log-order statistic, centred at its own population quantile $\log Q_X(\tau_{n_j}) = \log U(n_j/k_j)$: in the representation $\log X_{n_j-k_j:n_j,\,j} - \log U(n_j/k_j) = \gamma \log V_j + A(n_j/k_j)\, \Psi_\rho(V_j) + o(A(n_j/k_j))$ with $V_j = (k_j/n_j)\, U^{\leftarrow}(\,\cdot\,)$ satisfying $\sqrt{k_j}(V_j - 1) \xrightarrow{d} \mathcal{N}(0,1)$, the second-order term is $\sqrt{k_j}\, A(n_j/k_j)\, \Psi_\rho(V_j) = A(n_j/k_j)\, O_p(1) = o_p(1)$ because $\Psi_\rho(1) = 0$, so it leaves no bias at the $\sqrt{k_j}$ scale (Daouia, Girard, et al. 2018). This is in contrast to the Hill estimator, whose bias b_j^H arises precisely from *averaging* that second-order term across the top k_j order statistics, where it does not vanish. The load-bearing structural fact in (A.8) is the *zero off-diagonal entry*: the Hill estimator is a smooth functional of the centred log-spacings above $X_{n_j-k_j:n_j,\,j}$, while the intermediate-threshold fluctuation is the centred (k_j+1)-th upper log-order statistic, and the two decouple in the tail-empirical-process limit with common variance γ^2. The diagonal form $\gamma^2 \boldsymbol{I}_2$ is therefore essential for the simple $\boldsymbol{V_c}$-and-diag$(\boldsymbol{L}^\star)$ covariance structure of (3.12) used in the rest of the chapter.

Step (ii): linear lift to (Hill, log-Weissman). The Weissman decomposition (3.9), exact for every n, reads

$$\log \hat{q}_j^\star(1-\tau_n \mid k_j) - \log Q_X(\tau_n) \;=\; L_j(\tau_n)\,\big(\hat{\gamma}_j(k_j) - \gamma\big) + \big(\log X_{n_j-k_j:n_j,\,j} - \log Q_X(\tau_{n_j})\big) + r_j(\tau_n), \tag{A.9}$$

where $r_j(\tau_n) \coloneqq \gamma\, L_j(\tau_n) + \log Q_X(\tau_{n_j}) - \log Q_X(\tau_n)$ is a deterministic remainder. Identity (A.9) follows by inserting the definition $\hat{q}_j^\star(1-\tau_n \mid k_j) = (k_j/(n_j(1-\tau_n)))^{\hat{\gamma}_j(k_j)} X_{n_j-k_j:n_j,\,j}$ and adding and subtracting $\gamma\, L_j(\tau_n)$ and $\log Q_X(\tau_{n_j})$.

The remainder $r_j(\tau_n)$ measures the deterministic gap between the population log-quantile at the chapter level τ_n and its first-order regular-variation approximation pivoted at τ_{n_j}. By the second-order condition $\mathcal{C}_2(\gamma, \rho, A)$ applied to $U = (1/\overline{F})^{\leftarrow}$ at base point $t = n_j/k_j$ and ratio $y = k_j/(n_j(1-\tau_n))$, one has $\log U(t\,y) - \log U(t) - \gamma \log y = A(t)\, \Psi_\rho(y) + o(A(t))$ uniformly on compact y-sets, with $\Psi_\rho(y) = (y^\rho - 1)/\rho$. Therefore

$$r_j(\tau_n) \;=\; -A(n_j/k_j)\,\Psi_\rho\bigl(k_j/(n_j(1-\tau_n))\bigr) \;+\; o(A(n_j/k_j)), \tag{A.10}$$

and the per-sample bias rate $\sqrt{k_j}\,A(n_j/k_j) \to \lambda_j$ together with $L_j(\tau_n) \to L_j^\star \in \mathbb{R}$ yields $\sqrt{k_j}\,r_j(\tau_n) \to -\lambda_j\,\Psi_\rho(e^{L_j^\star}) =: b_j^R$, a finite deterministic shift. Since $r_j(\tau_n)$ is deterministic, it shifts only the limiting mean and does not enter the covariance: the asymptotic covariance below is determined entirely by the two random inputs $\hat\gamma_j(k_j) - \gamma$ and $\log X_{n_j-k_j:n_j,j} - \log Q_X(\tau_{n_j})$ of (A.8).

Write (A.9) in matrix form,

$$\sqrt{k_j}\begin{pmatrix}\hat\gamma_j(k_j)-\gamma\\ \log\hat q_j^\star(1-\tau_n\mid k_j) - \log Q_X(\tau_n)\end{pmatrix} = \underbrace{\begin{pmatrix}1 & 0\\ L_j(\tau_n) & 1\end{pmatrix}}_{=:\,\boldsymbol{M}_j(\tau_n)}\sqrt{k_j}\begin{pmatrix}\hat\gamma_j(k_j)-\gamma\\ \log X_{n_j-k_j:n_j,j} - \log Q_X(\tau_{n_j})\end{pmatrix} + \begin{pmatrix}0\\ \sqrt{k_j}\,r_j(\tau_n)\end{pmatrix}$$

The deterministic matrix $\boldsymbol{M}_j(\tau_n) \to \boldsymbol{M}_j^\star := (1, 0;\, L_j^\star, 1)$ entrywise, and the deterministic shift converges to $(0, b_j^R)^\top$. The continuous mapping theorem and Slutsky combine (A.8) with these deterministic convergences to deliver

$$\sqrt{k_j}\begin{pmatrix}\hat\gamma_j(k_j)-\gamma\\ \log\hat q_j^\star(1-\tau_n\mid k_j) - \log Q_X(\tau_n)\end{pmatrix} \xrightarrow{d} \mathcal{N}\left(\boldsymbol{\mu}_j^\star,\; \gamma^2\begin{pmatrix}1 & L_j^\star\\ L_j^\star & 1+(L_j^\star)^2\end{pmatrix}\right), \tag{A.11}$$

with mean $\boldsymbol{\mu}_j^\star = \boldsymbol{M}_j^\star(b_j^H, 0)^\top + (0, b_j^R)^\top = \bigl(b_j^H,\, L_j^\star\, b_j^H + b_j^R\bigr)^\top$, the vanishing order-statistic bias of (A.8) carrying through. The covariance is $\boldsymbol{M}_j^\star\,(\gamma^2\boldsymbol{I}_2)\,(\boldsymbol{M}_j^\star)^\top = \gamma^2(1, L_j^\star;\, L_j^\star, 1+(L_j^\star)^2)$, which is precisely (3.10).

Step (iii): cross-sample independence and rescaling to $\sqrt{k}$. The m samples are independent, so the $2m$ random variables $\bigl(\hat\gamma_j(k_j)-\gamma,\, \log\hat q_j^\star(1-\tau_n\mid k_j) - \log Q_X(\tau_n)\bigr)_{j=1}^m$ are organised as m independent 2-blocks. Stacking them into the column vector

$$\boldsymbol{Z}_n \;:=\; \bigl(\hat\gamma_1-\gamma,\,\ldots,\,\hat\gamma_m-\gamma,\; \log\hat q_1^\star - \log Q_X(\tau_n),\,\ldots,\,\log\hat q_m^\star - \log Q_X(\tau_n)\bigr)^\top \in \mathbb{R}^{2m}$$

(Hill components stacked first, log-Weissman components second), the joint CLT of $\sqrt{k_j}\boldsymbol{Z}_n^{(j)}$ from (A.11) aggregates across j under independence to a joint $2m$-dimensional Gaussian limit. To express the limit on the common $\sqrt{k}$ scale, multiply each per-sample $\sqrt{k_j}$ input by the deterministic factor $\sqrt{k/k_j}$, which converges to a finite limit:

$$\frac{k}{k_j} \;=\; \frac{\sum_{i=1}^m k_i}{k_j} \;=\; \sum_{i=1}^m \frac{k_1/k_j}{k_1/k_i} \;\longrightarrow\; c_j\sum_{i=1}^m c_i^{-1},$$

using (2.12). The per-sample within-block covariance (A.11) scales by the factor k/k_j, so its diagonal entry $(\sqrt{k}/\sqrt{k_j})^2\cdot\gamma^2 \to \gamma^2\,c_j\,\sum_i c_i^{-1} = (\boldsymbol{V_c})_{j,j}$, the diagonal entry of the diagonal pooled-Hill covariance (2.14) for the independent-samples specialisation. We therefore obtain

$$\sqrt{k}\,\boldsymbol{Z}_n \xrightarrow{d} \mathcal{N}(\boldsymbol{\mu}_Z,\, \boldsymbol{\Sigma}_Z), \tag{A.12}$$

where the $2m \times 2m$ covariance $\boldsymbol{\Sigma}_Z$ has the block-structured form

$$\boldsymbol{\Sigma}_Z \;=\; \begin{pmatrix} \boldsymbol{V_c} & \operatorname{diag}(\boldsymbol{L}^\star)\,\boldsymbol{V_c} \\ \operatorname{diag}(\boldsymbol{L}^\star)\,\boldsymbol{V_c} & \boldsymbol{V_c} + \operatorname{diag}(\boldsymbol{L}^\star)^2\,\boldsymbol{V_c} \end{pmatrix}, \tag{A.13}$$

expressed in $m \times m$ blocks. Each block is diagonal: the upper-left block is $\boldsymbol{V_c}$ itself (with diagonal entries $(\boldsymbol{V_c})_{j,j} = \gamma^2\, c_j \sum_i c_i^{-1}$); the off-diagonal blocks have jth diagonal entry $L_j^\star(\boldsymbol{V_c})_{j,j}$; the lower-right block has jth diagonal entry $(1 + (L_j^\star)^2)(\boldsymbol{V_c})_{j,j}$. All non-diagonal entries within each block vanish because of cross-sample independence. The mean $\boldsymbol{\mu}_Z$ is the corresponding linear functional of the per-sample bias parameters b_j^H and b_j^R, rescaled to the $\sqrt{k}$ scale: explicitly, $(\boldsymbol{\mu}_Z)_j = \sqrt{c_j \sum_i c_i^{-1}}\, b_j^H = (\boldsymbol{B_c})_j$ for $j = 1, \dots, m$, and $(\boldsymbol{\mu}_Z)_{m+j} = \sqrt{c_j \sum_i c_i^{-1}}\, (L_j^\star\, b_j^H + b_j^R) =: (\boldsymbol{B_c^\star})_j$, the order-statistic bias being absent by (A.8). The first half of $\boldsymbol{\mu}_Z$ is exactly the pooled-Hill bias vector $\boldsymbol{B_c}$ of Theorem 2.15; the second half is the pooled-Weissman bias vector $\boldsymbol{B_c^\star}$ of (3.14), specialised to the intermediate-level target.

Step (iv): linear pooling map. By definitions (3.4) and (3.5),

$$\hat{\gamma}_n(\boldsymbol{\omega}_\gamma) - \gamma \;=\; \sum_{j=1}^m \omega_{\gamma,j}\,(\hat{\gamma}_j(k_j) - \gamma), \qquad \log \widehat{q}_n^\star(1-\tau_n \mid \boldsymbol{\omega}_q) - \log Q_X(\tau_n) \;=\; \sum_{j=1}^m \omega_{q,j}\,\big(\log \hat{q}_j^\star - \log Q_X(\tau_n)\big),$$

where the second identity uses $\boldsymbol{\omega}_q^\top \boldsymbol{1} = 1$ so that subtracting the common term $\log Q_X(\tau_n)$ commutes with the weighted sum. Both pooled coordinates are linear in $\boldsymbol{Z}_n$, and the relation is exactly

$$\begin{pmatrix} \hat{\gamma}_n(\boldsymbol{\omega}_\gamma) - \gamma \\ \log \widehat{q}_n^\star(1 - \tau_n \mid \boldsymbol{\omega}_q) - \log Q_X(\tau_n) \end{pmatrix} \;=\; \boldsymbol{A}\,\boldsymbol{Z}_n, \qquad \boldsymbol{A} \;:=\; \begin{pmatrix} \boldsymbol{\omega}_\gamma^\top & \boldsymbol{0}_m^\top \\ \boldsymbol{0}_m^\top & \boldsymbol{\omega}_q^\top \end{pmatrix} \;\in \mathbb{R}^{2\times 2m}.$$

Applying $\boldsymbol{A}$ to (A.12),

$$\sqrt{k}\,\boldsymbol{A}\,\boldsymbol{Z}_n \xrightarrow{d} \mathcal{N}(\boldsymbol{A}\,\boldsymbol{\mu}_Z,\, \boldsymbol{A}\,\boldsymbol{\Sigma}_Z\,\boldsymbol{A}^\top).$$

The mean follows from the block structure of $\boldsymbol{\mu}_Z$ identified in Step (iii): $\boldsymbol{A}\,\boldsymbol{\mu}_Z = (\boldsymbol{\omega}_\gamma^\top \boldsymbol{B_c},\, \boldsymbol{\omega}_q^\top \boldsymbol{B_c^\star})^\top$, which is the chapter's $\boldsymbol{\mu}(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q)$.

For the covariance, partition $\boldsymbol{A}$ along the same Hill / log-Weissman split as $\boldsymbol{\Sigma}_Z$ and compute block-by-block, using (A.13):

$$\begin{aligned} \boldsymbol{A}\,\boldsymbol{\Sigma}_Z\,\boldsymbol{A}^\top\big|_{1,1} &= \boldsymbol{\omega}_\gamma^\top\,\boldsymbol{V_c}\,\boldsymbol{\omega}_\gamma, \\ \boldsymbol{A}\,\boldsymbol{\Sigma}_Z\,\boldsymbol{A}^\top\big|_{1,2} &= \boldsymbol{\omega}_\gamma^\top\,\operatorname{diag}(\boldsymbol{L}^\star)\,\boldsymbol{V_c}\,\boldsymbol{\omega}_q, \\ \boldsymbol{A}\,\boldsymbol{\Sigma}_Z\,\boldsymbol{A}^\top\big|_{2,2} &= \boldsymbol{\omega}_q^\top\,(\boldsymbol{V_c} + \operatorname{diag}(\boldsymbol{L}^\star)^2\,\boldsymbol{V_c})\,\boldsymbol{\omega}_q, \end{aligned}$$

with the $(2,1)$ block equal to the $(1,2)$ block by symmetry of $\boldsymbol{\Sigma}_Z$. The diagonality of $\boldsymbol{V_c}$ (under independent sampling, (2.14)) ensures that $\boldsymbol{V_c}$ and $\operatorname{diag}(\boldsymbol{L}^\star)$ commute, so the off-diagonal block can equivalently be written $\boldsymbol{\omega}_\gamma^\top \boldsymbol{V_c}\, \operatorname{diag}(\boldsymbol{L}^\star)\, \boldsymbol{\omega}_q$ and is the entrywise sum $\sum_j \omega_{\gamma,j}\, \omega_{q,j}\, L_j^\star\, (\boldsymbol{V_c})_{j,j}$. This recovers the covariance (3.12) of the proposition.

Combining the two displays, the pooled 2-vector converges in distribution to a Gaussian with mean $\boldsymbol{\mu}(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q)$ and covariance $\boldsymbol{\Sigma}(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q)$, which is precisely (3.11). □

A.3 Proof of Theorem 3.3

We prove the pooled QB intermediate-expectile CLT (3.19) under the standing assumptions of Section 3.1: m independent samples with common marginal F, common tail index $\gamma \in (0, 1/2)$, the proportionality conditions (2.12), the second-order condition $\mathcal{C}_2(\gamma, \rho, A)$ with per-sample bias rate $\sqrt{k_j}\, A(n_j/k_j) \to \lambda_j \in \mathbb{R}$, an intermediate target $\tau_n \to 1$ with $n(1-\tau_n) \to \infty$, and the log-rate convergence $L_j(\tau_n) \to L_j^\star \in \mathbb{R}$ for every j. The number of samples m is fixed. We work for arbitrary deterministic weight vectors $\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q \in \mathbb{R}^m$ satisfying $\boldsymbol{\omega}_\gamma^\top \mathbf{1} = \boldsymbol{\omega}_q^\top \mathbf{1} = 1$; the extension to estimated weights is collected in Corollary 3.7.

The proof composes four steps: (i) a deterministic identity on the log scale that decomposes the expectile log-ratio into a random Hill increment, a random log-Weissman increment, and a deterministic expectile–quantile gap; (ii) a Taylor linearisation of $\log \psi$ at γ that converts the Hill increment into a linear functional of $\hat{\gamma}_n - \gamma$ up to a $\sqrt{k}$-negligible remainder; (iii) the second-order expectile–quantile expansion that controls the deterministic gap; (iv) the multivariate delta method applied to Proposition 3.2 to read off the asymptotic mean and covariance, followed by the continuous-mapping theorem applied to exp.

Step (i): the deterministic log-ratio decomposition. By the QB factorisation (3.8),

$$\log \widehat{\xi}^{\,\text{pool}}_{\tau_n}(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q) \;=\; \log \psi\big(\hat{\gamma}_n(\boldsymbol{\omega}_\gamma)\big) \;+\; \log \widehat{q}^{\,\star}_n(1-\tau_n \mid \boldsymbol{\omega}_q).$$

Subtract $\log \xi_{\tau_n}$ from both sides and insert $\pm \log \psi(\gamma) \pm \log Q_X(\tau_n)$:

$$\begin{aligned}
\log \widehat{\xi}^{\,\text{pool}}_{\tau_n} - \log \xi_{\tau_n} \;=\; & \underbrace{\big[\log \psi(\hat{\gamma}_n(\boldsymbol{\omega}_\gamma)) - \log \psi(\gamma)\big]}_{\text{(H)}} \\
&+ \underbrace{\big[\log \widehat{q}^{\,\star}_n(1-\tau_n \mid \boldsymbol{\omega}_q) - \log Q_X(\tau_n)\big]}_{\text{(W)}} \\
&+ \underbrace{\big[\log \psi(\gamma) + \log Q_X(\tau_n) - \log \xi_{\tau_n}\big]}_{\text{(G)}}.
\end{aligned} \tag{A.14}$$

The identity (A.14) is exact for every n. Terms (H) and (W) are random and centred at zero by Proposition 3.2; term (G) is deterministic and depends on the target level τ_n only through population quantities. The sign in (G) is chosen so that the rescaled $\sqrt{k}\,$(G) converges directly to $b^{\xi/Q}$ in (3.18), eliminating the bookkeeping of an intermediate minus sign in Step (iii).

Step (ii): Taylor linearisation of $\log\psi$. The argument is identical in spirit to that of Appendix A.1. The map $g \mapsto \log\psi(g) = -g\log(g^{-1}-1)$ is C^∞ on $(0,1)$ with first derivative $m(g)$ defined in (3.15) and second derivative

$$m'(g) \;=\; \frac{1}{(1-g)^2} + \frac{1}{1-g} + \frac{1}{g} \;>\; 0, \qquad g \in (0,1).$$

By Theorem 2.5 and (3.2), $\hat\gamma_n(\boldsymbol{\omega}_\gamma) \xrightarrow{\mathbb{P}} \gamma \in (0,1/2) \subset (0,1)$, so the event $E_n := \{\hat\gamma_n(\boldsymbol{\omega}_\gamma) \in (0,1)\}$ satisfies $\mathbb{P}(E_n) \to 1$ and we may work on it without affecting convergence in distribution. On E_n, Taylor's theorem with Lagrange remainder yields a random $\tilde\gamma_\star$ between $\hat\gamma_n(\boldsymbol{\omega}_\gamma)$ and γ such that

$$\log\psi(\hat\gamma_n(\boldsymbol{\omega}_\gamma)) - \log\psi(\gamma) \;=\; m(\gamma)\,(\hat\gamma_n(\boldsymbol{\omega}_\gamma)-\gamma) + \tfrac{1}{2}\,m'(\tilde\gamma_\star)\,(\hat\gamma_n(\boldsymbol{\omega}_\gamma)-\gamma)^2. \tag{A.15}$$

Proposition 3.2 gives $\hat\gamma_n(\boldsymbol{\omega}_\gamma) - \gamma = O_p(k^{-1/2})$, hence $(\hat\gamma_n-\gamma)^2 = O_p(k^{-1})$. The squeeze $\tilde\gamma_\star \xrightarrow{\mathbb{P}} \gamma$ together with continuity of m' on $(0,1)$ gives $m'(\tilde\gamma_\star) \xrightarrow{\mathbb{P}} m'(\gamma) = O_p(1)$. The Lagrange remainder in (A.15) is therefore $O_p(k^{-1}) = o_p(k^{-1/2})$, so

$$\sqrt{k}\,(\mathrm{H}) \;=\; m(\gamma)\,\sqrt{k}\,(\hat\gamma_n(\boldsymbol{\omega}_\gamma)-\gamma) \;+\; o_p(1). \tag{A.16}$$

Step (iii): the deterministic expectile–quantile gap. The first-order equivalence $\xi_\tau/Q_X(\tau) \to \psi(\gamma)$ of Proposition 2.9 sharpens, under the second-order condition $\mathcal{C}_2(\gamma,\rho,A)$ with the moment bound $\mathbb{E}\,|X_-|^{2+\delta} < \infty$ of (3.2), to the second-order expansion

$$\log\xi_\tau - \log\psi(\gamma) - \log Q_X(\tau) \;=\; c(\gamma,\rho)\,A\big((1-\tau)^{-1}\big) \;+\; o\big(A((1-\tau)^{-1})\big), \qquad \tau\uparrow 1, \tag{A.17}$$

the logarithm of the second-order expansion (2.7) of Daouia, Girard, et al. (2018), with the explicit constant $c(\gamma,\rho)$ recorded there. The additive log form (A.17) follows by a Taylor expansion of $\log(1+x)$ around zero, the ratio differing from $\psi(\gamma)$ by a vanishing quantity of order A; the moment-rate term $O(Q_X(\tau)^{-1})$ of (2.7) is $\sqrt{k}$-negligible under the standing condition (3.3) and is absorbed into the remainder. Only the finiteness of $c(\gamma,\rho)$ for $\gamma \in (0,1/2)$ and $\rho \le 0$ is used below.

To convert (A.17) into a $\sqrt{k}$-rescaled shift, we show that

$$\sqrt{k}\,A\big((1-\tau_n)^{-1}\big) \;\longrightarrow\; \Lambda^\bullet \;\in\; \mathbb{R}, \tag{A.18}$$

a finite limit determined by the chapter setup. Indeed, for any fixed j, the log-rate convergence $L_j(\tau_n) \to L_j^\star$ gives the ratio

$$\frac{(1-\tau_n)^{-1}}{n_j/k_j} \;=\; \frac{k_j}{n_j(1-\tau_n)} \;=\; e^{L_j(\tau_n)} \;\longrightarrow\; e^{L_j^\star} \;\in\; (0,\infty),$$

so by the regular variation $|A| \in \mathrm{RV}_\rho$ at infinity (Haan and Ferreira 2006, Theorem 2.3.9),

$$\frac{A((1-\tau_n)^{-1})}{A(n_j/k_j)} \;\longrightarrow\; (e^{L_j^\star})^\rho.$$

Multiplying by $\sqrt{k_j}\,A(n_j/k_j) \to \lambda_j$ and then by $\sqrt{k/k_j} \to \sqrt{c_j \sum_i c_i^{-1}}$, the limit (A.18) exists and equals

$$\Lambda^\bullet \;=\; \sqrt{c_j \textstyle\sum_{i=1}^m c_i^{-1}}\;\lambda_j\,(e^{L_j^\star})^\rho \qquad \text{for every } j = 1,\dots,m. \tag{A.19}$$

The right-hand side of (A.19) appears to depend on j, but the left-hand side $\sqrt{k}\,A((1-\tau_n)^{-1})$ does not: hence the per-j limits must coincide whenever they exist. This is a consistency identity implicit in the chapter setup — not an additional hypothesis — expressing the compatibility of the per-sample bias rates $\sqrt{k_j}\,A(n_j/k_j) \to \lambda_j$ and log-rate limits $L_j(\tau_n) \to L_j^\star$ across j under a single global $A \in \mathrm{RV}_\rho$. Inserting (A.18) into (A.17) at $\tau = \tau_n$ and using the sign convention of (G) chosen in (A.14),

$$\sqrt{k}\,(\mathrm{G}) \;=\; \sqrt{k}\,\big[\log\psi(\gamma) + \log Q_X(\tau_n) - \log\xi_{\tau_n}\big] \;\longrightarrow\; -c(\gamma,\rho)\,\Lambda^\bullet \;=\; b^{\xi/Q}, \tag{A.20}$$

which is the finite deterministic shift announced in (3.18).

Step (iv): the delta method and the limit. Insert (A.16) and the trivial identity $\sqrt{k}\,(\mathrm{W}) = \sqrt{k}\,(\mathrm{W})$ into the $\sqrt{k}$-rescaled version of (A.14), and recall (A.20):

$$\sqrt{k}\,\big[\log\widehat{\xi}^{\,\mathrm{pool}}_{\tau_n} - \log\xi_{\tau_n}\big] \;=\; (m(\gamma),\,1)\,\sqrt{k}\begin{pmatrix}\hat\gamma_n(\boldsymbol{\omega}_\gamma) - \gamma \\ \log\widehat{q}^{\,\star}_n(1-\tau_n \mid \boldsymbol{\omega}_q) - \log Q_X(\tau_n)\end{pmatrix} \;+\; \sqrt{k}\,(\mathrm{G}) \;+\; o_p(1). \tag{A.21}$$

By Proposition 3.2 and the continuous-mapping theorem applied to the deterministic linear functional $(g,x) \mapsto m(\gamma)\,g + x$, the leading term on the right of (A.21) converges in distribution to a univariate Gaussian with mean $(m(\gamma),1)\,\boldsymbol{\mu}(\boldsymbol{\omega}_\gamma,\boldsymbol{\omega}_q)$ and variance $(m(\gamma),1)\,\boldsymbol{\Sigma}(\boldsymbol{\omega}_\gamma,\boldsymbol{\omega}_q)\,(m(\gamma),1)^\top$. Reading off from (3.13) and (3.12),

$$(m(\gamma),1)\,\boldsymbol{\mu}(\boldsymbol{\omega}_\gamma,\boldsymbol{\omega}_q) \;=\; m(\gamma)\,\boldsymbol{\omega}_\gamma^\top \boldsymbol{B_c} \;+\; \boldsymbol{\omega}_q^\top \boldsymbol{B}^\star_{\boldsymbol{c}},$$

and the quadratic form expands to

$$\begin{aligned}(m(\gamma),1)\,\boldsymbol{\Sigma}(\boldsymbol{\omega}_\gamma,\boldsymbol{\omega}_q)\,(m(\gamma),1)^\top \;&=\; m(\gamma)^2\,\boldsymbol{\omega}_\gamma^\top \boldsymbol{V_c}\,\boldsymbol{\omega}_\gamma \;+\; 2\,m(\gamma)\,\boldsymbol{\omega}_\gamma^\top\,\mathrm{diag}(\boldsymbol{L}^\star)\,\boldsymbol{V_c}\,\boldsymbol{\omega}_q \\ &\quad+\; \boldsymbol{\omega}_q^\top(\boldsymbol{V_c} + \mathrm{diag}(\boldsymbol{L}^\star)^2\,\boldsymbol{V_c})\boldsymbol{\omega}_q \;=\; \sigma^\xi(\boldsymbol{\omega}_\gamma,\boldsymbol{\omega}_q)^2.\end{aligned}$$

The deterministic shift $\sqrt{k}\,(\mathrm{G})$ in (A.21) converges to $b^{\xi/Q}$ by (A.20). Combining via Slutsky's theorem,

$$\sqrt{k}\,\big[\log\widehat{\xi}^{\,\mathrm{pool}}_{\tau_n} - \log\xi_{\tau_n}\big] \;\xrightarrow{d}\; \mathcal{N}\big(\mu^\xi(\boldsymbol{\omega}_\gamma,\boldsymbol{\omega}_q),\,\sigma^\xi(\boldsymbol{\omega}_\gamma,\boldsymbol{\omega}_q)^2\big), \tag{A.22}$$

with $\mu^\xi(\boldsymbol{\omega}_\gamma,\boldsymbol{\omega}_q) \;=\; m(\gamma)\,\boldsymbol{\omega}_\gamma^\top\boldsymbol{B_c} + \boldsymbol{\omega}_q^\top\boldsymbol{B}^\star_{\boldsymbol{c}} + b^{\xi/Q}$ as in (3.21): the random part contributes $m(\gamma)\,\boldsymbol{\omega}_\gamma^\top\boldsymbol{B_c} + \boldsymbol{\omega}_q^\top\boldsymbol{B}^\star_{\boldsymbol{c}}$ to the mean and the deterministic shift contributes $+b^{\xi/Q}$, both with the same sign as in (A.21).

To convert the log-ratio statement (A.22) into the ratio statement (3.19), observe that (A.22) forces $\sqrt{k}\,\log(\widehat{\xi}^{\,\mathrm{pool}}_{\tau_n}/\xi_{\tau_n}) \;=\; O_p(1)$, equivalently $\log(\widehat{\xi}^{\,\mathrm{pool}}_{\tau_n}/\xi_{\tau_n}) \;=\; o_p(1)$. A first-order Taylor expansion of $u \mapsto e^u - 1$ around $u = 0$ then gives the exact identity

$$\frac{\widehat{\xi}^{\text{pool}}_{\tau_n}}{\xi_{\tau_n}} - 1 \;=\; \log \frac{\widehat{\xi}^{\text{pool}}_{\tau_n}}{\xi_{\tau_n}} \cdot (1 + o_p(1)), \tag{A.23}$$

since the next term in the expansion is squared and thus $O_p(1/k) = o_p(1/\sqrt{k})$. Multiplying through by $\sqrt{k}$ and applying Slutsky's theorem,

$$\sqrt{k}\left(\frac{\widehat{\xi}^{\text{pool}}_{\tau_n}}{\xi_{\tau_n}} - 1\right) \;=\; \sqrt{k}\,\log \frac{\widehat{\xi}^{\text{pool}}_{\tau_n}}{\xi_{\tau_n}} \;+\; o_p(1) \;\xrightarrow{d}\; \mathcal{N}(\mu^{\xi},\, \sigma^{\xi,\,2}),$$

which is (3.19). $\square$

A.4 Proof of Propositions 3.4 and 3.5

We prove Proposition 3.5 (the general case). The equal-fraction Proposition 3.4 is recovered as the specialisation $\boldsymbol{L}^\star = L^\star \mathbf{1}$ at the end of the section.

We minimise the asymptotic variance $\sigma^{\xi}(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q)^2$ of (3.20) over deterministic weight vectors $\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q \in \mathbb{R}^m$ satisfying $\boldsymbol{\omega}_\gamma^\top \mathbf{1} = \boldsymbol{\omega}_q^\top \mathbf{1} = 1$, under the standing assumptions of Section 3.1. Recall $\boldsymbol{D} := \operatorname{diag}(\boldsymbol{L}^\star)$ and the scalars a, b, c, Δ defined in (3.27).

Step (i): the change of variables. The factorisation (3.24),

$$\sigma^{\xi}(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q)^2 \;=\; \boldsymbol{u}^\top \boldsymbol{V_c} \boldsymbol{u} + \boldsymbol{\omega}_q^\top \boldsymbol{V_c} \boldsymbol{\omega}_q, \qquad \boldsymbol{u} := m(\gamma)\,\boldsymbol{\omega}_\gamma + \boldsymbol{D}\,\boldsymbol{\omega}_q,$$

relies only on the diagonality of $\boldsymbol{V_c}$ and is verified by direct expansion. For $m(\gamma) \neq 0$, the linear map $(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q) \mapsto (\boldsymbol{u}, \boldsymbol{\omega}_q)$ is a bijection of $\mathbb{R}^m \times \mathbb{R}^m$ onto itself, with inverse $\boldsymbol{\omega}_\gamma = m(\gamma)^{-1}(\boldsymbol{u} - \boldsymbol{D}\,\boldsymbol{\omega}_q)$. The simplex constraints translate accordingly: from $\boldsymbol{\omega}_\gamma^\top \mathbf{1} = 1$ and $\boldsymbol{D}\mathbf{1} = \boldsymbol{L}^\star$,

$$\boldsymbol{u}^\top \mathbf{1} - \boldsymbol{\omega}_q^\top \boldsymbol{L}^\star \;=\; m(\gamma), \qquad \boldsymbol{\omega}_q^\top \mathbf{1} \;=\; 1. \tag{A.24}$$

The optimisation (3.24) reduces to a strictly convex quadratic programme on $\mathbb{R}^{2m}$, with the block-diagonal positive-definite Hessian $\operatorname{diag}(\boldsymbol{V_c}, \boldsymbol{V_c})$ and the two linear constraints (A.24).

Step (ii): the dual variables. Introduce Lagrange multipliers $\mu, \nu \in \mathbb{R}$ and form

$$\mathcal{L}(\boldsymbol{u}, \boldsymbol{\omega}_q; \mu, \nu) \;=\; \boldsymbol{u}^\top \boldsymbol{V_c} \boldsymbol{u} + \boldsymbol{\omega}_q^\top \boldsymbol{V_c} \boldsymbol{\omega}_q - \mu\,(\boldsymbol{u}^\top \mathbf{1} - \boldsymbol{\omega}_q^\top \boldsymbol{L}^\star - m(\gamma)) - \nu\,(\boldsymbol{\omega}_q^\top \mathbf{1} - 1).$$

The first-order conditions read

$$2\boldsymbol{V_c}\,\boldsymbol{u} \;=\; \mu\,\mathbf{1}, \qquad 2\boldsymbol{V_c}\,\boldsymbol{\omega}_q \;=\; \nu\,\mathbf{1} - \mu\,\boldsymbol{L}^\star. \tag{A.25}$$

Both equations admit the unique solutions

$$\boldsymbol{u} \;=\; (\mu/2)\,\boldsymbol{V_c}^{-1}\mathbf{1}, \qquad \boldsymbol{\omega}_q \;=\; (\nu/2)\,\boldsymbol{V_c}^{-1}\mathbf{1} - (\mu/2)\,\boldsymbol{V_c}^{-1}\boldsymbol{L}^\star, \tag{A.26}$$

displaying $\boldsymbol{u}$ in $\operatorname{span}\{\boldsymbol{V_c}^{-1}\mathbf{1}\}$ and $\boldsymbol{\omega}_q$ in $\operatorname{span}\{\boldsymbol{V_c}^{-1}\mathbf{1}, \boldsymbol{V_c}^{-1}\boldsymbol{L}^\star\}$ at the optimum.

Step (iii): solving for the multipliers. Substitute (A.26) into the constraints (A.24). Recall $a = \mathbf{1}^\top \boldsymbol{V_c}^{-1}\mathbf{1}$, $b = \mathbf{1}^\top \boldsymbol{V_c}^{-1}\boldsymbol{L}^\star = (\boldsymbol{L}^\star)^\top \boldsymbol{V_c}^{-1}\mathbf{1}$ (using symmetry of $\boldsymbol{V_c}^{-1}$), and $c = (\boldsymbol{L}^\star)^\top \boldsymbol{V_c}^{-1}\boldsymbol{L}^\star$:

$$\boldsymbol{u}^\top\mathbf{1} = (\mu/2)\,a, \qquad \boldsymbol{\omega}_q^\top\boldsymbol{L}^\star = (\nu/2)\,b - (\mu/2)\,c, \qquad \boldsymbol{\omega}_q^\top\mathbf{1} = (\nu/2)\,a - (\mu/2)\,b.$$

The constraints (A.24) become the linear system

$$\begin{pmatrix} a+c & -b \\ -b & a \end{pmatrix}\begin{pmatrix} \mu/2 \\ \nu/2 \end{pmatrix} = \begin{pmatrix} m(\gamma) \\ 1 \end{pmatrix}. \tag{A.27}$$

The coefficient matrix has determinant $a(a+c) - b^2 = \Delta$. Cauchy–Schwarz in the $\boldsymbol{V_c}^{-1}$-inner product gives $b^2 \le ac$, with equality iff $\boldsymbol{L}^\star \in \operatorname{span}\{\mathbf{1}\}$ (which, since all components of $\boldsymbol{V_c}^{-1}\mathbf{1}$ are strictly positive, forces $\boldsymbol{L}^\star = L^\star\mathbf{1}$). Therefore $ac - b^2 \ge 0$ and, since $a > 0$ (diagonal positive entries of $\boldsymbol{V_c}^{-1}$),

$$\Delta \;=\; a^2 + (ac - b^2) \;>\; 0, \tag{A.28}$$

strictly, with $\Delta = a^2$ exactly under the distributed-inference setting $\boldsymbol{L}^\star = L^\star\mathbf{1}$. Cramer's rule applied to (A.27) yields

$$\mu/2 \;=\; \frac{m(\gamma)\,a + b}{\Delta}, \qquad \nu/2 \;=\; \frac{a + c + m(\gamma)\,b}{\Delta}. \tag{A.29}$$

Step (iv): the primal solution. Inserting (A.29) into (A.26) gives the optimal $\boldsymbol{\omega}_q$ of (3.28). For $\boldsymbol{\omega}_\gamma$, invert the change of variables: $\boldsymbol{\omega}_\gamma^\star = m(\gamma)^{-1}(\boldsymbol{u}^\star - \boldsymbol{D}\,\boldsymbol{\omega}_q^\star)$. Using the diagonality identities $\boldsymbol{D}\boldsymbol{V_c}^{-1}\mathbf{1} = \boldsymbol{V_c}^{-1}\boldsymbol{L}^\star$ and $\boldsymbol{D}\boldsymbol{V_c}^{-1}\boldsymbol{L}^\star = \boldsymbol{V_c}^{-1}\boldsymbol{D}^2\mathbf{1}$ (both consequences of $\boldsymbol{V_c}, \boldsymbol{D}$ being diagonal and hence commuting), one obtains

$$\boldsymbol{u}^\star - \boldsymbol{D}\,\boldsymbol{\omega}_q^\star \;=\; \frac{m(\gamma)\,a + b}{\Delta}\,\boldsymbol{V_c}^{-1}\mathbf{1} \;-\; \frac{a + c + m(\gamma)\,b}{\Delta}\,\boldsymbol{V_c}^{-1}\boldsymbol{L}^\star \;+\; \frac{m(\gamma)\,a + b}{\Delta}\,\boldsymbol{V_c}^{-1}\boldsymbol{D}^2\mathbf{1},$$

which is (3.29) after the factor $m(\gamma)^{-1}$. The minimum variance follows from Lagrangian duality: at the optimum of a strictly convex quadratic minimised under linear constraints, the optimal value equals $\frac{1}{2}(\mu^\star\,d_1 + \nu^\star\,d_2)$ where $d_1 = m(\gamma)$, $d_2 = 1$ are the constraint right-hand sides. Substituting (A.29):

$$\sigma^\xi(\boldsymbol{\omega}_\gamma^{\xi,\mathrm{Var}}, \boldsymbol{\omega}_q^{\xi,\mathrm{Var}})^2 \;=\; \frac{m(\gamma)\,[m(\gamma)\,a + b] + [a + c + m(\gamma)\,b]}{\Delta} \;=\; \frac{(m(\gamma)^2 + 1)\,a + 2\,m(\gamma)\,b + c}{\Delta},$$

which is (3.30).

Step (v): the departure formula. A direct computation gives

$$\boldsymbol{\omega}_q^{\xi,\mathrm{Var}} - \boldsymbol{\omega}^{\mathrm{Var}} \;=\; \frac{(a + c + m(\gamma)\,b)\,\boldsymbol{V_c}^{-1}\mathbf{1} - (m(\gamma)\,a + b)\,\boldsymbol{V_c}^{-1}\boldsymbol{L}^\star}{\Delta} \;-\; \frac{\boldsymbol{V_c}^{-1}\mathbf{1}}{a}.$$

Bring to a common denominator $a\Delta$ and collect the $\boldsymbol{V_c}^{-1}\mathbf{1}$ coefficient: $a(a+c+m(\gamma)\,b)-\Delta = a^2+ac+m(\gamma)\,ab-a^2-ac+b^2 = b\,(m(\gamma)\,a+b)$. Hence

$$\boldsymbol{\omega}_q^{\xi,\mathrm{Var}} - \boldsymbol{\omega}^{\mathrm{Var}} \;=\; \frac{m(\gamma)\,a+b}{a\,\Delta}\left[b\,\boldsymbol{V_c}^{-1}\mathbf{1} - a\,\boldsymbol{V_c}^{-1}\boldsymbol{L}^\star\right] \;=\; -\,\frac{m(\gamma)\,a+b}{\Delta}\,\boldsymbol{V_c}^{-1}(\boldsymbol{L}^\star - \bar{L}^\star\mathbf{1}),$$

with $\bar{L}^\star = b/a$, which is (3.31).

Specialisation to the distributed-inference setting. Substituting $\boldsymbol{L}^\star = L^\star\mathbf{1}$ gives $b = L^\star a$, $c = (L^\star)^2 a$, $\Delta = a^2$. From (A.29):

$$\mu/2 \;=\; \frac{(m(\gamma)+L^\star)\,a}{a^2} \;=\; \frac{m(\gamma)+L^\star}{a}, \qquad \nu/2 \;=\; \frac{(1+L^{\star 2}+m(\gamma)\,L^\star)\,a}{a^2} \;=\; \frac{1+L^{\star 2}+m(\gamma)\,L^\star}{a},$$

and $\boldsymbol{\omega}_q^\star = (\nu/2-(\mu/2)L^\star)\,\boldsymbol{V_c}^{-1}\mathbf{1} = ((1+L^{\star 2}+m(\gamma)L^\star)-L^\star(m(\gamma)+L^\star))\boldsymbol{V_c}^{-1}\mathbf{1}/a = \boldsymbol{V_c}^{-1}\mathbf{1}/a = \boldsymbol{\omega}^{\mathrm{Var}}$. A symmetric calculation gives $\boldsymbol{\omega}_\gamma^\star = \boldsymbol{\omega}^{\mathrm{Var}}$, and the minimum-variance value reduces to $[(m(\gamma)+L^\star)^2+1]/a$, recovering Proposition 3.4.

The degenerate case $m(\gamma)=0$. When $m(\gamma)=0$, the change-of-variables map is no longer a bijection: $\boldsymbol{u} = \boldsymbol{D}\boldsymbol{\omega}_q$ depends only on $\boldsymbol{\omega}_q$ and $\boldsymbol{\omega}_\gamma$ disappears from the variance. The factorisation collapses to $\sigma^\xi(\boldsymbol{\omega}_\gamma,\boldsymbol{\omega}_q)^2 = \boldsymbol{\omega}_q^\top\boldsymbol{V_c}(\boldsymbol{I}+\boldsymbol{D}^2)\boldsymbol{\omega}_q$, a strictly convex quadratic in $\boldsymbol{\omega}_q$ alone (since $\boldsymbol{V_c}(\boldsymbol{I}+\boldsymbol{D}^2)$ is the product of two positive-definite diagonal matrices, hence positive definite). Minimisation under $\boldsymbol{\omega}_q^\top\mathbf{1}=1$ gives, by the standard Lagrangian calculation for a single linear constraint,

$$\boldsymbol{\omega}_q^{\xi,\mathrm{Var}}\big|_{m(\gamma)=0} \;=\; \frac{(\boldsymbol{I}+\boldsymbol{D}^2)^{-1}\boldsymbol{V_c}^{-1}\mathbf{1}}{\mathbf{1}^\top(\boldsymbol{I}+\boldsymbol{D}^2)^{-1}\boldsymbol{V_c}^{-1}\mathbf{1}}, \tag{A.30}$$

with $\boldsymbol{\omega}_\gamma$ indeterminate. In the distributed-inference setting $\boldsymbol{D} = L^\star\boldsymbol{I}$, $(\boldsymbol{I}+\boldsymbol{D}^2)^{-1}$ is a scalar multiple of $\boldsymbol{I}$ and (A.30) reduces to $\boldsymbol{\omega}^{\mathrm{Var}}$; in general it differs from $\boldsymbol{\omega}^{\mathrm{Var}}$ along directions in $\mathrm{span}\{\mathbf{1},\boldsymbol{L}^\star,\ldots\}$. $\square$

A.5 Proof of Proposition 3.6

We prove existence, uniqueness, and the closed form (3.34) of the AMSE minimiser under the standing assumptions of Section 3.1 and $m(\gamma)\neq 0$. Recall the stacked notation $\boldsymbol{w} = (\boldsymbol{\omega}_\gamma^\top,\boldsymbol{\omega}_q^\top)^\top$, the bias gradient and variance Hessian $\boldsymbol{h},\boldsymbol{Q}$ of (3.33), the constraint matrix $\boldsymbol{A}\in\mathbb{R}^{2m\times 2}$ with columns $\mathbf{1}_g = (\mathbf{1}^\top,\mathbf{0}^\top)^\top$ and $\mathbf{1}_q = (\mathbf{0}^\top,\mathbf{1}^\top)^\top$, and the matrix $\boldsymbol{M} := \boldsymbol{Q}+\boldsymbol{h}\boldsymbol{h}^\top$.

Step (i): rewriting the AMSE as a quadratic form. By construction $\mu^\xi(\boldsymbol{\omega}_\gamma,\boldsymbol{\omega}_q) = \boldsymbol{h}^\top\boldsymbol{w} + b^{\xi/Q}$ and $\sigma^\xi(\boldsymbol{\omega}_\gamma,\boldsymbol{\omega}_q)^2 = \boldsymbol{w}^\top\boldsymbol{Q}\boldsymbol{w}$, so

$$\begin{aligned} \mathrm{AMSE}^\xi(\boldsymbol{w}) \;&=\; (\boldsymbol{h}^\top\boldsymbol{w}+b^{\xi/Q})^2 + \boldsymbol{w}^\top\boldsymbol{Q}\boldsymbol{w} \\ &=\; \boldsymbol{w}^\top\boldsymbol{M}\,\boldsymbol{w} \;+\; 2\,b^{\xi/Q}\,\boldsymbol{h}^\top\boldsymbol{w} \;+\; (b^{\xi/Q})^2, \end{aligned}$$

recognising $(\boldsymbol{h}^\top\boldsymbol{w})^2 = \boldsymbol{w}^\top(\boldsymbol{h}\boldsymbol{h}^\top)\boldsymbol{w}$ and $\boldsymbol{M} = \boldsymbol{Q}+\boldsymbol{h}\boldsymbol{h}^\top$. The additive constant $(b^{\xi/Q})^2$ does not influence the argmin and is dropped below.

Step (ii): positive-definiteness of $\boldsymbol{M}$. $\boldsymbol{Q}$ is positive definite. Indeed, the variance factorisation $\boldsymbol{w}^\top \boldsymbol{Q}\boldsymbol{w} = \boldsymbol{u}^\top \boldsymbol{V_c}\boldsymbol{u} + \boldsymbol{\omega}_q^\top \boldsymbol{V_c}\boldsymbol{\omega}_q$ with $\boldsymbol{u} = m(\gamma)\boldsymbol{\omega}_\gamma + \boldsymbol{D}\boldsymbol{\omega}_q$ exhibits $\boldsymbol{Q}$ as the pull-back of the block-diagonal positive-definite matrix $\mathrm{diag}(\boldsymbol{V_c}, \boldsymbol{V_c})$ under the linear bijection $\boldsymbol{w} \mapsto (\boldsymbol{u}, \boldsymbol{\omega}_q)$ (valid for $m(\gamma) \neq 0$, see Appendix A.4). Hence $\boldsymbol{w}^\top \boldsymbol{Q}\boldsymbol{w} \geq 0$ with equality iff $\boldsymbol{u} = 0$ and $\boldsymbol{\omega}_q = 0$, which forces $\boldsymbol{w} = 0$. The rank-one positive semi-definite update $\boldsymbol{h}\boldsymbol{h}^\top$ preserves positive definiteness, so $\boldsymbol{M} = \boldsymbol{Q} + \boldsymbol{h}\boldsymbol{h}^\top$ is positive definite. Strict convexity of $\mathrm{AMSE}^\xi(\boldsymbol{w}) - (b^{\xi/Q})^2$ on $\mathbb{R}^{2m}$, and a fortiori on the closed affine constraint set $\{\boldsymbol{A}^\top \boldsymbol{w} = \mathbf{1}_2\}$, follows.

Step (iii): existence and uniqueness via the Lagrangian. Form

$$\mathcal{L}(\boldsymbol{w}; \boldsymbol{\nu}) \;=\; \boldsymbol{w}^\top \boldsymbol{M}\,\boldsymbol{w} + 2\,b^{\xi/Q}\,\boldsymbol{h}^\top \boldsymbol{w} - \boldsymbol{\nu}^\top (\boldsymbol{A}^\top \boldsymbol{w} - \mathbf{1}_2),$$

with $\boldsymbol{\nu} \in \mathbb{R}^2$. The first-order conditions read

$$2\,\boldsymbol{M}\,\boldsymbol{w} + 2\,b^{\xi/Q}\,\boldsymbol{h} - \boldsymbol{A}\,\boldsymbol{\nu} \;=\; \mathbf{0}, \tag{A.31}$$

together with $\boldsymbol{A}^\top \boldsymbol{w} = \mathbf{1}_2$. From (A.31),

$$\boldsymbol{w} \;=\; \boldsymbol{M}^{-1}\Big[\tfrac{1}{2}\boldsymbol{A}\,\boldsymbol{\nu} - b^{\xi/Q}\,\boldsymbol{h}\Big]. \tag{A.32}$$

Imposing the constraint $\boldsymbol{A}^\top \boldsymbol{w} = \mathbf{1}_2$:

$$\tfrac{1}{2}\,\boldsymbol{A}^\top \boldsymbol{M}^{-1}\boldsymbol{A}\,\boldsymbol{\nu} - b^{\xi/Q}\,\boldsymbol{A}^\top \boldsymbol{M}^{-1}\boldsymbol{h} \;=\; \mathbf{1}_2.$$

The matrix $\boldsymbol{A}^\top \boldsymbol{M}^{-1}\boldsymbol{A} \in \mathbb{R}^{2\times 2}$ is positive definite: $\boldsymbol{M}^{-1}$ is PD, and $\boldsymbol{A}$ has full column rank since its two columns $\mathbf{1}_g, \mathbf{1}_q$ have disjoint nonzero supports. Hence

$$\boldsymbol{\nu} \;=\; 2\,(\boldsymbol{A}^\top \boldsymbol{M}^{-1}\boldsymbol{A})^{-1}(\mathbf{1}_2 \;+\; b^{\xi/Q}\,\boldsymbol{A}^\top \boldsymbol{M}^{-1}\boldsymbol{h}). \tag{A.33}$$

Substituting (A.33) back into (A.32) delivers the closed form (3.34). Strict convexity of the objective on the affine constraint set together with the linear FOC (A.31) forces this $\boldsymbol{w}$ to be the unique constrained minimiser.

Step (iv): equivalence with the block FOC system. Expand (A.31) block-by-block. The product $\boldsymbol{M}\boldsymbol{w} = \boldsymbol{Q}\boldsymbol{w} + (\boldsymbol{h}^\top \boldsymbol{w})\,\boldsymbol{h} = \boldsymbol{Q}\boldsymbol{w} + (\mu^\xi - b^{\xi/Q})\boldsymbol{h}$, using $\mu^\xi = \boldsymbol{h}^\top \boldsymbol{w} + b^{\xi/Q}$. Substituting, (A.31) becomes

$$2\,\boldsymbol{Q}\boldsymbol{w} \;+\; 2\,(\mu^\xi - b^{\xi/Q})\,\boldsymbol{h} \;+\; 2\,b^{\xi/Q}\,\boldsymbol{h} \;=\; \boldsymbol{A}\,\boldsymbol{\nu} \quad\iff\quad \boldsymbol{Q}\boldsymbol{w} \;+\; \mu^\xi\,\boldsymbol{h} \;=\; \tfrac{1}{2}\,\boldsymbol{A}\,\boldsymbol{\nu}.$$

Writing $\boldsymbol{\nu} = (\lambda_\gamma, \lambda_q)^\top$ and reading off the upper and lower blocks of size m:

$$\begin{aligned}
\big[\boldsymbol{Q}\boldsymbol{w}\big]_{1,\ldots,m} \;&=\; m(\gamma)^2\,\boldsymbol{V_c}\,\boldsymbol{\omega}_\gamma + m(\gamma)\,\boldsymbol{D}\,\boldsymbol{V_c}\,\boldsymbol{\omega}_q,\\
\big[\boldsymbol{Q}\boldsymbol{w}\big]_{m+1,\ldots,2m} \;&=\; m(\gamma)\,\boldsymbol{D}\,\boldsymbol{V_c}\,\boldsymbol{\omega}_\gamma + (\boldsymbol{V_c} + \boldsymbol{D}^2\boldsymbol{V_c})\,\boldsymbol{\omega}_q,
\end{aligned}$$

$[\boldsymbol{h}]_{1,\ldots,m} = m(\gamma)\boldsymbol{B_c}$, $[\boldsymbol{h}]_{m+1,\ldots,2m} = \boldsymbol{B_c^\star}$, $[\boldsymbol{A\nu}]_{1,\ldots,m} = \lambda_\gamma \mathbf{1}$, $[\boldsymbol{A\nu}]_{m+1,\ldots,2m} = \lambda_q \mathbf{1}$. The block decomposition reproduces (3.35). Uniqueness from strict convexity, combined with the consistency identity $\mu^\xi = m(\gamma)\,\boldsymbol{\omega}_\gamma^\top \boldsymbol{B_c} + \boldsymbol{\omega}_q^\top \boldsymbol{B_c^\star} + b^{\xi/Q}$ at the minimiser, completes the equivalence. □

A.6 Proof of Corollary 3.7

We prove the asymptotic-coverage statement of Corollary 3.7 under the standing assumptions of Section 3.1, the undersmoothing condition (3.39), and the consistency assumptions (3.40) on the estimated weight pair $(\hat{\boldsymbol{\omega}}_\gamma, \hat{\boldsymbol{\omega}}_q)$, the plug-in covariance $\widehat{\boldsymbol{V}}_{\boldsymbol{c}}$, and the plug-in QB Jacobian $\hat{m}$. We retain the simplex constraints $\hat{\boldsymbol{\omega}}_\gamma^\top \mathbf{1} = \hat{\boldsymbol{\omega}}_q^\top \mathbf{1} = 1$ *exactly*, which holds for the variance- and AMSE-optimal plug-ins of Propositions 2.16, 3.5 and 3.6 by construction.

The proof composes four steps: (i) recast the conclusion of Theorem 3.3 on the log scale; (ii) lift the deterministic-weight CLT to the estimated weights via the exact simplex constraint and Slutsky; (iii) substitute the plug-in variance to obtain the asymptotic pivot under undersmoothing; (iv) transport coverage to the original scale by strict monotonicity of exp.

Step (i): log-scale recasting. By the QB factorisation (3.8),

$$\log \widehat{\xi}^{\text{pool}}_{\tau_n}(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q) \;=\; \log \psi(\hat{\gamma}_n(\boldsymbol{\omega}_\gamma)) \;+\; \log \widehat{q}_n^\star(1-\tau_n \mid \boldsymbol{\omega}_q). \tag{A.34}$$

Theorem 3.3 delivers $\sqrt{k}\,(\widehat{\xi}^{\text{pool}}_{\tau_n}(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q)/\xi_{\tau_n} - 1) \xrightarrow{d} \mathcal{N}(\mu^\xi, \sigma^{\xi,2})$. The displayed log-CLT (A.22) in the proof of Theorem 3.3 (Appendix A.3) is the same statement on the log scale,

$$\sqrt{k}\,(\log \widehat{\xi}^{\text{pool}}_{\tau_n}(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q) - \log \xi_{\tau_n}) \;\xrightarrow{d}\; \mathcal{N}(\mu^\xi(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q),\, \sigma^\xi(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q)^2). \tag{A.35}$$

Under undersmoothing (3.39), every entry of $\boldsymbol{B_c}, \boldsymbol{B_c^\star}$ and the offset $b^{\xi/Q}$ vanishes (inspection of (3.14), (3.18), and the formula for $\boldsymbol{B_c}$ in Theorem 2.15, each of which is a continuous linear combination of the per-sample bias rates $\lambda_j = \lim \sqrt{k_j}\,A(n_j/k_j)$). Hence $\mu^\xi(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q) = 0$ and the limit in (A.35) reduces to $\mathcal{N}(0, \sigma^\xi(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q)^2)$.

Step (ii): estimated-weight lift. We show that replacing $(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q)$ by $(\hat{\boldsymbol{\omega}}_\gamma, \hat{\boldsymbol{\omega}}_q)$ in (A.34) changes the left-hand side of (A.35) by $o_p(1)$. Decompose this change into a Hill and a Weissman contribution,

$$\begin{aligned} \Delta_n^\gamma &:= \log \psi(\hat{\gamma}_n(\hat{\boldsymbol{\omega}}_\gamma)) - \log \psi(\hat{\gamma}_n(\boldsymbol{\omega}_\gamma)), \\ \Delta_n^q &:= \log \widehat{q}_n^\star(1-\tau_n \mid \hat{\boldsymbol{\omega}}_q) - \log \widehat{q}_n^\star(1-\tau_n \mid \boldsymbol{\omega}_q). \end{aligned}$$

Hill contribution. Linearity of the pooled Hill in its weights ((3.4)) gives

$$\hat{\gamma}_n(\hat{\boldsymbol{\omega}}_\gamma) - \hat{\gamma}_n(\boldsymbol{\omega}_\gamma) \;=\; (\hat{\boldsymbol{\omega}}_\gamma - \boldsymbol{\omega}_\gamma)^\top \hat{\boldsymbol{\gamma}}_n \;=\; (\hat{\boldsymbol{\omega}}_\gamma - \boldsymbol{\omega}_\gamma)^\top (\hat{\boldsymbol{\gamma}}_n - \gamma \mathbf{1}),$$

the deterministic offset $\gamma\mathbf{1}$ vanishing by the exact simplex identity $(\hat{\boldsymbol{\omega}}_\gamma - \boldsymbol{\omega}_\gamma)^\top \mathbf{1} = 1 - 1 = 0$ on both the plug-in and the limit weight vectors. The per-sample Hill CLT (within-sample component of (3.10)) gives $\sqrt{k_j}(\hat\gamma_j-\gamma)=O_p(1)$, hence $\sqrt{k}(\hat\gamma_j-\gamma)=\sqrt{k/k_j}\,O_p(1)=O_p(1)$ via $k_j/k\to c_j\in(0,1)$ from (2.12). Therefore $\sqrt{k}(\hat{\boldsymbol\gamma}_n-\gamma\mathbf{1})=O_p(1)$, and the Cauchy–Schwarz inequality combined with $\hat{\boldsymbol\omega}_\gamma-\boldsymbol\omega_\gamma=o_p(1)$ yields

$$\sqrt{k}\,\big(\hat\gamma_n(\hat{\boldsymbol\omega}_\gamma)-\hat\gamma_n(\boldsymbol\omega_\gamma)\big)\;=\;(\hat{\boldsymbol\omega}_\gamma-\boldsymbol\omega_\gamma)^\top\sqrt{k}(\hat{\boldsymbol\gamma}_n-\gamma\mathbf{1})\;=\;o_p(1)\cdot O_p(1)\;=\;o_p(1). \tag{A.36}$$

The map $\log\psi$ is continuously differentiable on $(0,1/2)$ with derivative $m(\,\cdot\,)$ defined in (3.15); on the good event $\{\hat\gamma_n(\boldsymbol\omega_\gamma),\hat\gamma_n(\hat{\boldsymbol\omega}_\gamma)\in(0,1)\}$ — of probability $\to1$ by the consistency $\hat\gamma_n(\hat{\boldsymbol\omega}_\gamma),\hat\gamma_n(\boldsymbol\omega_\gamma)\xrightarrow{\mathbb{P}}\gamma$ — the mean value theorem yields a random $\tilde\gamma_n$ between the two pooled values such that

$$\Delta_n^\gamma\;=\;m(\tilde\gamma_n)\,\big[\hat\gamma_n(\hat{\boldsymbol\omega}_\gamma)-\hat\gamma_n(\boldsymbol\omega_\gamma)\big].$$

Squeeze gives $\tilde\gamma_n\xrightarrow{\mathbb{P}}\gamma$ and, by continuity of $m(\,\cdot\,)$, $m(\tilde\gamma_n)\xrightarrow{\mathbb{P}}m(\gamma)$. Combining with (A.36), $\sqrt{k}\,\Delta_n^\gamma=O_p(1)\cdot o_p(1)=o_p(1)$.

Weissman contribution. Weighted-geometric linearity of $\log\hat q_n^\star$ in $\boldsymbol\omega_q$ ((3.5)) gives

$$\Delta_n^q\;=\;(\hat{\boldsymbol\omega}_q-\boldsymbol\omega_q)^\top\big(\log\hat{\boldsymbol q}_n^\star-\log Q_X(\tau_n)\mathbf{1}\big),$$

where the deterministic offset $\log Q_X(\tau_n)\mathbf{1}$ cancels via the simplex identity $(\hat{\boldsymbol\omega}_q-\boldsymbol\omega_q)^\top\mathbf{1}=0$. The within-sample Weissman component of (3.10) delivers $\sqrt{k_j}(\log\hat q_j^\star-\log Q_X(\tau_n))=O_p(1)$ under $L_j(\tau_n)\to L_j^\star\in\mathbb{R}$, and the same $\sqrt{k/k_j}=O(1)$ rescaling promotes this to $\sqrt{k}(\log\hat{\boldsymbol q}_n^\star-\log Q_X(\tau_n)\mathbf{1})=O_p(1)$. Hence

$$\sqrt{k}\,\Delta_n^q\;=\;(\hat{\boldsymbol\omega}_q-\boldsymbol\omega_q)^\top\sqrt{k}\big(\log\hat{\boldsymbol q}_n^\star-\log Q_X(\tau_n)\mathbf{1}\big)\;=\;o_p(1)\cdot O_p(1)\;=\;o_p(1). \tag{A.37}$$

Assembly. Combining (A.35) (under undersmoothing, limit-mean zero) with $\sqrt{k}\,\Delta_n^\gamma+\sqrt{k}\,\Delta_n^q=o_p(1)$ and Slutsky's theorem,

$$\sqrt{k}\,\big(\log\hat\xi_{\tau_n}^{\mathrm{pool}}(\hat{\boldsymbol\omega}_\gamma,\hat{\boldsymbol\omega}_q)-\log\xi_{\tau_n}\big)\;\xrightarrow{d}\;\mathcal{N}\big(0,\,\sigma^\xi(\boldsymbol\omega_\gamma,\boldsymbol\omega_q)^2\big). \tag{A.38}$$

Step (iii): self-normalisation. The variance (3.20) is a polynomial in $(\boldsymbol V_{\boldsymbol c},\boldsymbol L^\star,m,\boldsymbol\omega_\gamma,\boldsymbol\omega_q)$, and $\sigma^\xi(\boldsymbol\omega_\gamma,\boldsymbol\omega_q)^2>0$ because $\boldsymbol V_{\boldsymbol c}$ is positive definite, $\boldsymbol\omega_q^\top\mathbf{1}=1$ forces $\boldsymbol\omega_q\neq\mathbf{0}$, and the structurally non-negative contribution $\boldsymbol\omega_q^\top\boldsymbol V_{\boldsymbol c}\boldsymbol\omega_q$ to (3.20) is strictly positive. The plug-in substitutes $\widehat{\boldsymbol V_{\boldsymbol c}},\hat L_j^\star:=L_j(\tau_n),\hat m,\hat{\boldsymbol\omega}_\gamma,\hat{\boldsymbol\omega}_q$ for their population analogues. The log-rate plug-in $\hat L_j^\star=L_j(\tau_n)=\log(k_j/(n_j(1-\tau_n)))$ is deterministic and converges to $L_j^\star$ by the chapter setup of Section 3.1; the remaining four convergences are the assumptions (3.40). The continuous-mapping theorem applied first to the polynomial (3.20) and then to $\sqrt{\cdot}$ on $(0,\infty)$ yields

$$\hat\sigma^\xi(\hat{\boldsymbol\omega}_\gamma,\hat{\boldsymbol\omega}_q)\;\xrightarrow{\mathbb{P}}\;\sigma^\xi(\boldsymbol\omega_\gamma,\boldsymbol\omega_q)\;>\;0.$$

Slutsky's theorem applied to the ratio of (A.38) and this consistent variance estimator produces the asymptotic pivot

$$Z_n \;:=\; \frac{\sqrt{k}\,\big(\log \widehat{\xi}^{\,\mathrm{pool}}_{\tau_n}(\hat{\boldsymbol{\omega}}_\gamma, \hat{\boldsymbol{\omega}}_q) - \log \xi_{\tau_n}\big)}{\hat{\sigma}^{\xi}(\hat{\boldsymbol{\omega}}_\gamma, \hat{\boldsymbol{\omega}}_q)} \;\xrightarrow{d}\; \mathcal{N}(0,1). \tag{A.39}$$

For every $\alpha \in (0,1)$, $\mathbb{P}(|Z_n| \le z_{1-\alpha/2}) \to 1-\alpha$. Inverting the inequality $|Z_n| \le z_{1-\alpha/2}$ for $\log \xi_{\tau_n}$ recovers exactly the event $\log \xi_{\tau_n} \in I_n^{\log}$ with $I_n^{\log}$ defined in (3.41), so $\mathbb{P}(\log \xi_{\tau_n} \in I_n^{\log}) \to 1-\alpha$.

Step (iv): transport to the original scale. The map $x \mapsto e^x$ is strictly increasing and continuous on $\mathbb{R}$, so for the random closed interval $I_n^{\log} = [a_n, b_n]$,

$$\log \xi_{\tau_n} \in [a_n, b_n] \quad\Longleftrightarrow\quad \xi_{\tau_n} \in [e^{a_n}, e^{b_n}] \;=\; \exp(I_n^{\log}) \;=\; I_n.$$

The two events coincide pointwise on the underlying probability space, hence $\mathbb{P}(\xi_{\tau_n} \in I_n) = \mathbb{P}(\log \xi_{\tau_n} \in I_n^{\log}) \to 1-\alpha$. □

A.7 Proof of Theorem 3.9

We prove the expectile-tail-homogeneity test of Theorem 3.9 under the standing assumptions of Section 3.1, with $\hat{\boldsymbol{\Sigma}}_n \xrightarrow{\mathbb{P}} \boldsymbol{\Sigma}$ and $\boldsymbol{\Sigma}$ positive definite. The argument composes six steps: (i) the per-sample QB log-expectile CLT, obtained by the single-sample specialisation of Appendix A.3; (ii) aggregation across the independent samples and rescaling to the common $\sqrt{k}$ scale, which yields the joint limit (3.43) with covariance (3.44) and mean (3.45); (iii) the GLS algebra reducing the statistic to a quadratic form in a single asymptotically Gaussian vector; (iv) the resulting noncentral $\chi^2_{m-1}(\delta)$ limit; (v) the two regimes in which $\delta = 0$; (vi) consistency under the alternative. Throughout, $S := \sum_{i=1}^m c_i^{-1}$, so that $(\boldsymbol{V_c})_{j,j} = \gamma^2\, c_j\, S$ by (2.14) and $k/k_j \to c_j\, S$ by (2.12), as established in Appendix A.2.

Step (i): per-sample QB log-expectile CLT. Fix $j \in \{1,\dots,m\}$. The per-sample QB log-expectile (3.42) decomposes exactly, as in (A.14) with $m=1$ and unit weight, into

$$\log \widehat{\xi}_{\tau_n,j} - \log \xi_{\tau_n} \;=\; \underbrace{\big[\log \psi(\hat{\gamma}_j(k_j)) - \log \psi(\gamma)\big]}_{(\mathrm{H})_j} \;+\; \underbrace{\big[\log \hat{q}^\star_j(1-\tau_n \mid k_j) - \log Q_X(\tau_n)\big]}_{(\mathrm{W})_j} \;+\; (\mathrm{G}),$$

with $(\mathrm{G}) = \log \psi(\gamma) + \log Q_X(\tau_n) - \log \xi_{\tau_n}$ the deterministic expectile–quantile gap, which is common to all samples. The Taylor linearisation of Step (ii) of Appendix A.3 gives $(\mathrm{H})_j = m(\gamma)(\hat{\gamma}_j(k_j) - \gamma) + O_p(k_j^{-1})$, so on the $\sqrt{k_j}$ scale

$$\sqrt{k_j}\,\big(\log \widehat{\xi}_{\tau_n,j} - \log \xi_{\tau_n}\big) \;=\; (m(\gamma),\,1)\,\sqrt{k_j}\begin{pmatrix} \hat{\gamma}_j(k_j) - \gamma \\ \log \hat{q}^\star_j(1-\tau_n \mid k_j) - \log Q_X(\tau_n) \end{pmatrix} + \sqrt{k_j}\,(\mathrm{G}) + o_p(1).$$

The within-sample 2-vector obeys the joint CLT (3.10) (proved as (A.11)), with mean $\boldsymbol{\mu}^\star_j = \big(b^H_j,\, L^\star_j\, b^H_j + b^R_j\big)^\top$, $b^R_j = -\lambda_j\, \Psi_\rho(e^{L^\star_j})$ (the order-statistic bias being zero by (A.8)), and covariance $\gamma^2(1, L^\star_j;\, L^\star_j, 1 + (L^\star_j)^2)$. The deterministic gap satisfies, by the second-order expansion (A.17) and the regular variation of A used in (A.18),

$$\sqrt{k_j}\,(\mathrm{G}) \;=\; -c(\gamma,\rho)\,\sqrt{k_j}\,A((1-\tau_n)^{-1})+o(1) \;\longrightarrow\; -c(\gamma,\rho)\,\lambda_j\,(e^{L_j^\star})^{\rho} \;=:\; g_j.$$

Reading the row $(m(\gamma),1)$ through the within-sample covariance gives the scalar variance $\gamma^2[(m(\gamma)+L_j^\star)^2+1]$ (the computation of (3.20) at $m=1$, unit weight), so Slutsky's theorem yields

$$\sqrt{k_j}\,(\log\widehat{\xi}_{\tau_n,j}-\log\xi_{\tau_n}) \;\xrightarrow{d}\; \mathcal{N}\Big(\beta_j^\circ,\;\gamma^2[(m(\gamma)+L_j^\star)^2+1]\Big), \qquad \beta_j^\circ \;=\; (m(\gamma)+L_j^\star)\,b_j^H+b_j^R+g_j. \tag{A.40}$$

Step (ii): aggregation and rescaling to $\sqrt{k}$. The m samples are independent (3.1), so the scalars in (A.40) are independent across j, and the vector $\sqrt{k}(\hat{\boldsymbol{\Xi}}_n-\log\xi_{\tau_n}\mathbf{1})$ has independent (hence jointly Gaussian, in the limit) coordinates obtained by multiplying (A.40) by $\sqrt{k/k_j}\to\sqrt{c_jS}$. The jth limiting variance is

$$c_jS\,\gamma^2[(m(\gamma)+L_j^\star)^2+1] \;=\; [(m(\gamma)+L_j^\star)^2+1]\,(\boldsymbol{V_c})_{j,j},$$

which is the jth diagonal entry of $\boldsymbol{\Sigma}$ in (3.44); the off-diagonal entries vanish by independence, so the limiting covariance is exactly $\boldsymbol{\Sigma}$. The jth limiting mean is $\sqrt{c_jS}\,\beta_j^\circ$. Splitting β_j° as in (A.40) and using the $\sqrt{k}$-rescaled bias identities of Appendix A.2,

$$\sqrt{c_jS}\,b_j^H=(\boldsymbol{B_c})_j, \qquad \sqrt{c_jS}\,(L_j^\star\,b_j^H+b_j^R)=(\boldsymbol{B_c^\star})_j,$$

the first from Theorem 2.15 and the second from (3.14), we obtain

$$\sqrt{c_jS}\,\beta_j^\circ \;=\; \sqrt{c_jS}\,[m(\gamma)\,b_j^H+(L_j^\star\,b_j^H+b_j^R)] \;+\; \sqrt{c_jS}\,g_j \;=\; m(\gamma)\,(\boldsymbol{B_c})_j+(\boldsymbol{B_c^\star})_j \;+\; \sqrt{c_jS}\,g_j.$$

For the last term, the per-j identity (A.19) reads $\sqrt{c_jS}\,\lambda_j\,(e^{L_j^\star})^\rho=\Lambda^\bullet$, so

$$\sqrt{c_jS}\,g_j \;=\; -c(\gamma,\rho)\,\sqrt{c_jS}\,\lambda_j\,(e^{L_j^\star})^\rho \;=\; -c(\gamma,\rho)\,\Lambda^\bullet \;=\; b^{\xi/Q}$$

by (3.18), *independently of j*. Hence $\sqrt{c_jS}\,\beta_j^\circ=m(\gamma)\,(\boldsymbol{B_c})_j+(\boldsymbol{B_c^\star})_j+b^{\xi/Q}$, i.e. the limiting mean vector is $\boldsymbol{\beta}=m(\gamma)\boldsymbol{B_c}+\boldsymbol{B_c^\star}+b^{\xi/Q}\mathbf{1}$ of (3.45). This proves (3.43).

Step (iii): GLS reduction to a quadratic form. Write $\boldsymbol{Y}_n:=\sqrt{k}(\hat{\boldsymbol{\Xi}}_n-\log\xi_{\tau_n}\mathbf{1})$, so $\boldsymbol{Y}_n\xrightarrow{d}\boldsymbol{Y}\sim\mathcal{N}(\boldsymbol{\beta},\boldsymbol{\Sigma})$ by Step (ii). Introduce the random oblique projection

$$\boldsymbol{P}_n \;:=\; \boldsymbol{I}-\frac{\mathbf{1}\,\mathbf{1}^\top\hat{\boldsymbol{\Sigma}}_n^{-1}}{\mathbf{1}^\top\hat{\boldsymbol{\Sigma}}_n^{-1}\mathbf{1}},$$

which satisfies $\boldsymbol{P}_n\mathbf{1}=\mathbf{0}$ and, from the definition of $\hat{\nu}_n$ in (3.46), $\hat{\boldsymbol{\Xi}}_n-\hat{\nu}_n\mathbf{1}=\boldsymbol{P}_n\hat{\boldsymbol{\Xi}}_n=\boldsymbol{P}_n(\hat{\boldsymbol{\Xi}}_n-\log\xi_{\tau_n}\mathbf{1})=k^{-1/2}\,\boldsymbol{P}_n\boldsymbol{Y}_n$, the second equality because $\boldsymbol{P}_n\mathbf{1}=\mathbf{0}$ removes the common centring. Substituting into (3.46),

$$\Lambda_n^\xi \;=\; k\,(k^{-1/2}\boldsymbol{P}_n\boldsymbol{Y}_n)^\top\hat{\boldsymbol{\Sigma}}_n^{-1}(k^{-1/2}\boldsymbol{P}_n\boldsymbol{Y}_n) \;=\; \boldsymbol{Y}_n^\top\,\boldsymbol{P}_n^\top\hat{\boldsymbol{\Sigma}}_n^{-1}\boldsymbol{P}_n\,\boldsymbol{Y}_n. \tag{A.41}$$

By hypothesis $\hat{\boldsymbol{\Sigma}}_n \xrightarrow{\mathbb{P}} \boldsymbol{\Sigma}$, so $\boldsymbol{P}_n \xrightarrow{\mathbb{P}} \boldsymbol{P} := \boldsymbol{I} - \mathbf{1}\mathbf{1}^\top\boldsymbol{\Sigma}^{-1}/(\mathbf{1}^\top\boldsymbol{\Sigma}^{-1}\mathbf{1})$ and $\boldsymbol{P}_n^\top\hat{\boldsymbol{\Sigma}}_n^{-1}\boldsymbol{P}_n \xrightarrow{\mathbb{P}} \boldsymbol{M} := \boldsymbol{P}^\top\boldsymbol{\Sigma}^{-1}\boldsymbol{P}$, both by the continuous-mapping theorem. A direct computation, writing $\boldsymbol{w} := \boldsymbol{\Sigma}^{-1}\mathbf{1}$ and $s := \mathbf{1}^\top\boldsymbol{\Sigma}^{-1}\mathbf{1} = \mathbf{1}^\top\boldsymbol{w}$, gives the standard GLS identity

$$\boldsymbol{M} \;=\; \boldsymbol{P}^\top\boldsymbol{\Sigma}^{-1}\boldsymbol{P} \;=\; \boldsymbol{\Sigma}^{-1} \;-\; \frac{\boldsymbol{\Sigma}^{-1}\mathbf{1}\,\mathbf{1}^\top\boldsymbol{\Sigma}^{-1}}{\mathbf{1}^\top\boldsymbol{\Sigma}^{-1}\mathbf{1}}, \tag{A.42}$$

since the three cross terms in the expansion of $(\boldsymbol{I} - \boldsymbol{w}\mathbf{1}^\top/s)\boldsymbol{\Sigma}^{-1}(\boldsymbol{I} - \mathbf{1}\boldsymbol{w}^\top/s)$ each reduce to $\boldsymbol{w}\boldsymbol{w}^\top/s$ (using $\mathbf{1}^\top\boldsymbol{w} = s$ and symmetry of $\boldsymbol{\Sigma}^{-1}$), leaving $\boldsymbol{\Sigma}^{-1} - \boldsymbol{w}\boldsymbol{w}^\top/s$. The matrix $\boldsymbol{M}$ is symmetric positive semi-definite and satisfies $\boldsymbol{M}\mathbf{1} = \boldsymbol{w} - \boldsymbol{w}(s/s) = \mathbf{0}$. By (A.41), Slutsky's theorem, and the continuous-mapping theorem applied to the (jointly continuous) quadratic-form map $(\boldsymbol{A}, \boldsymbol{y}) \mapsto \boldsymbol{y}^\top\boldsymbol{A}\boldsymbol{y}$,

$$\Lambda_n^\xi \xrightarrow{d} \boldsymbol{Y}^\top\boldsymbol{M}\,\boldsymbol{Y}, \qquad \boldsymbol{Y} \sim \mathcal{N}(\boldsymbol{\beta}, \boldsymbol{\Sigma}). \tag{A.43}$$

Step (iv): the noncentral chi-square limit. Standardise $\boldsymbol{Z} := \boldsymbol{\Sigma}^{-1/2}\boldsymbol{Y} \sim \mathcal{N}(\boldsymbol{\Sigma}^{-1/2}\boldsymbol{\beta}, \boldsymbol{I}_m)$, where $\boldsymbol{\Sigma}^{-1/2}$ is the symmetric positive-definite square root of $\boldsymbol{\Sigma}^{-1}$. Then $\boldsymbol{Y}^\top\boldsymbol{M}\boldsymbol{Y} = \boldsymbol{Z}^\top\widetilde{\boldsymbol{M}}\boldsymbol{Z}$ with

$$\widetilde{\boldsymbol{M}} \;:=\; \boldsymbol{\Sigma}^{1/2}\boldsymbol{M}\,\boldsymbol{\Sigma}^{1/2} \;=\; \boldsymbol{I}_m - \frac{\boldsymbol{v}\boldsymbol{v}^\top}{\boldsymbol{v}^\top\boldsymbol{v}}, \qquad \boldsymbol{v} := \boldsymbol{\Sigma}^{-1/2}\mathbf{1},$$

using (A.42) and $\boldsymbol{\Sigma}^{1/2}\boldsymbol{w} = \boldsymbol{\Sigma}^{-1/2}\mathbf{1} = \boldsymbol{v}$, $s = \boldsymbol{v}^\top\boldsymbol{v}$. The matrix $\widetilde{\boldsymbol{M}}$ is the orthogonal projection onto $\boldsymbol{v}^\perp$: it is symmetric, idempotent, and of rank $m-1$. A quadratic form $\boldsymbol{Z}^\top\widetilde{\boldsymbol{M}}\boldsymbol{Z}$ in a unit-covariance Gaussian $\boldsymbol{Z} \sim \mathcal{N}(\boldsymbol{\mu}_Z, \boldsymbol{I}_m)$ with $\widetilde{\boldsymbol{M}}$ a rank-r orthogonal projection is noncentral χ^2_r with noncentrality $\boldsymbol{\mu}_Z^\top\widetilde{\boldsymbol{M}}\boldsymbol{\mu}_Z$. Here $r = m-1$ and $\boldsymbol{\mu}_Z = \boldsymbol{\Sigma}^{-1/2}\boldsymbol{\beta}$, so

$$\Lambda_n^\xi \xrightarrow{d} \chi^2_{m-1}(\delta), \qquad \delta \;=\; \boldsymbol{\beta}^\top\boldsymbol{\Sigma}^{-1/2}\widetilde{\boldsymbol{M}}\boldsymbol{\Sigma}^{-1/2}\boldsymbol{\beta} \;=\; \boldsymbol{\beta}^\top\boldsymbol{M}\boldsymbol{\beta} \;=\; \boldsymbol{\beta}^\top\boldsymbol{\Sigma}^{-1}\boldsymbol{\beta} - \frac{(\mathbf{1}^\top\boldsymbol{\Sigma}^{-1}\boldsymbol{\beta})^2}{\mathbf{1}^\top\boldsymbol{\Sigma}^{-1}\mathbf{1}},$$

the last equality by (A.42). This is (3.48). Since $\widetilde{\boldsymbol{M}}$ is positive semi-definite, $\delta \geq 0$, with $\delta = 0$ iff $\widetilde{\boldsymbol{M}}\boldsymbol{\Sigma}^{-1/2}\boldsymbol{\beta} = \mathbf{0}$, i.e. $\boldsymbol{\Sigma}^{-1/2}\boldsymbol{\beta} \in \operatorname{span}\{\boldsymbol{v}\} = \operatorname{span}\{\boldsymbol{\Sigma}^{-1/2}\mathbf{1}\}$, equivalently $\boldsymbol{\beta} \in \operatorname{span}\{\mathbf{1}\}$.

Step (v): the two regimes with $\delta = 0$. *(i) Undersmoothing.* Under (3.39), $\lambda_j = 0$ for every j, so the per-sample $\sqrt{k_j}$-biases $b_j^H = \lambda_j/(1-\rho)$ and $b_j^R = -\lambda_j\Psi_\rho(e^{L_j^\star})$ both vanish (the order-statistic bias being identically zero), and $\Lambda^\bullet = 0$, whence $\boldsymbol{B_c} = \boldsymbol{B_c^\star} = \mathbf{0}$ and $b^{\xi/Q} = 0$. Therefore $\boldsymbol{\beta} = \mathbf{0} \in \operatorname{span}\{\mathbf{1}\}$ and $\delta = 0$.

(ii) Equal-fraction distributed inference. Here $L_j^\star = L^\star$ is sample-free. The two per-sample $\sqrt{k_j}$-biases are proportional to λ_j with coefficients depending only on the common parameters $(\gamma, \rho, L^\star)$: $b_j^H = \lambda_j/(1-\rho)$ and $b_j^R = -\lambda_j\,\Psi_\rho(e^{L^\star})$, the order-statistic bias being identically zero. The $\sqrt{k}$-rescaling factor satisfies, by (A.19) at $L_j^\star = L^\star$,

$$\sqrt{c_j S}\,\lambda_j \;=\; \Lambda^\bullet\,(e^{L^\star})^{-\rho}, \qquad \text{sample-free.}$$

Consequently $(\boldsymbol{B_c})_j = \sqrt{c_j S}\,b_j^H = \Lambda^\bullet(e^{L^\star})^{-\rho}/(1-\rho)$ and $(\boldsymbol{B_c^\star})_j = \sqrt{c_j S}\,(L^\star b_j^H + b_j^R) = \Lambda^\bullet(e^{L^\star})^{-\rho}[L^\star/(1-\rho) - \Psi_\rho(e^{L^\star})]$ are both sample-free, and $b^{\xi/Q}\mathbf{1}$ is collinear with $\mathbf{1}$ by construction. Hence every coordinate of $\boldsymbol{\beta} = m(\gamma)\boldsymbol{B_c} + \boldsymbol{B_c^\star} + b^{\xi/Q}\mathbf{1}$ equals the common constant

$$\beta_0 \;=\; \Lambda^\bullet(e^{L^\star})^{-\rho}\left[\frac{m(\gamma)+L^\star}{1-\rho} - \Psi_\rho(e^{L^\star})\right] + b^{\xi/Q},$$

so $\boldsymbol{\beta} = \beta_0\mathbf{1} \in \operatorname{span}\{\mathbf{1}\}$ and $\delta = 0$. This is the expectile-level counterpart of the bias-collinearity that renders the tail-index test Theorem 2.19 asymptotically central under the same distributed-inference regime.

Step (vi): consistency under the alternative. Write $\boldsymbol{\Xi}_n^\star := (\log\xi_{\tau_n,1},\dots,\log\xi_{\tau_n,m})^\top$ for the vector of *population* per-sample log-expectiles, and abbreviate $\boldsymbol{D}_n := \hat{\boldsymbol{\Sigma}}_n^{-1} \xrightarrow{\mathbb{P}} \boldsymbol{\Sigma}^{-1} \succ \mathbf{0}$ with the induced seminorm $\|\boldsymbol{x}\|_{\boldsymbol{D}_n}^2 := \boldsymbol{x}^\top\boldsymbol{D}_n\boldsymbol{x}$. Each $\widehat{\xi}_{\tau_n,j}$ is consistent for its own $\xi_{\tau_n,j}$, so the per-sample CLT (A.40) recentred at $\boldsymbol{\Xi}_n^\star$ gives $\hat{\boldsymbol{\Xi}}_n - \boldsymbol{\Xi}_n^\star = O_p(k^{-1/2})$. Because the GLS residual $\hat{\boldsymbol{\Xi}}_n - \hat{\nu}_n\mathbf{1} = \boldsymbol{P}_n\hat{\boldsymbol{\Xi}}_n$ is, by definition of $\hat{\nu}_n$, the minimiser of $c \mapsto \|\hat{\boldsymbol{\Xi}}_n - c\mathbf{1}\|_{\boldsymbol{D}_n}^2$, the statistic is the squared GLS distance of $\hat{\boldsymbol{\Xi}}_n$ from $\operatorname{span}\{\mathbf{1}\}$:

$$\Lambda_n^\xi \;=\; k\,\min_{c\in\mathbb{R}}\big\|\hat{\boldsymbol{\Xi}}_n - c\mathbf{1}\big\|_{\boldsymbol{D}_n}^2.$$

By the triangle inequality for $\|\cdot\|_{\boldsymbol{D}_n}$ and the bound $\min_c\|\boldsymbol{x}-c\mathbf{1}\|_{\boldsymbol{D}_n} \ge \lambda_{\min}(\boldsymbol{D}_n)^{1/2}\min_c\|\boldsymbol{x}-c\mathbf{1}\|$ (valid because, at the $\|\cdot\|_{\boldsymbol{D}_n}$-optimal c, the entrywise inequality $\boldsymbol{x}^\top\boldsymbol{D}_n\boldsymbol{x} \ge \lambda_{\min}(\boldsymbol{D}_n)\|\boldsymbol{x}\|^2$ applies),

$$\sqrt{\Lambda_n^\xi} \;=\; \sqrt{k}\,\min_c\big\|\hat{\boldsymbol{\Xi}}_n - c\mathbf{1}\big\|_{\boldsymbol{D}_n} \;\ge\; \lambda_{\min}(\boldsymbol{D}_n)^{1/2}\sqrt{k}\,\min_c\big\|\boldsymbol{\Xi}_n^\star - c\mathbf{1}\big\| \;-\; \sqrt{k}\,\big\|\hat{\boldsymbol{\Xi}}_n - \boldsymbol{\Xi}_n^\star\big\|_{\boldsymbol{D}_n}.$$

The subtracted term is $O_p(1)$ since $\sqrt{k}(\hat{\boldsymbol{\Xi}}_n - \boldsymbol{\Xi}_n^\star) = O_p(1)$ and $\boldsymbol{D}_n = O_p(1)$. Under the stated alternative $\sqrt{k}\,\min_c\|\boldsymbol{\Xi}_n^\star - c\mathbf{1}\| \to \infty$, while $\lambda_{\min}(\boldsymbol{D}_n) \xrightarrow{\mathbb{P}} \lambda_{\min}(\boldsymbol{\Sigma}^{-1}) > 0$ by positive definiteness of $\boldsymbol{\Sigma}$. The first term on the right therefore diverges in probability and dominates the $O_p(1)$ correction, so $\Lambda_n^\xi \xrightarrow{\mathbb{P}} \infty$. $\square$

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